HANDS-ON LAB GUIDE

Building a Virtual Lab with Active Directory

A Step-by-Step Guide to Deploying
Windows Server 2025 & Windows 10 in VirtualBox

PLATFORM	SERVER OS	CLIENT OS
Oracle VirtualBox	**Windows Server 2025**	**Windows 10**

IT Students · System Administrators · Cybersecurity Professionals

Author: Ryan Adams
Date: July 2025

Overview

What you will build and why it matters

This guide provides step-by-step instructions for building a Windows Server 2025 domain environment using Oracle VirtualBox. Designed for IT students, system administrators, and cybersecurity professionals, this lab simulates real-world enterprise network infrastructure in a safe, isolated virtual environment.

What You Will Build

- Windows Server 2025 Domain Controller with Active Directory Domain Services
- Windows 10 Client Systems that join and authenticate to the domain
- DHCP and DNS Services for automated network configuration and name resolution
- NAT Configuration providing internet connectivity for domain clients
- Structured Network Design with IP addressing (static: 1-100, dynamic: 101-200)

Learning Objectives

- Virtualization Technology - VirtualBox configuration and VM management
- Windows Server Administration - Install, configure, and manage Server 2025
- Active Directory Fundamentals - Directory services implementation
- Network Infrastructure - Enterprise network topology and IP addressing
- Service Configuration - Deploy DHCP, DNS, and Remote Access services
- Domain Management - Centralized authentication and domain operations
- Troubleshooting Skills - Diagnose and resolve common issues

Why This Lab Matters

Understanding Windows Server and Active Directory is crucial for system administrators, cybersecurity professionals, IT support specialists, and students pursuing IT certifications. This lab provides hands-on experience that directly applies to real-world enterprise scenarios.

Let's begin building your virtual enterprise network!
Follow each section in order - every step builds on the last.

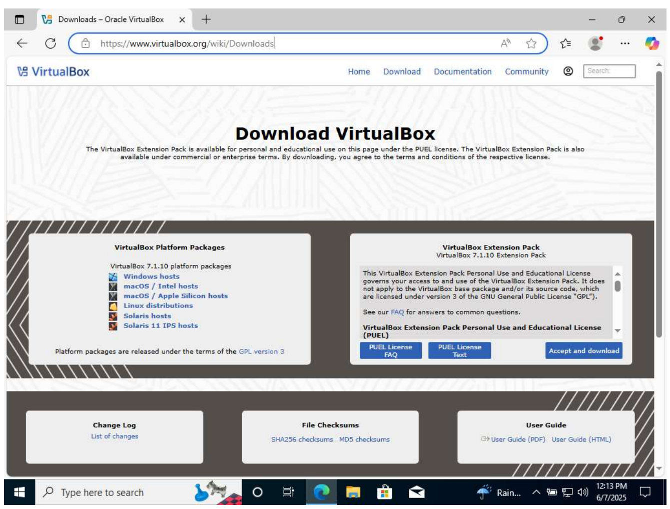
Table of Contents

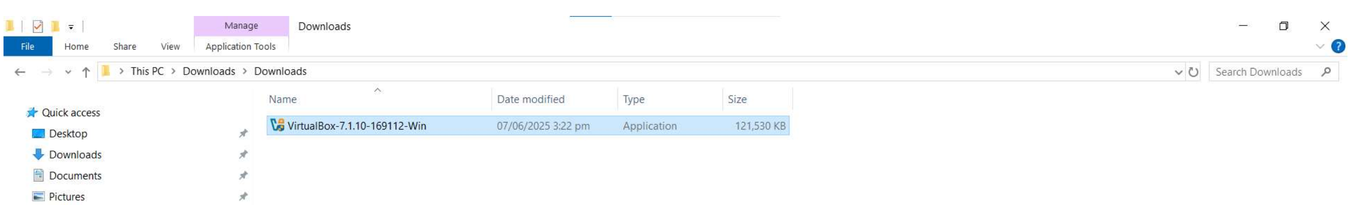
Step-by-step lab sections

4. Installing VirtualBox

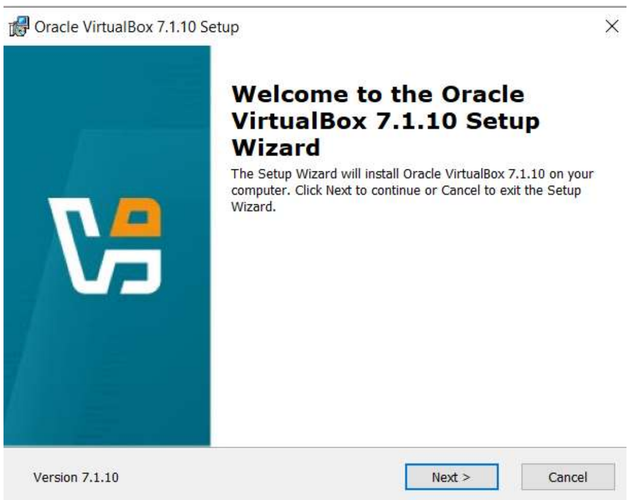
Install VirtualBox

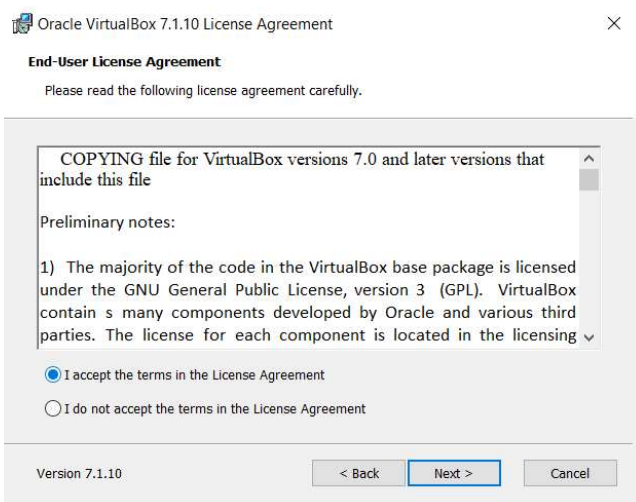
- Open your web browser and navigate to https://www.virtualbox.org/wiki/Downloads and click on your platform package, the download should automatically start.
- Navigate to the folder where you downloaded your file and double click it to start the installation process.



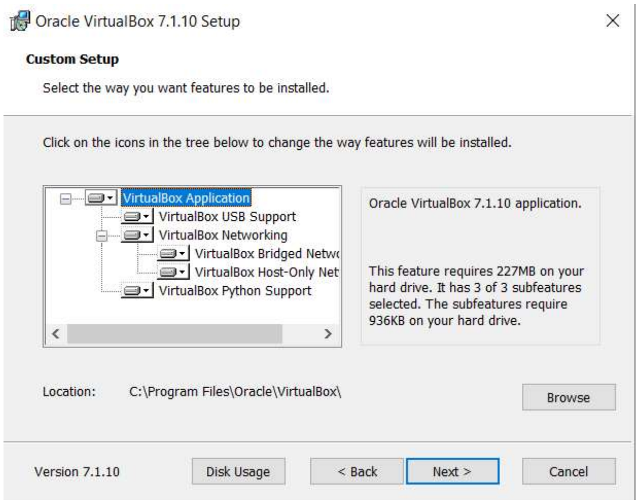


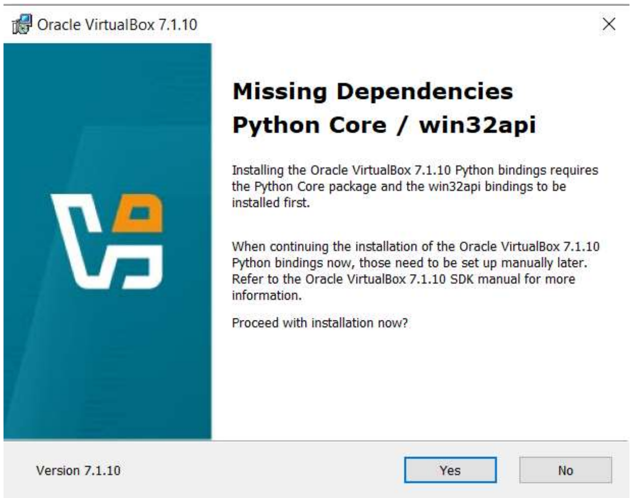
- Click Next.
- Accept licensing terms and Click Next.



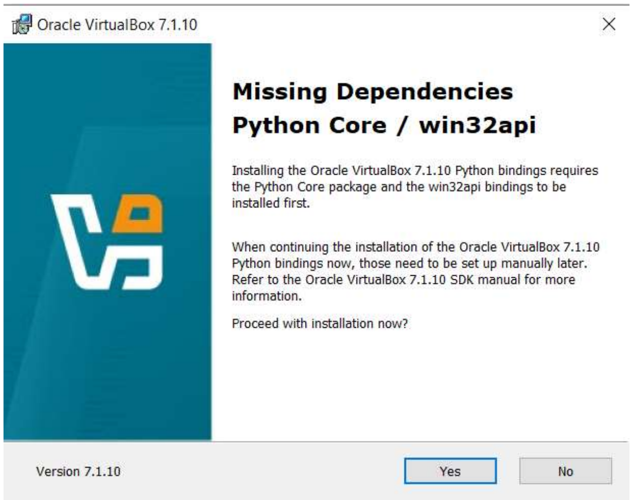


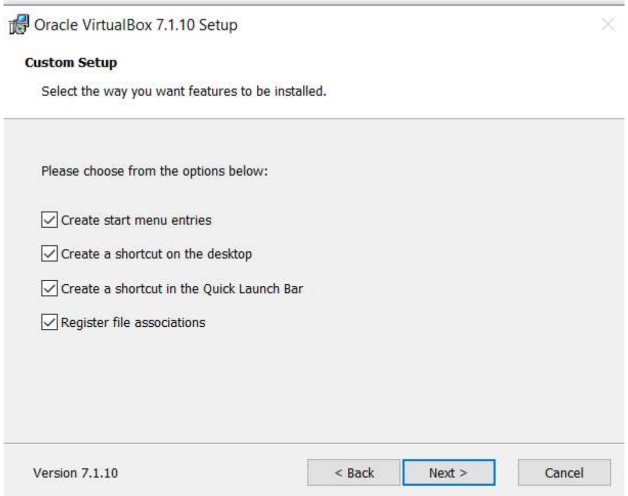
- Click Next.
- Click Yes to Warning.



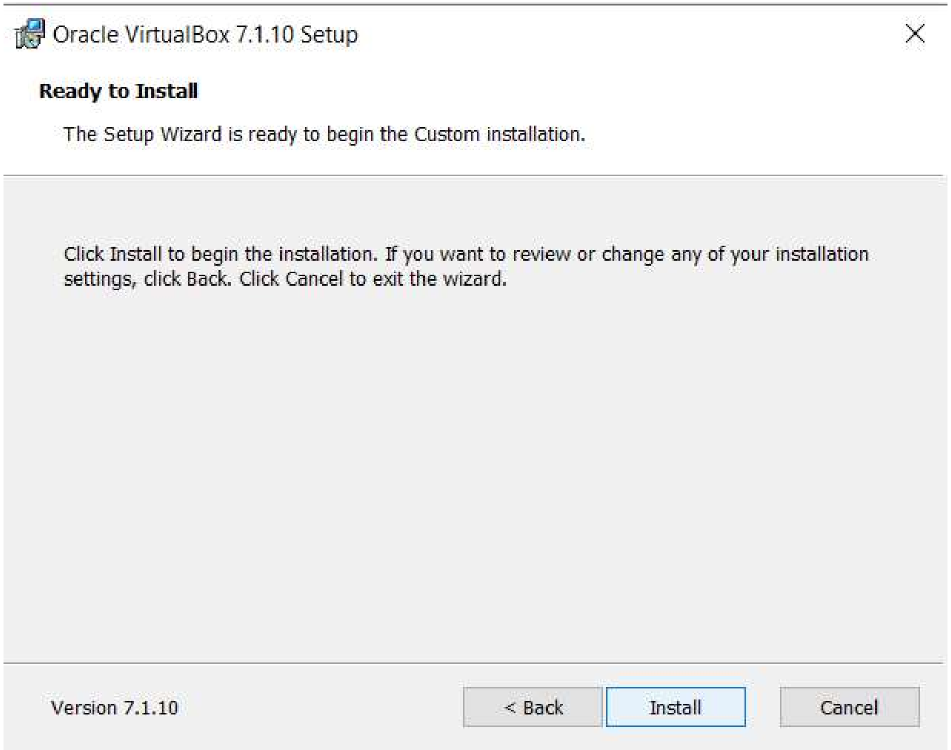


- Click Yes to Proceed with the installation.
- Click Next





Click continue to Install



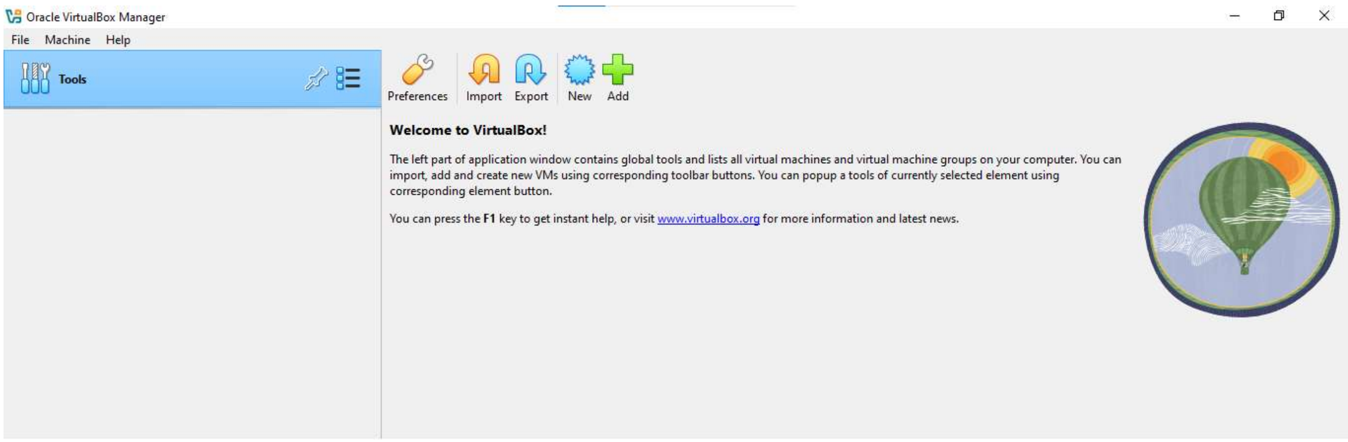
5. Creating Virtual Machines

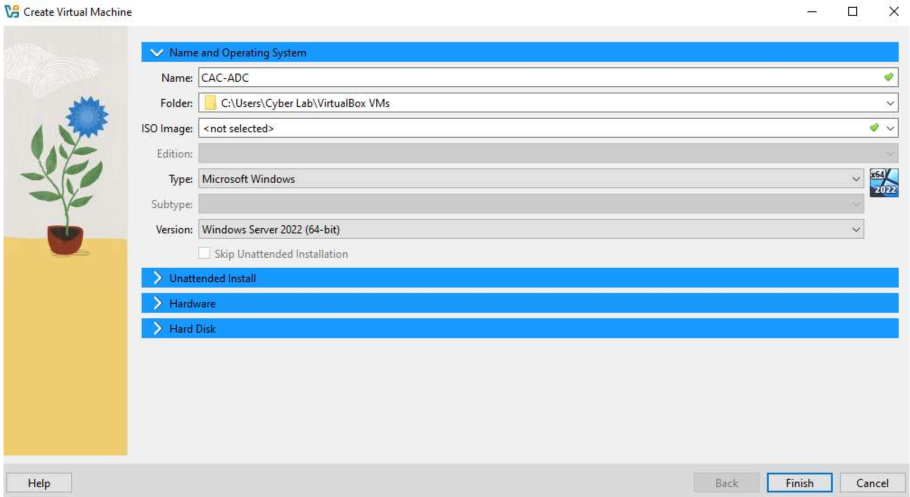
Windows Server 2025

- Open VirtualBox
- Click on the New button and a box will pop up.

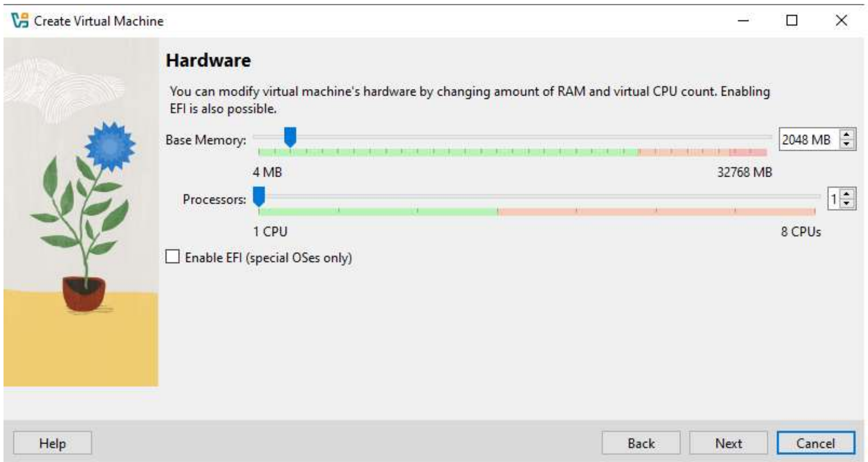
Enter a name for the virtual machine and select the appropriate Operating

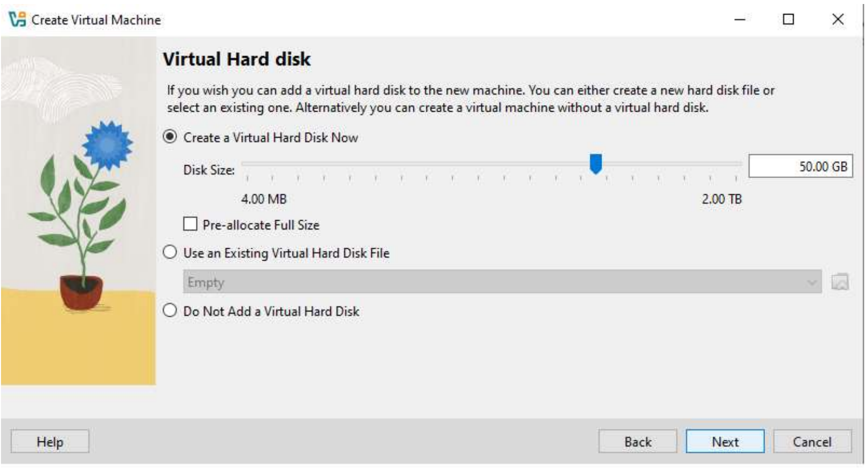
System type and version. Click on Next.



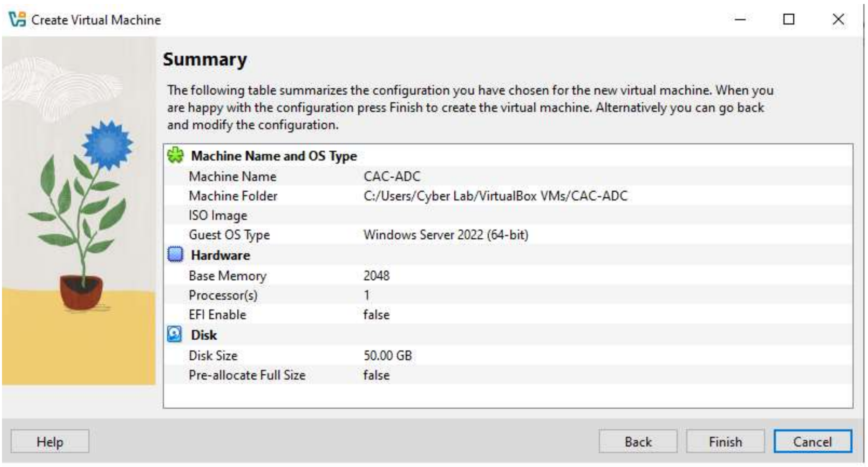


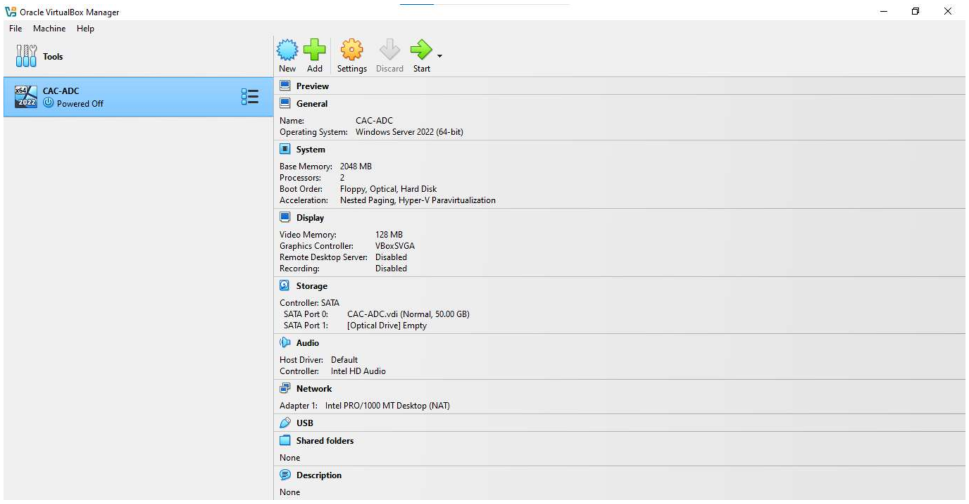
- Select the Base Memory and number of Processors.
- Select Create a Virtual Hard Disk Now. Click Next.



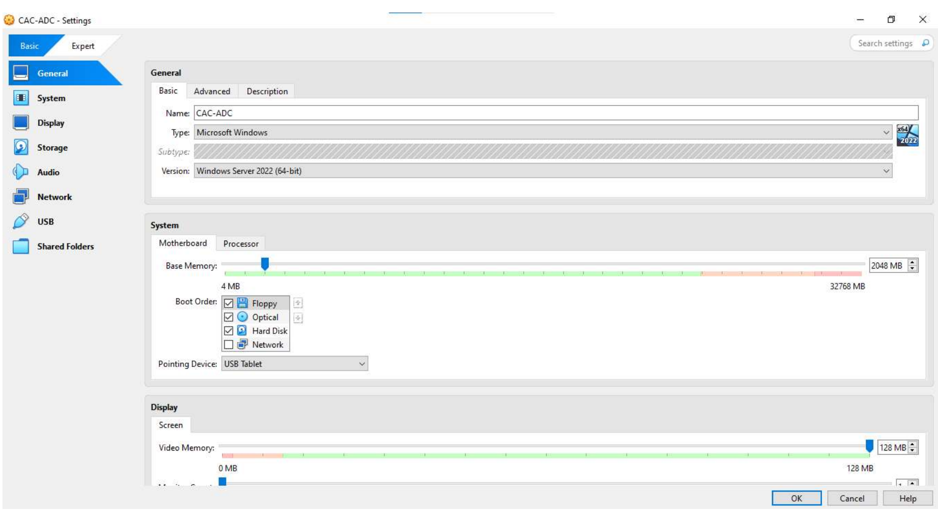


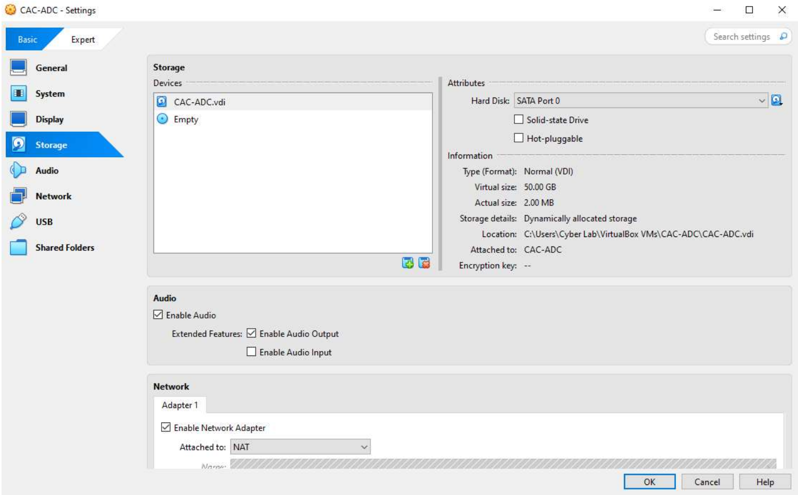
- Click on Finish.
- You should now see the following screen. Click on Settings.



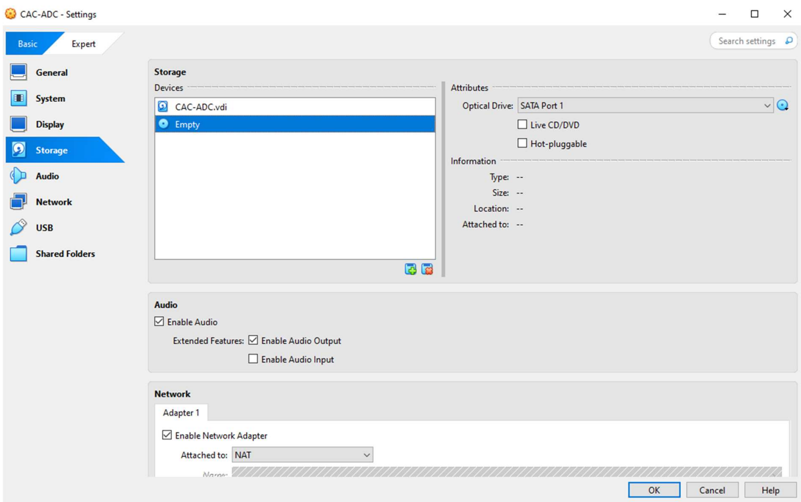


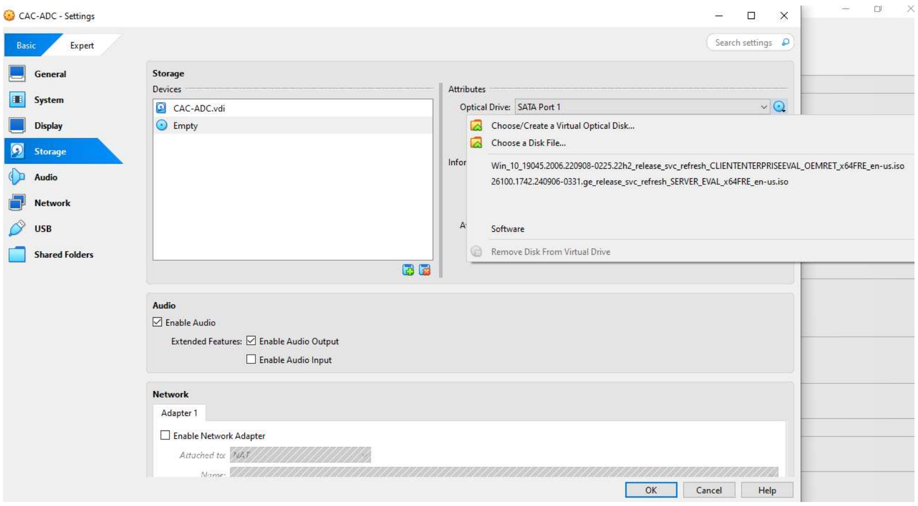
- After clicking on Setting, a box will pop up.
- Click Storage.





- Click on Empty.
- Click on the disk icon, a drop-down will pop up.

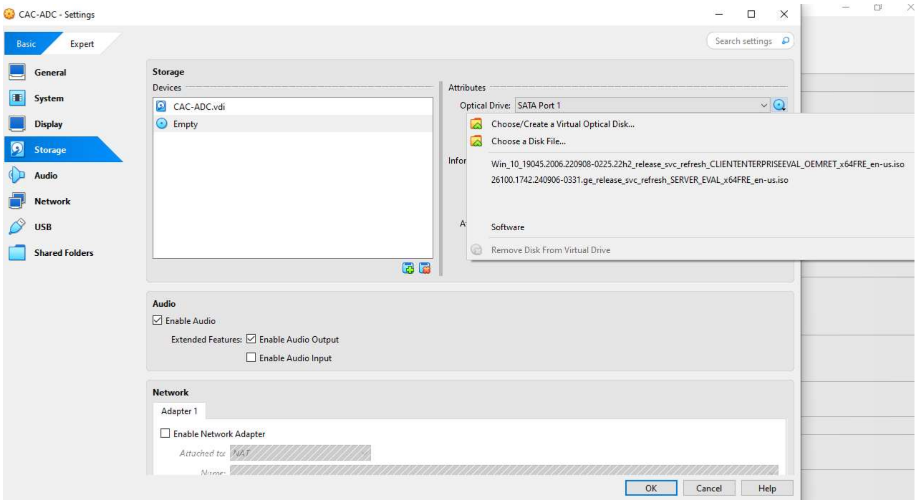


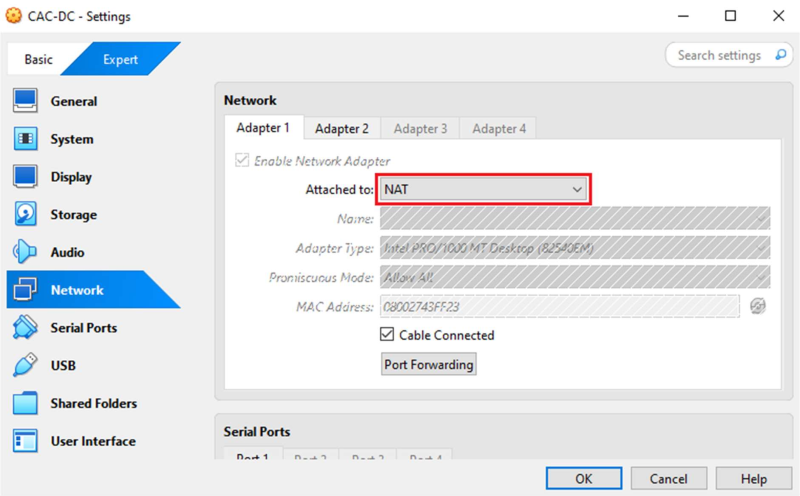


- Click on Choose a disk file, navigate to the folder where the Windows

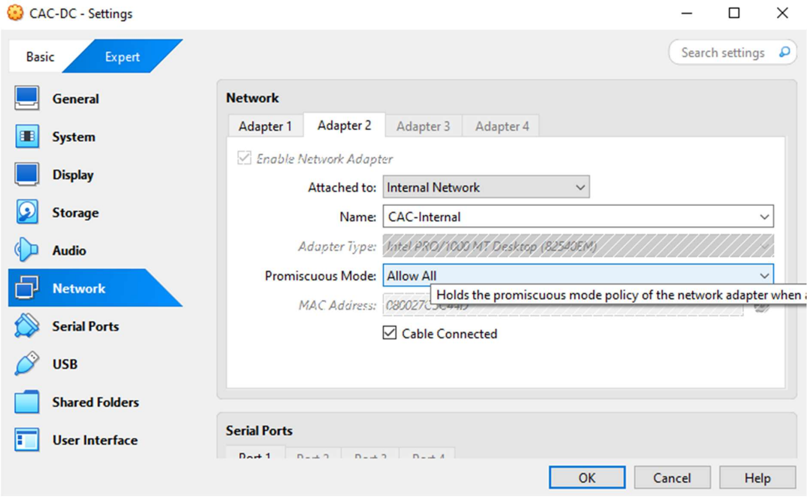
Server ISO file was saved and attach it. Click OK.

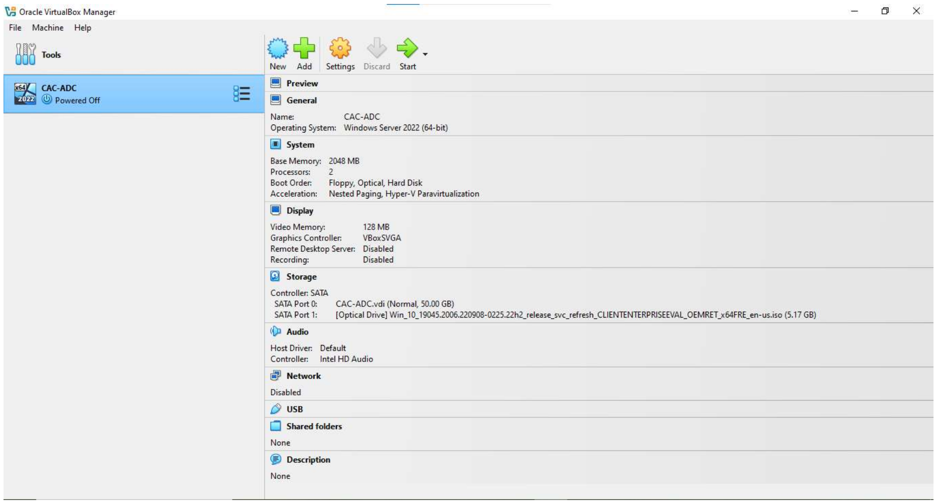
- Click Network and set Adapter 1 to Attached to: NAT.





- Enable Network Adapter 2 and set Attached to: Internal Network and type a Name for your network. I named mine CAC-Internal.
- Now the ISO file is attached to the virtual optical drive, Start the virtual machine to install Windows Server.

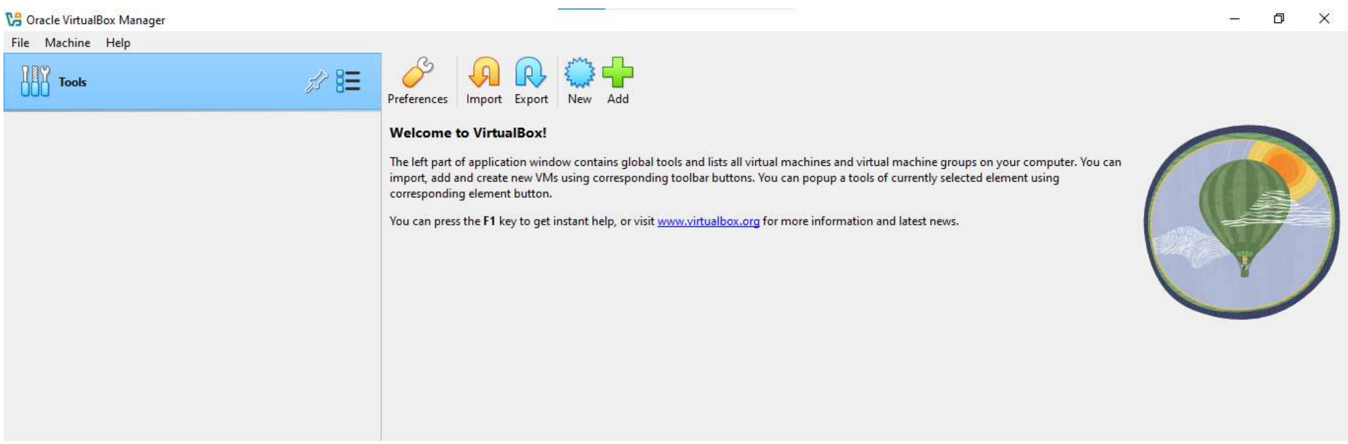


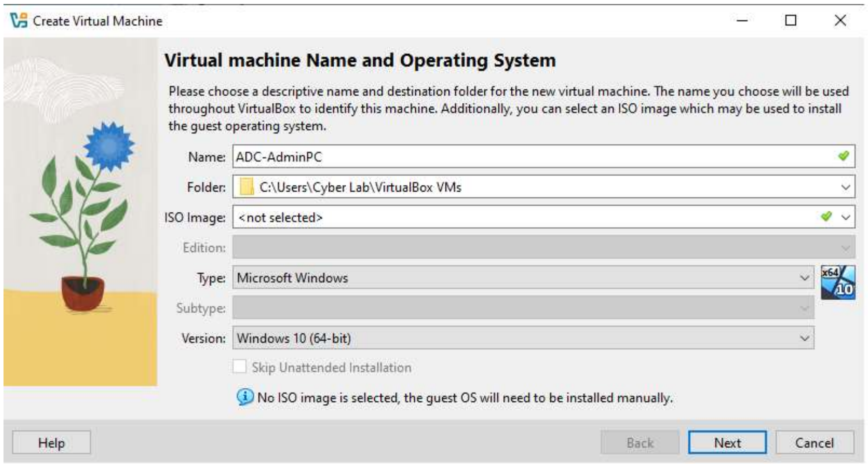


Windows 10

- Open VirtualBox
- Click on the New button and a box will pop up.
- Enter then name of the virtual machine and select the appropriate

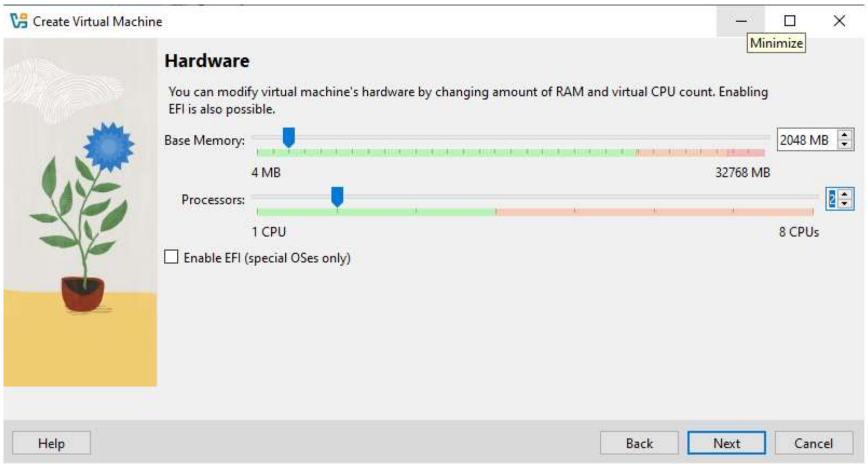
Operating System type and version. Click on Next.

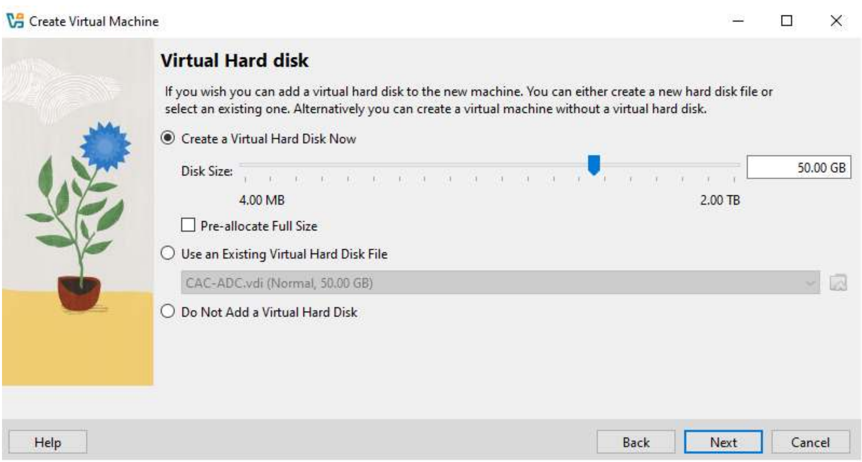




Select your Base Memory and Processors

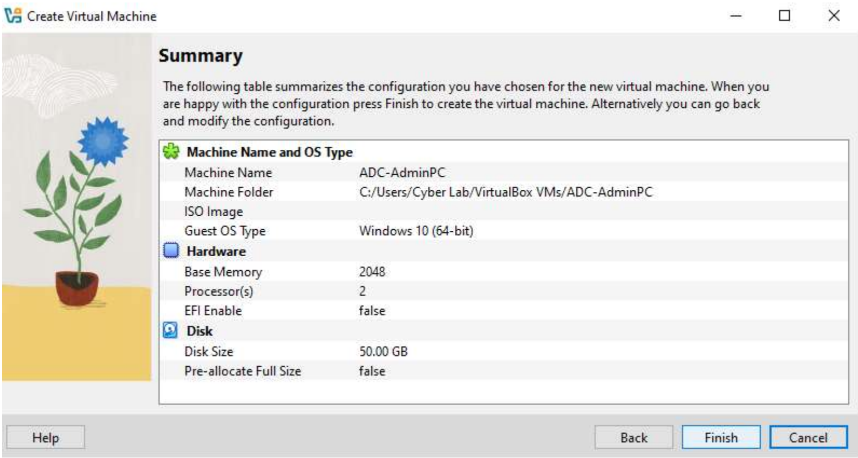
- Select Create a Virtual Hard Disk Now. Click Next.

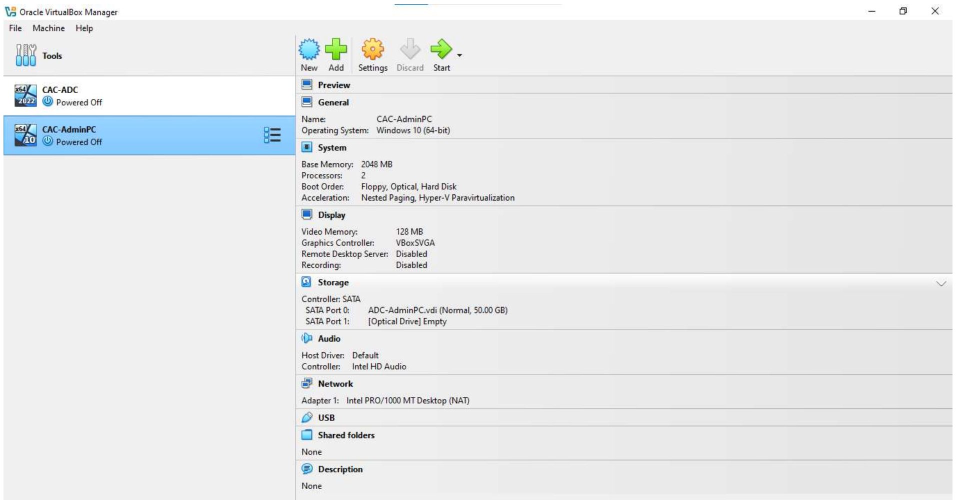




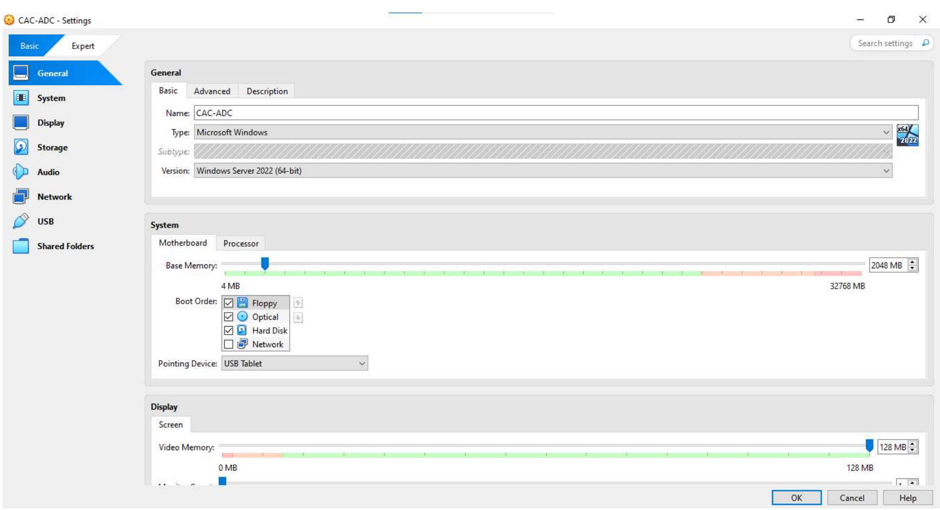
- Click on Finish.
- You should now see the following screen. Select Machine Click on

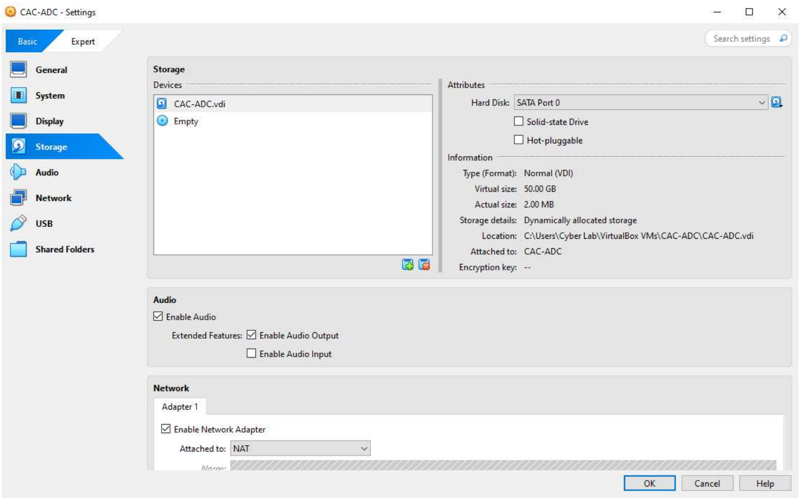
Settings.





- After clicking on Settings, a box will pop up.
- Click Storage.





- Click on Empty.
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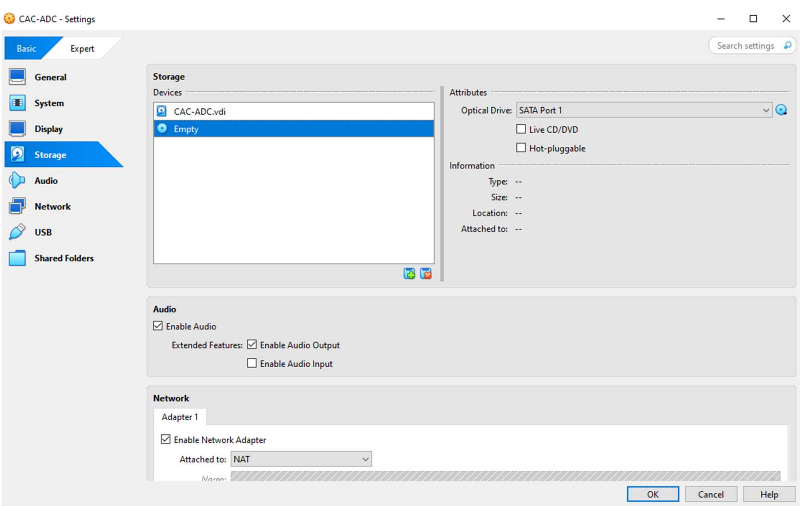
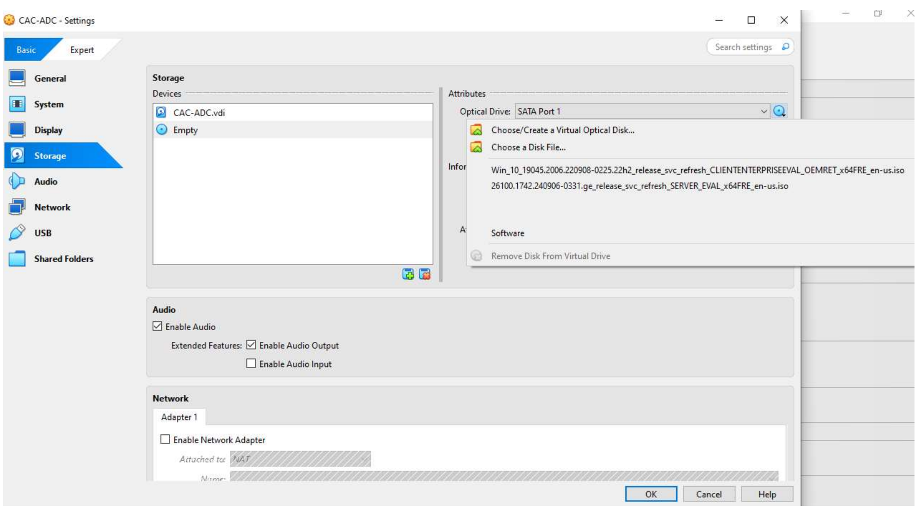
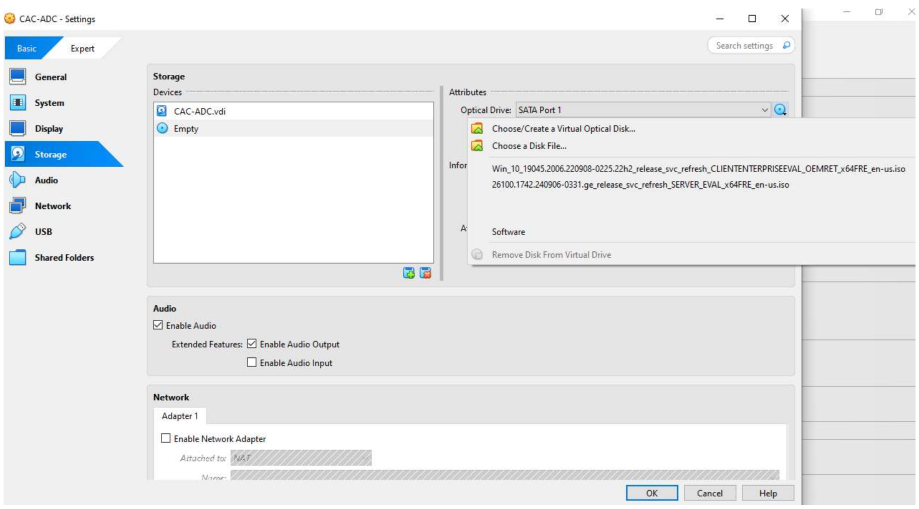
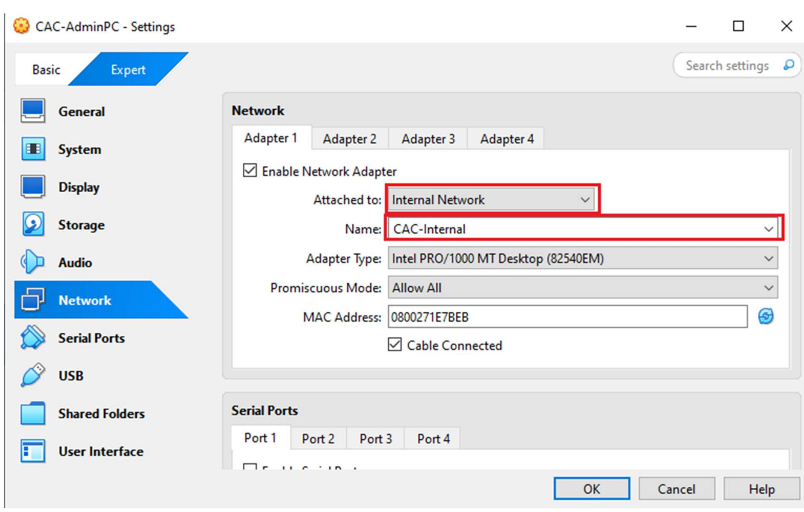


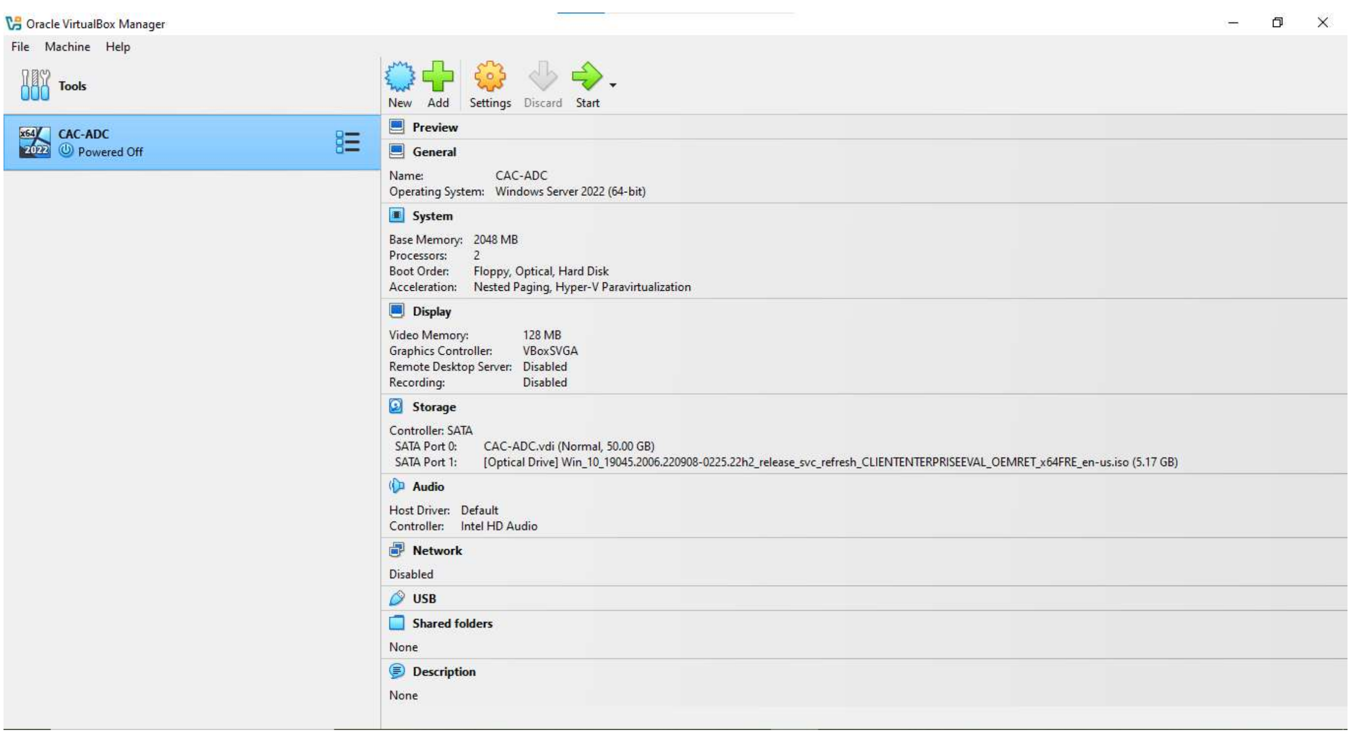


- Click on Choose a disk file, navigate to the folder where the Windows 10 ISO file was saved and attach it. Click OK.
- Click Network and set Attached to: Internal Network and select of type the Name of the Internal Network you created for Windows Server. (mine CAC-Internal)





- Now the ISO file is attached to the virtual optical drive, Start the virtual machine to install Windows Server.



6. VirtualBox GuestAdditions

VirtualBox Guest Additions are essential tools installed inside a virtual machine after its operating system is set up. They include drivers and applications that enhance the VM's performance and usability.

Key Features of Guest Additions

■ Mouse Pointer Integration: Provides seamless mouse control, eliminating the need to use the Host key to switch between the host and guest.

■ Shared Folders: Allows easy file exchange between the host and guest operating systems by treating host directories as network shares.

■ Improved Video Support: Delivers higher resolutions, non-standard video modes, and accelerated video performance. It also enables automatic window resizing for Windows, Linux, and Oracle Solaris guests.

■ Accelerated Graphics: Supports hardware-accelerated 3D graphics and 2D video for applications running in the guest.

■ Seamless Windows: Maps individual guest application windows directly onto the host's desktop, making them appear as if they're running natively on the host.

■ Host/Guest Communication Channels: Offers a mechanism for the host to control and monitor guest execution, including exchanging data and starting guest applications from the host.

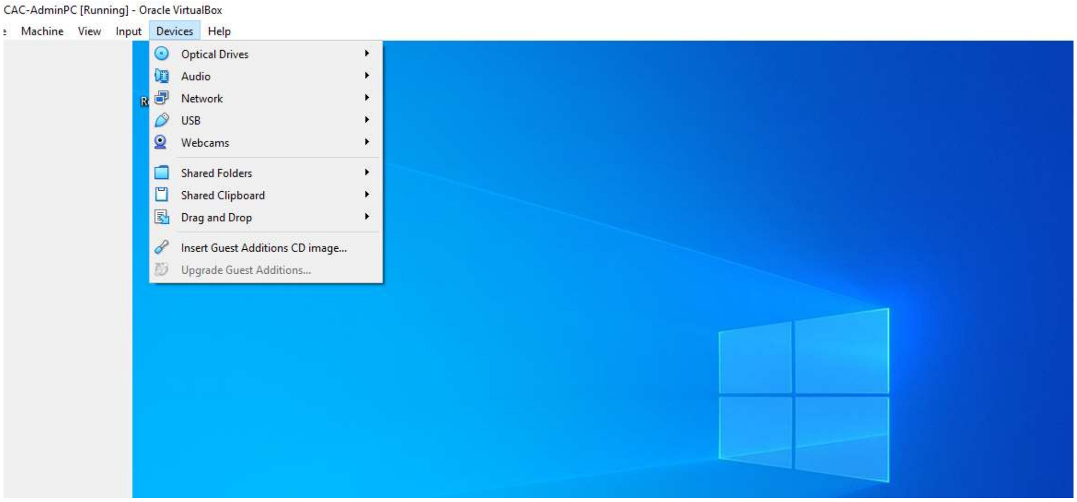
■ Time Synchronization: Keeps the guest's system time synchronized with the host, adjusting for differences caused by various factors like pausing the VM or restoring from a saved state.

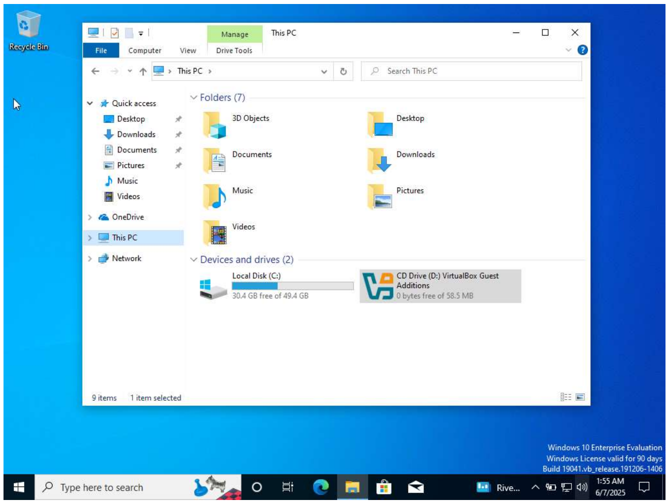
■ Shared Clipboard: Enables copying and pasting content between the host and guest operating systems.

■ Automated Logins: Also known as credentials passing, this feature simplifies the login process within the guest.

Installing Guest Additions

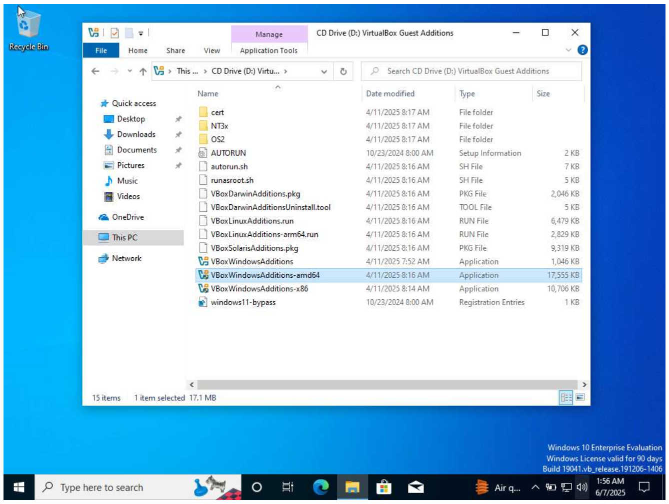
- Start your virtual machine. Once the guest OS has booted, go to the VirtualBox menu bar at the top of your virtual machine window and click Devices >> Insert Guest Additions CD Image.
- Navigate to This PC and double-click the VirtualBox Guest Additions CD drive.

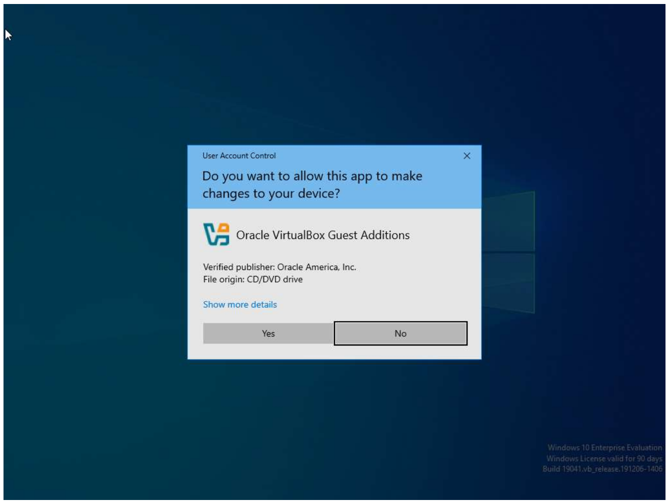




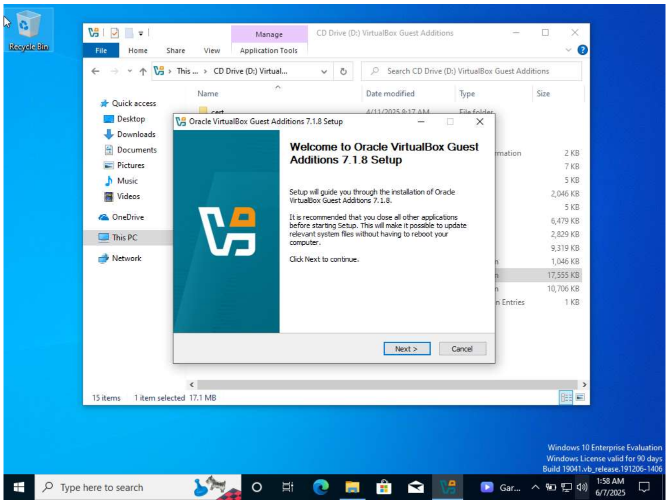
Run the installer by double-clicking VBoxWindowsAdditions-amd64

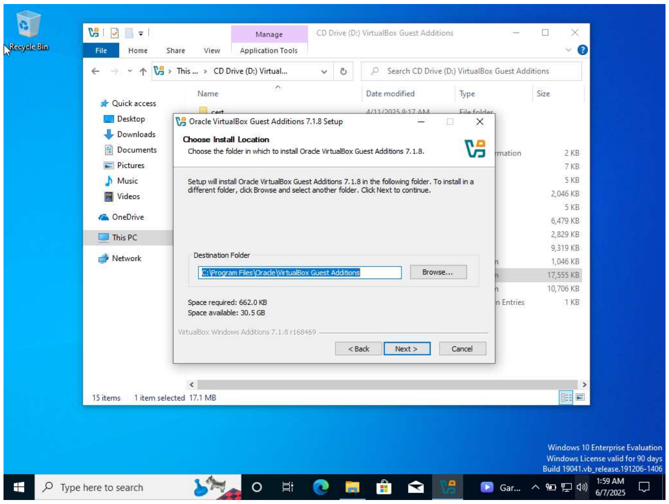
- Click Yes to continue.



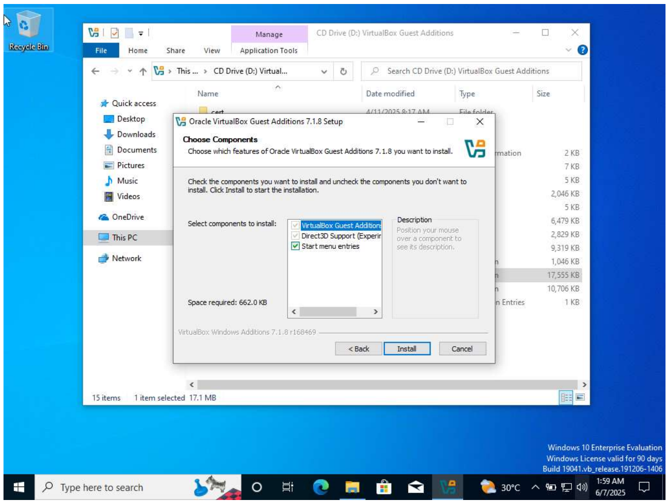


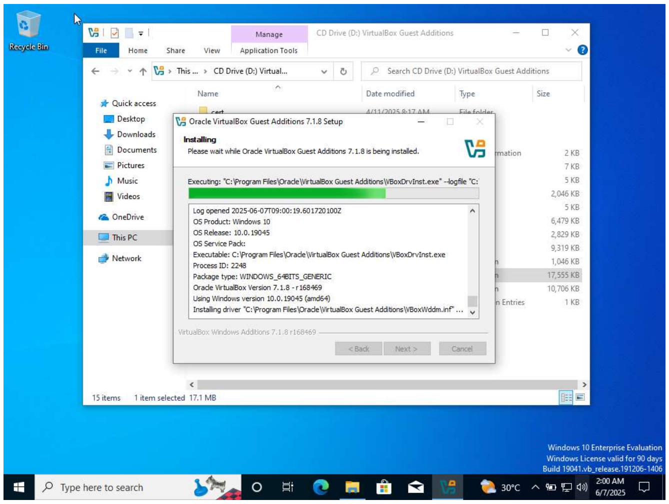
- Click Next to continue.
- Click Next to continue.





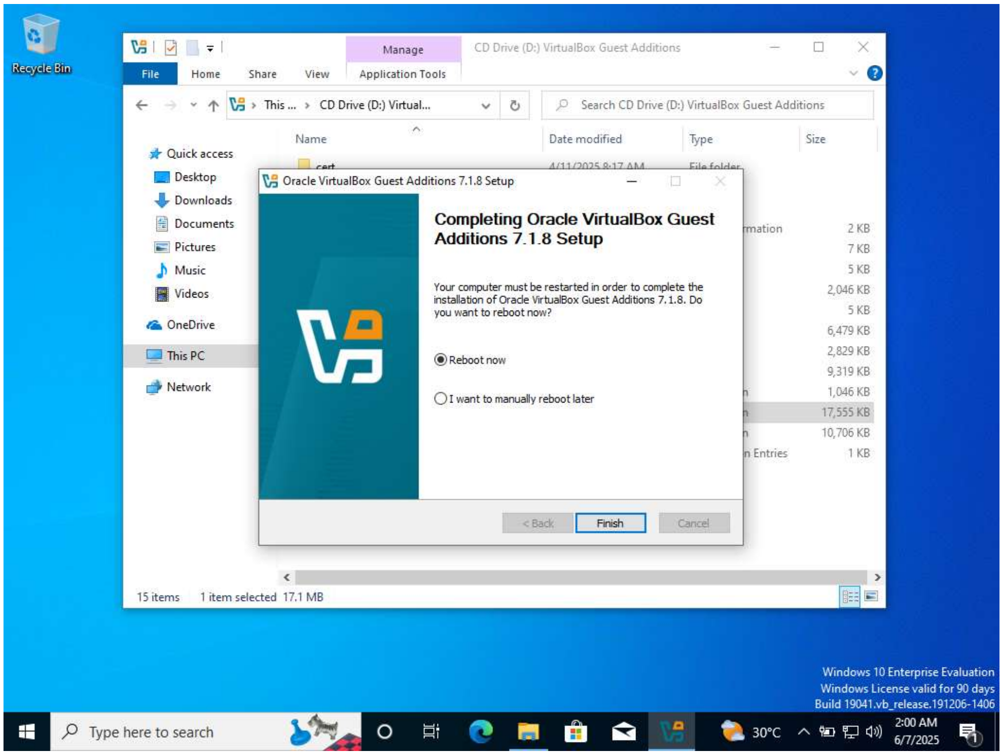
- Click Next to continue.
- Installing.





- Select Reboot now and Click Finish.

Install Guest Additions on both VMs post-OS setup to improve your experience.



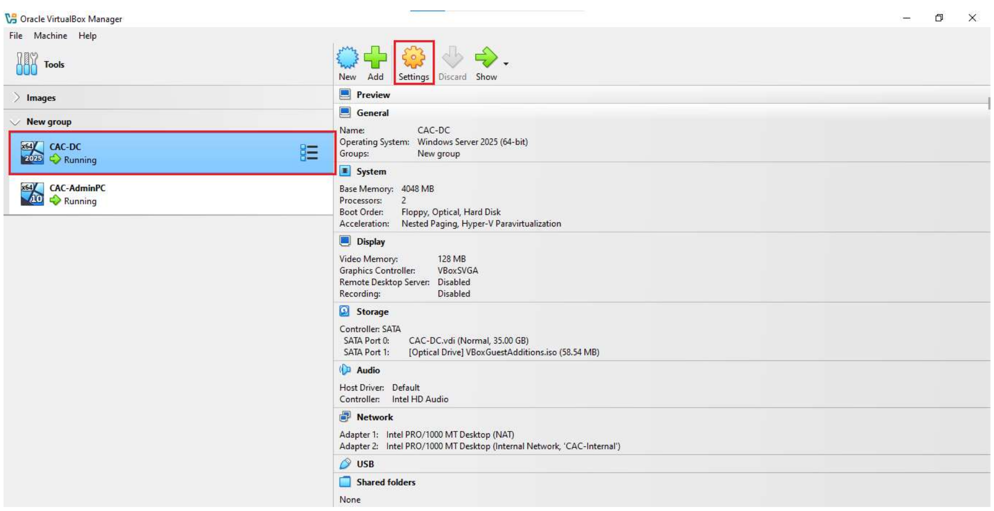
7. Network Configurations

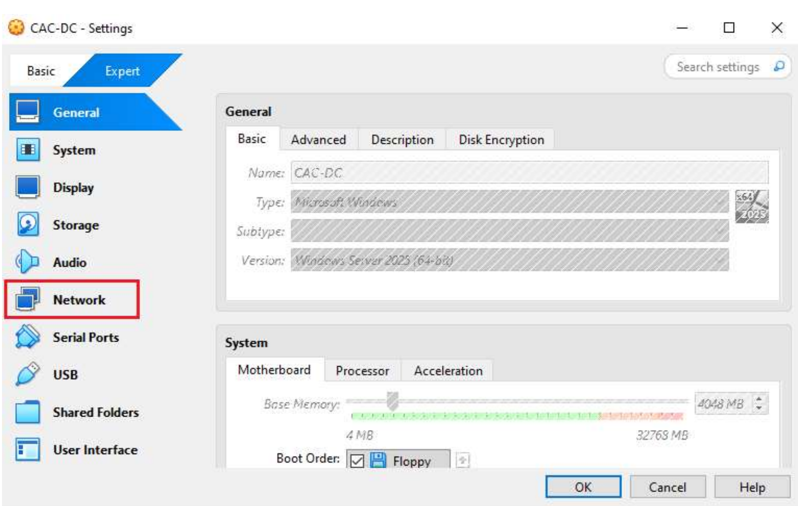
Domain Controller

- Select your Window Server VM (Domain Controller) in VirtualBox click

Settings.

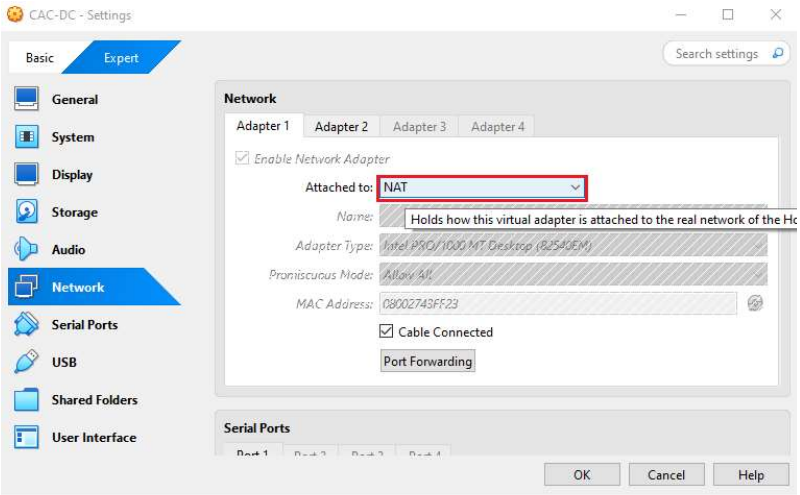
- Select Network.

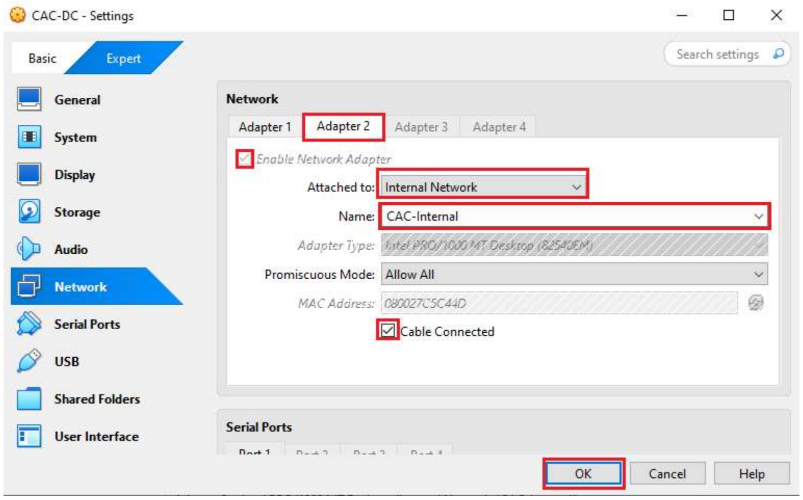




- On Adapter 1, Select NAT from the Attached to: drop-down menu.
- Next, Select Adapter 2. Select Internal Network from the Attached to

drop-down menu and create a name for your internal network.



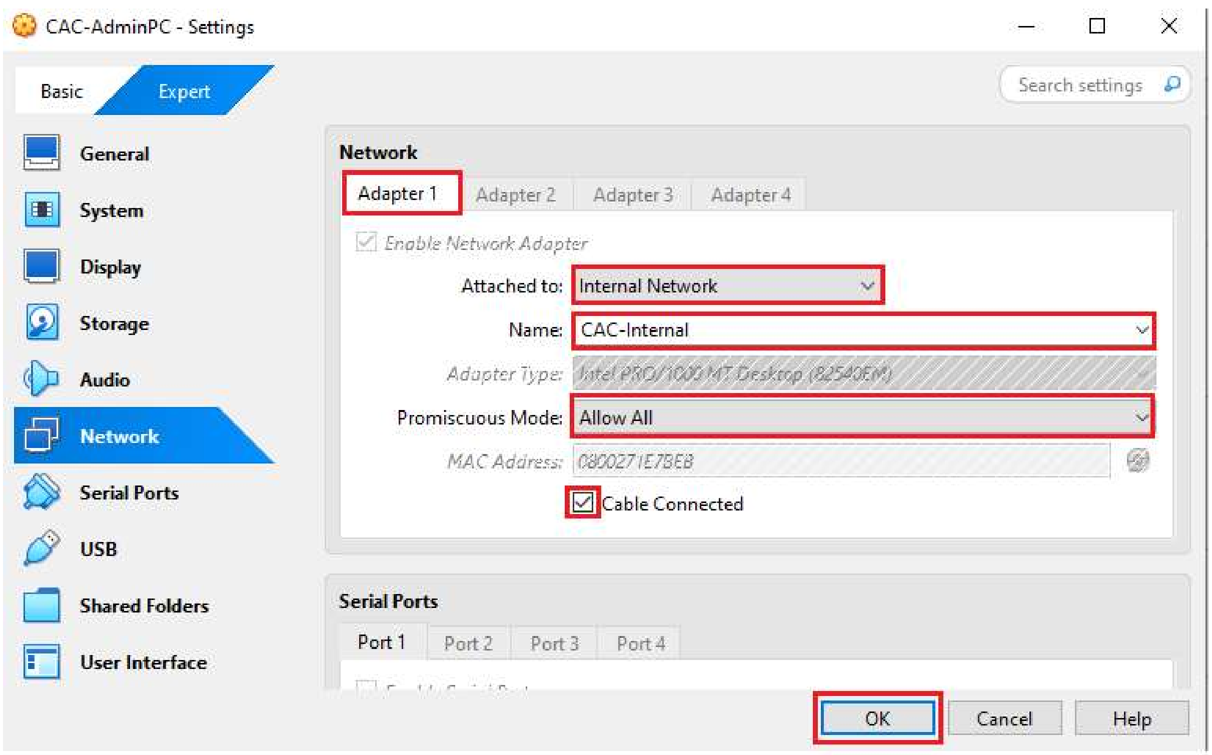


Client Machines

- Repeat the step to access the network setting on the Windows 10 VM.

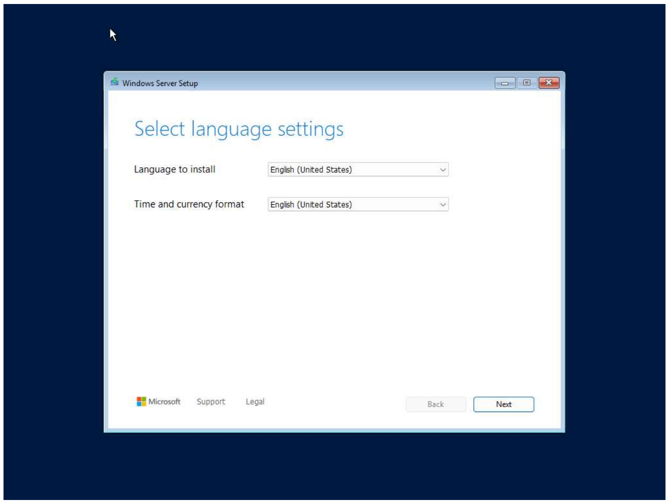
On Adapter 1, Select Internal Network from the Attached drop-down menu and select the Internal Network you previously created on the Windows Server VM.

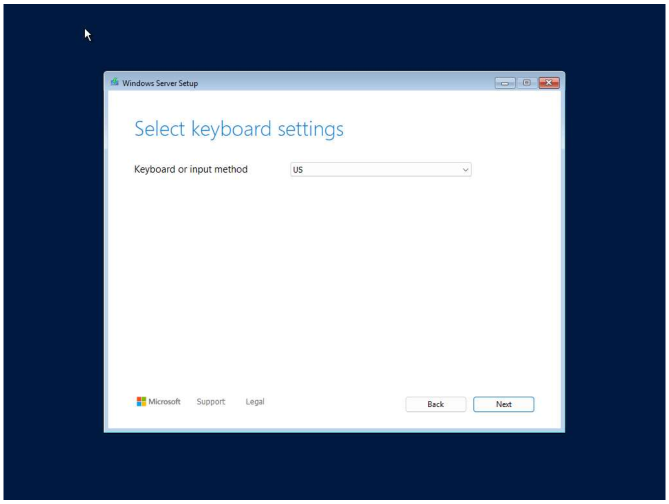
Repeat this process for the other client VMs that you want to communicate with on this network using only Adapter 1.



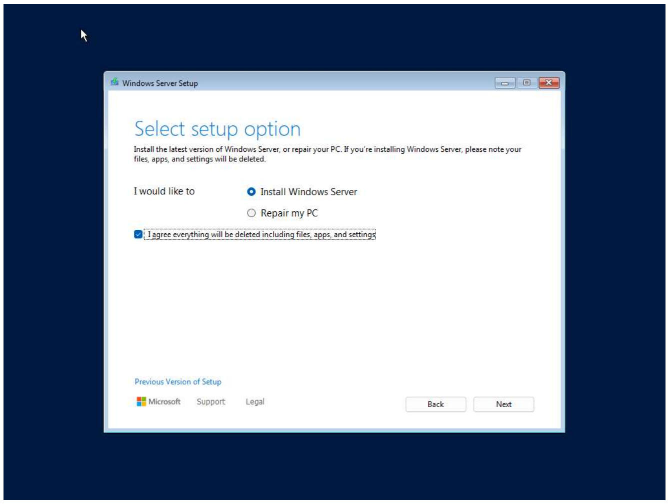
8. Installing Windows Server 2025

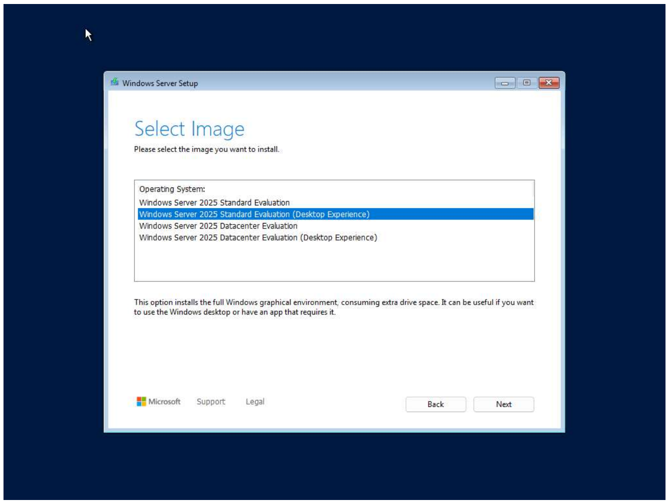
- After you start your machine, you will be prompted to select language, time/keyboard settings. When you are done, click on Next to continue.



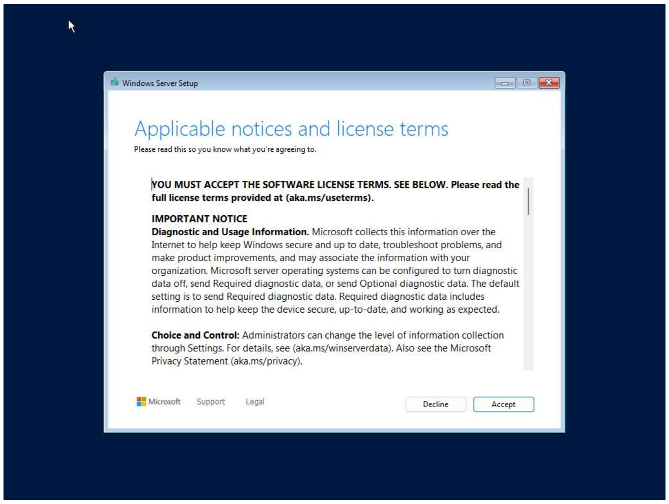


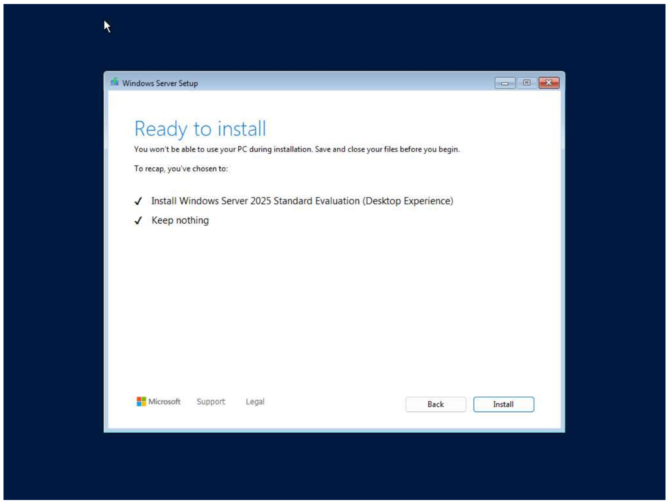
- Select Install Server, agree and click Next.
- Select Windows Server Desktop Experience and Click Next.





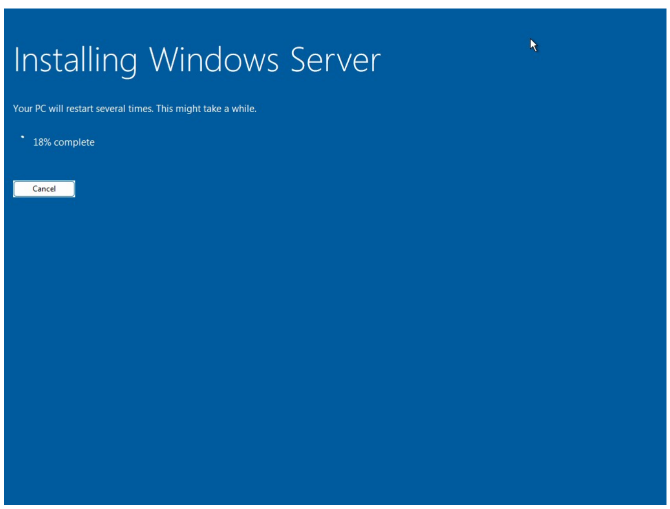
- Click Accept.
- Click Install

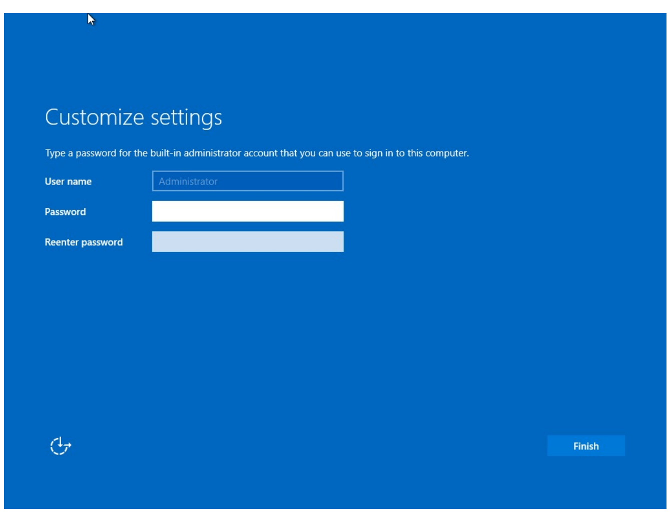




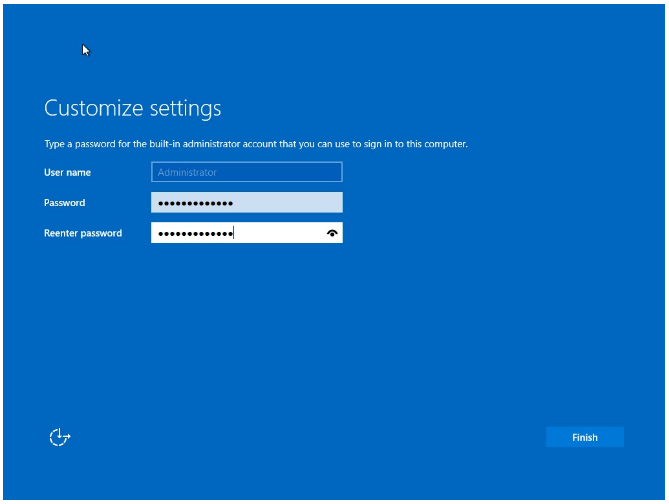
Installing Server

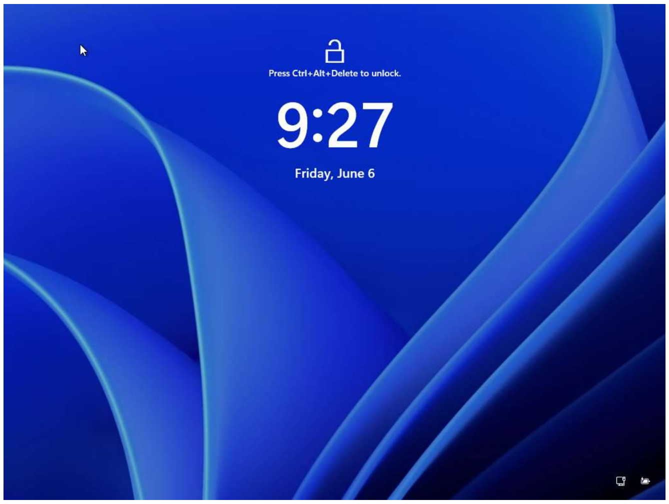
- Installation in progress, this may take a while. The installation may restart and will be met with the following.





- The username is Administrator by default. Set up the user account by entering your password. Click on Finish.
- After setting up the user account, you should be welcomed with this screen.

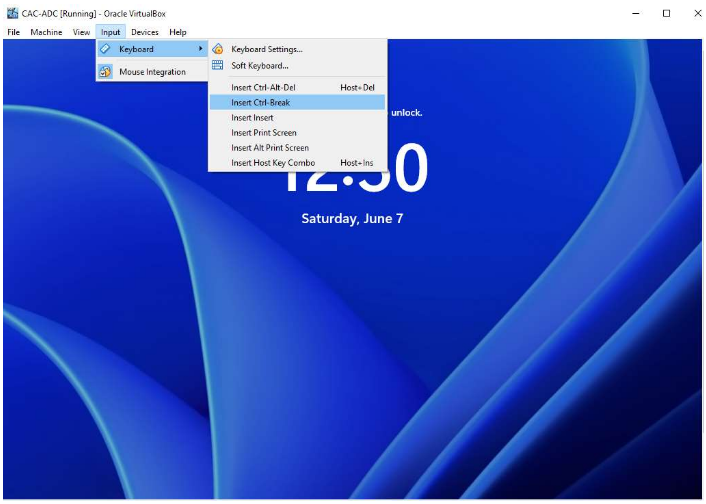


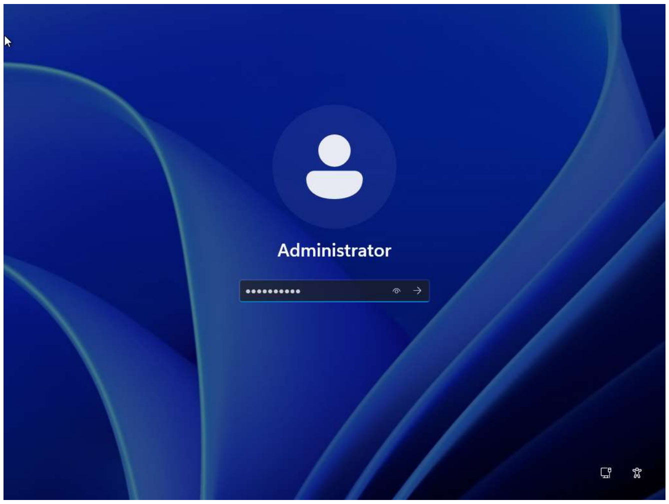


- To unlock the screen. Click on Input, a drop-down will pop up. Click

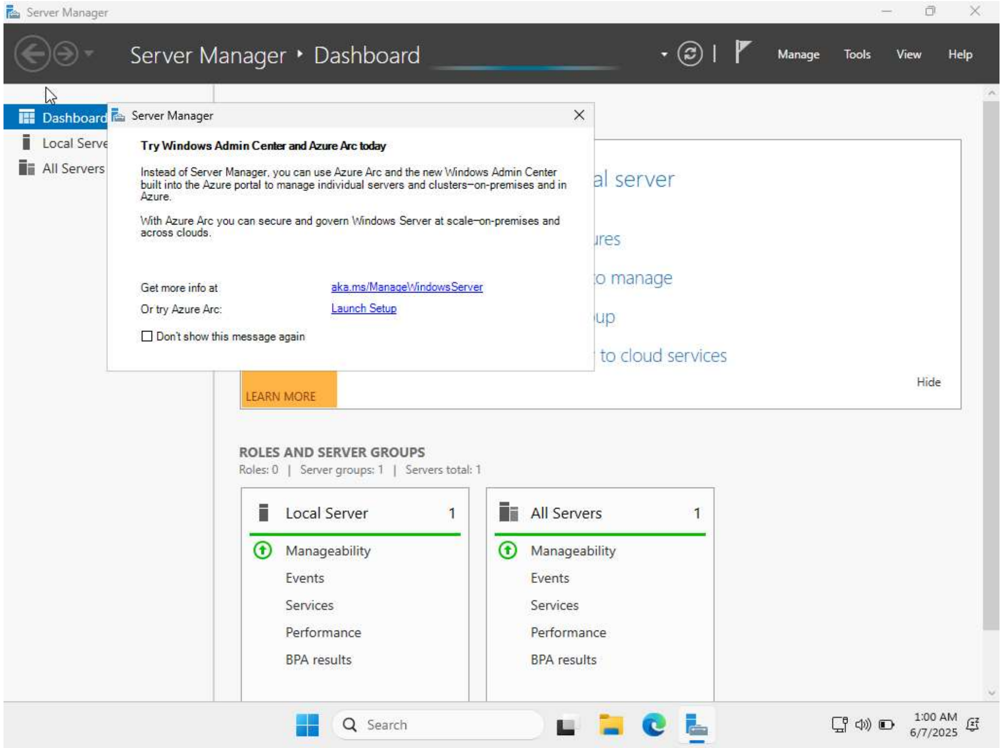
on Keyboard, and then click on Insert Ctrl-Alt-Del Host+Del.

- Log into your account.





■ You will be greeted by this screen. Congratulations!

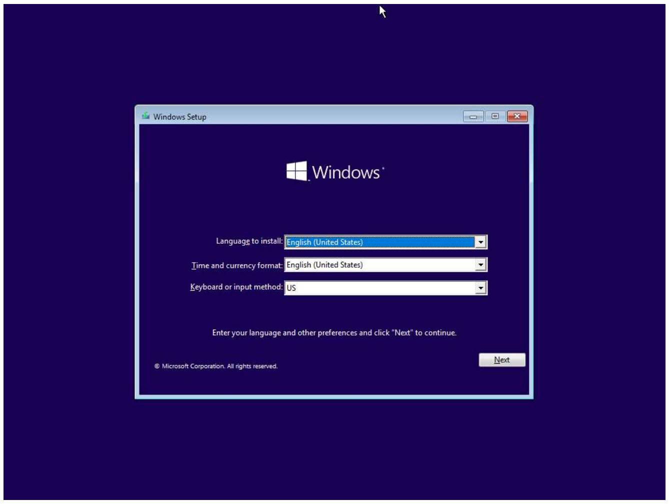


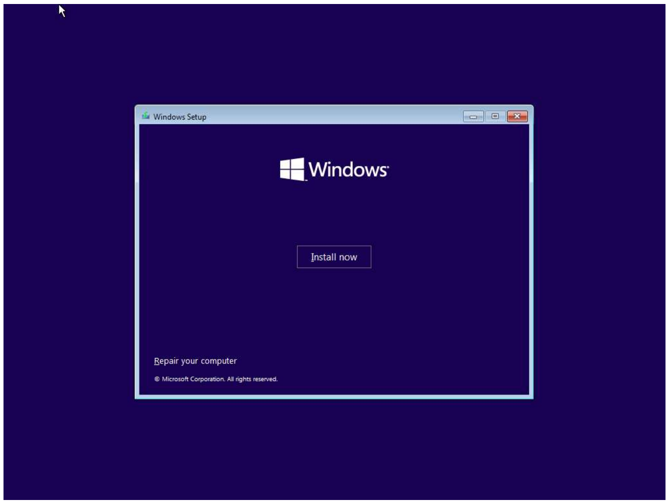
9. Installing Windows 10

- You will get a prompt to select language, time/key43board settings.

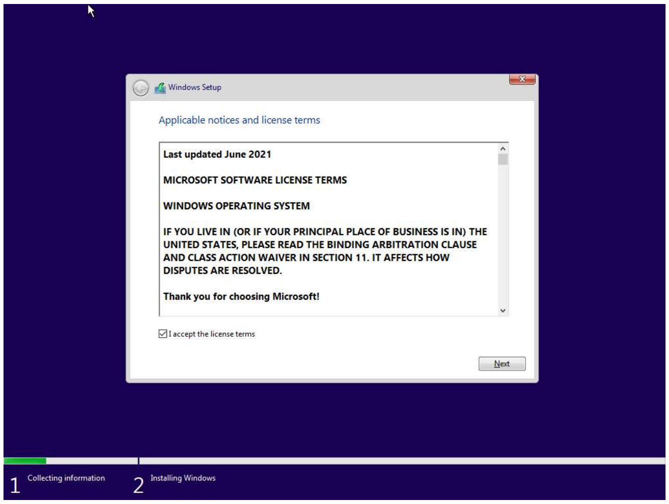
When you are done, click on Next to continue.

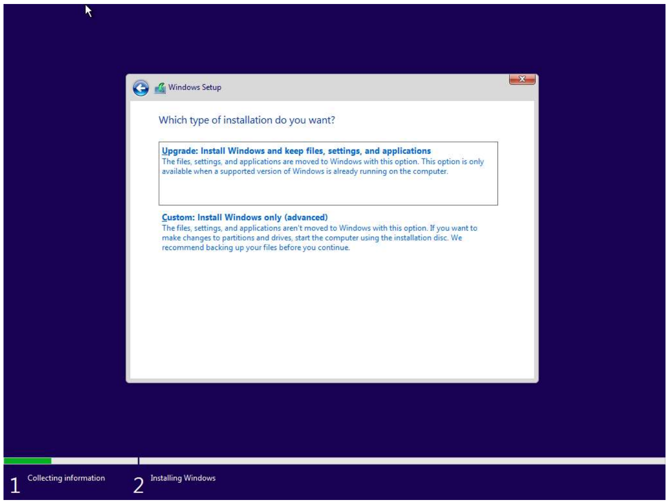
- Select Install Now.





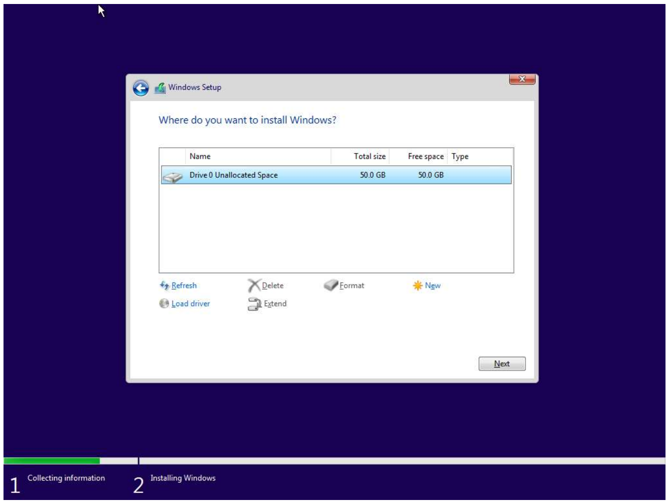
- Accept terms and Click Next.
- Select Custom: Install Window only.

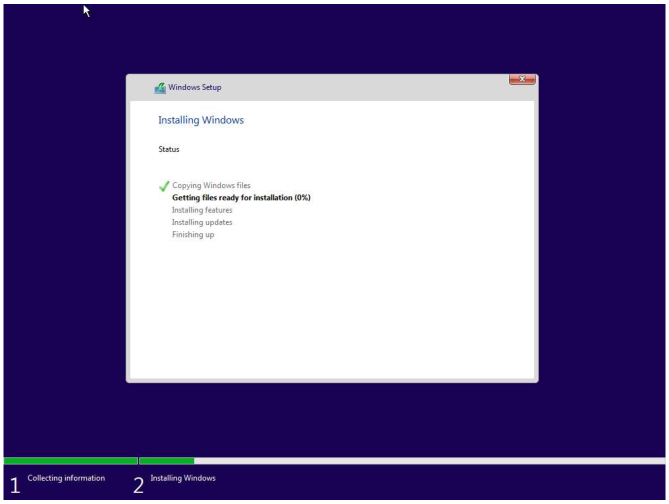




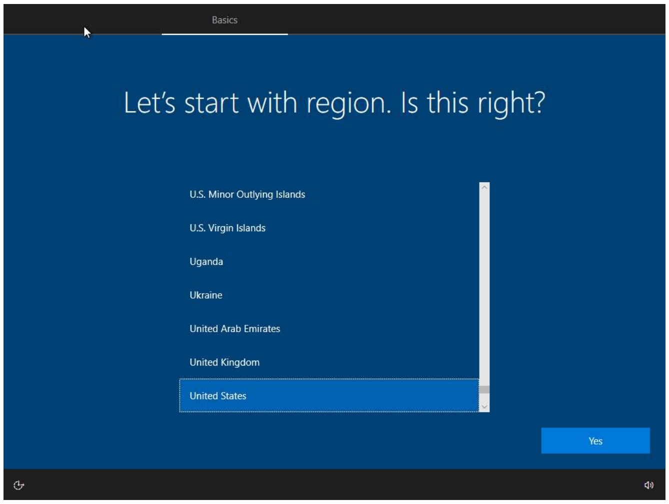
Select drive and click Next

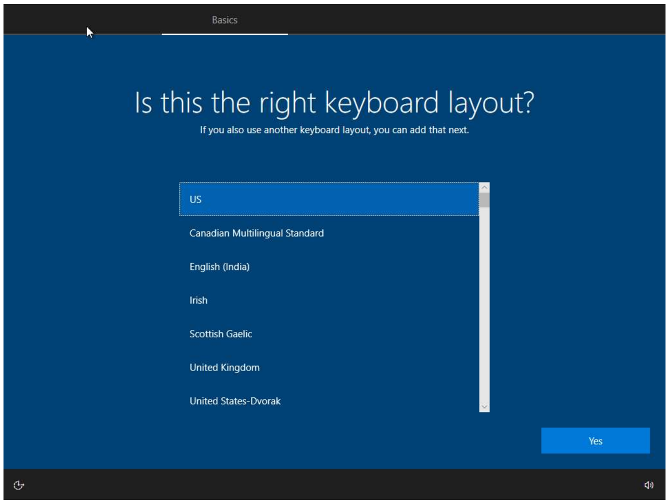
- Installing Windows 10



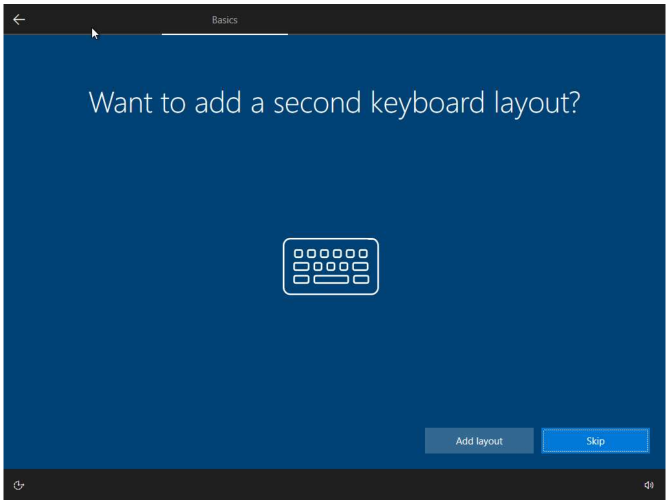


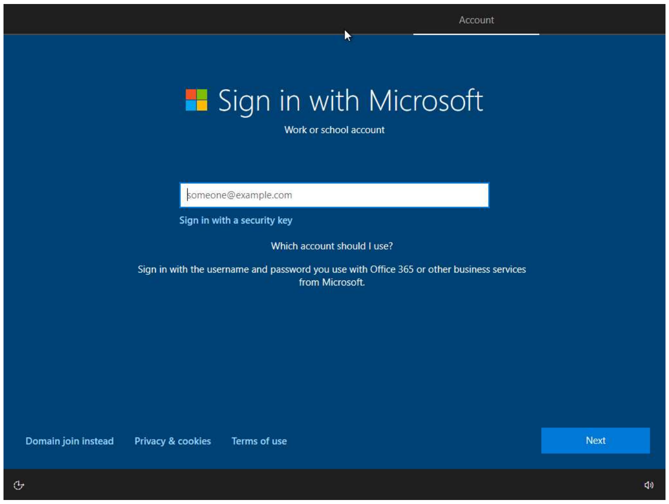
- When the installation is complete, you will see the following screen, select your region and click Yes.
- Select Keyboard Layout and click Yes



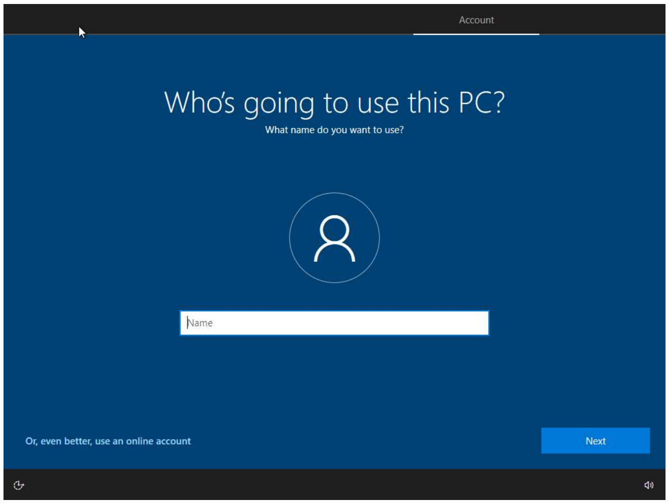


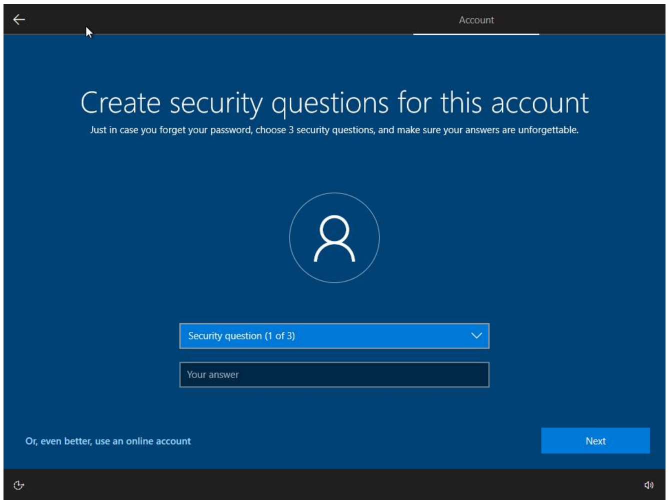
- Skip the Second Keyboard layout.
- Choose how you want to sign in. Click domain join instead.



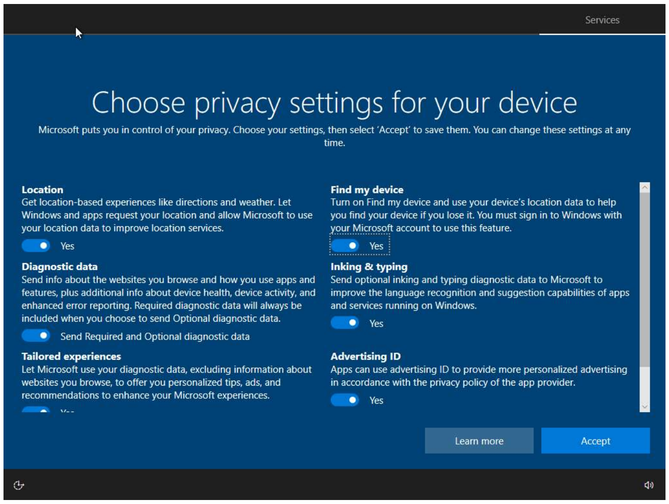


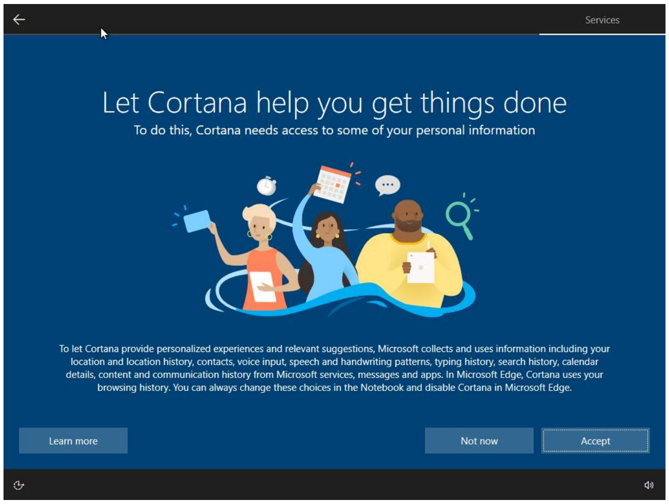
- Set user name and password and click Next.
- Set your security questions click Next.





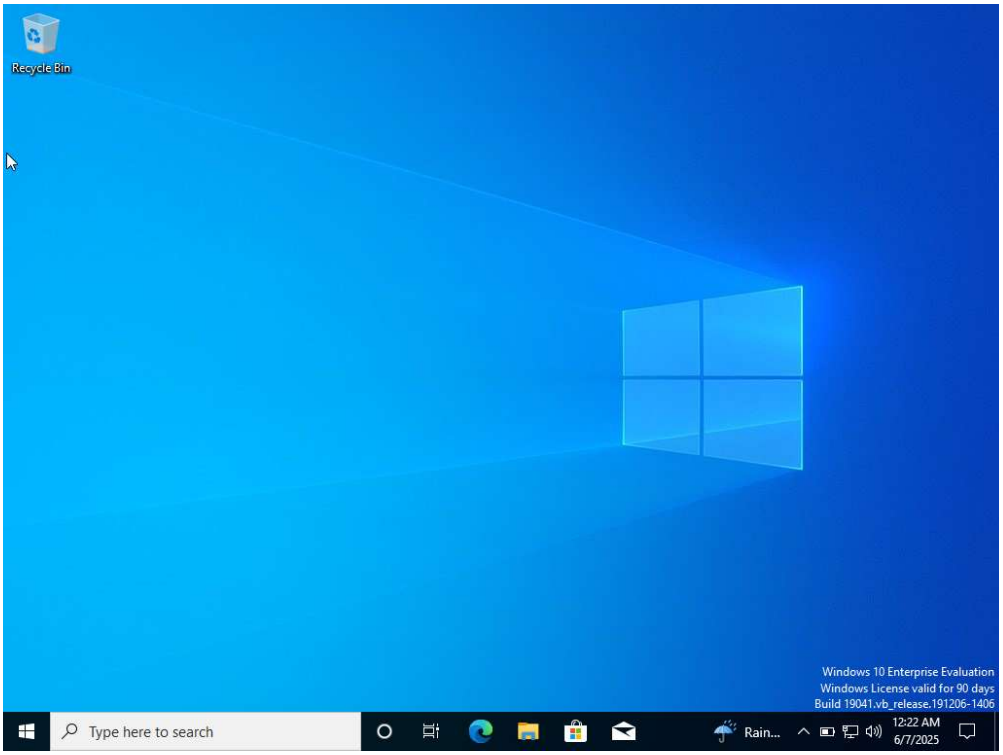
- Set your privacy setting and click Accept.
- Click Accept





- Next you will see Getting Windows Ready, then you will see your desktop load.

Congratulations, you are now ready to use your windows machine.

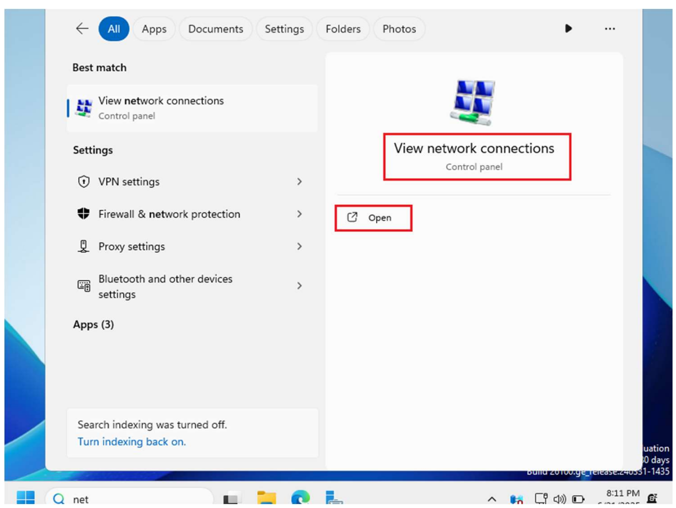


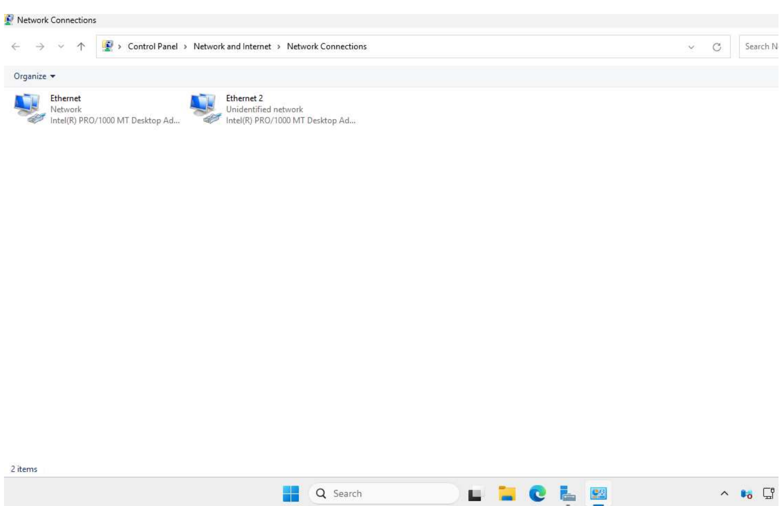
10. Configure Network Settings in Server and Clients

Windows Server

- On your Windows Server VM navigate to Settings
- A window will pop up showing the two Ethernet connections. Ethernet 1

is the NAT and Ethernet 2 is the Internal Network we created (CAC-Internal)



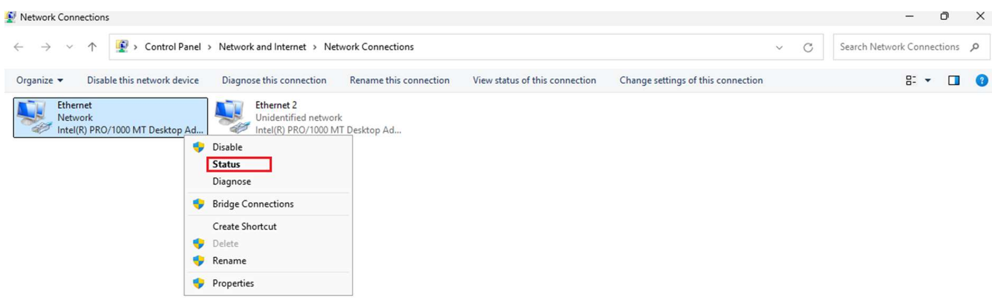


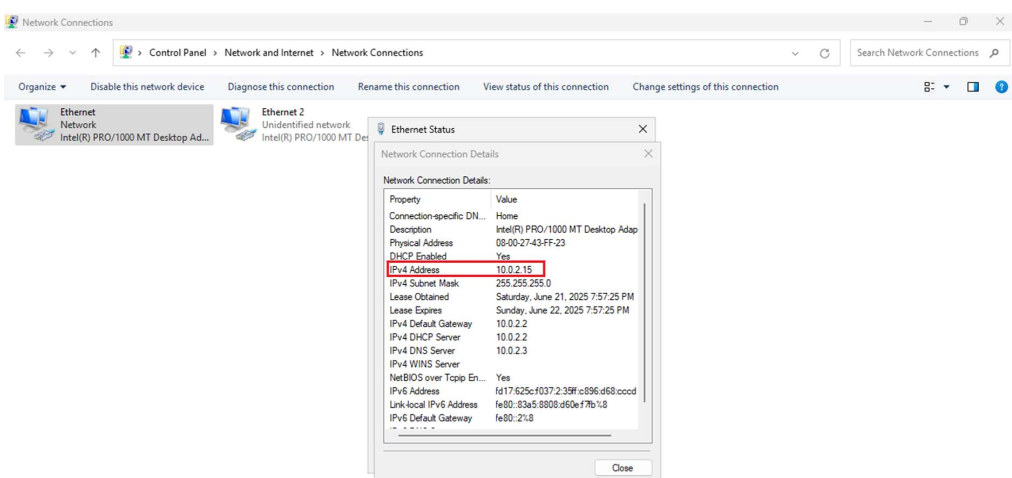
If you right-click on Ethernet 1 and click Status you will see the

connection details.

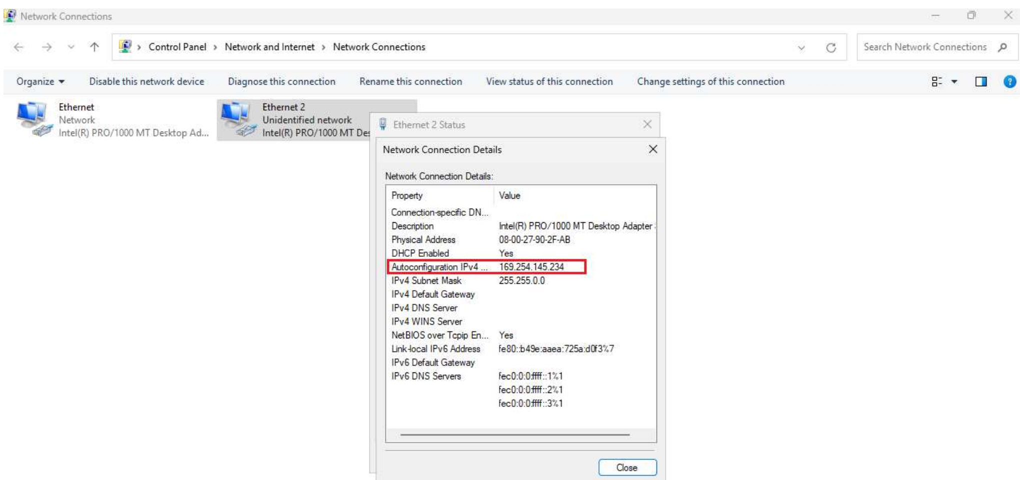
- The IPV4 Address of 10.10.2.15 is the address given out of the VBox

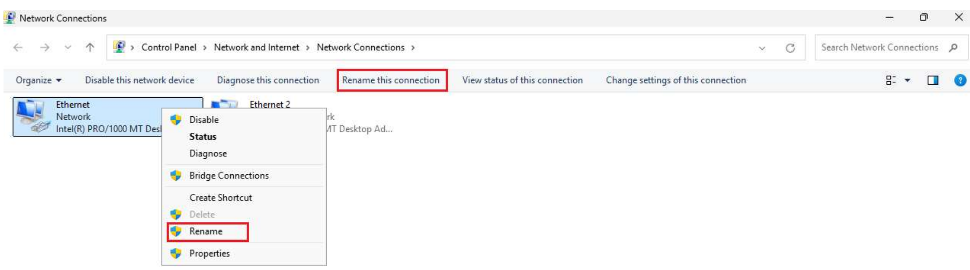
Network Adapter.





- If you were to do the same for Ethernet 2 you would see a 169.X.X.X address which is a private internal address.
- Rename Ethernet 1 to Internet and Ethernet 2 to the name of your Internal Network (CAC-Internal)





- Renamed Adapters.

IP Address Range Explanation

Static vs Dynamic IP Allocation Strategy

In this lab, we will implement a structured IP addressing scheme within the 192.168.2.0/24

network:

Static IP Range: 192.168.2.1 - 192.168.2.100

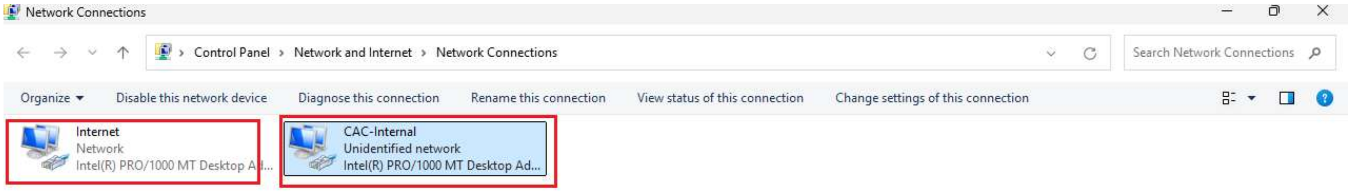
Purpose: Reserved for servers and network infrastructure

- Domain Controller: 192.168.2.10
- Additional Servers: 192.168.2.11-192.168.2.20 (if added later)
- Network Devices: 192.168.2.21-192.168.2.50 (printers, etc.)
- Reserved for Future Use: 192.168.2.51-192.168.2.100

Dynamic IP Range: 192.168.2.101 - 192.168.2.200

Purpose: DHCP pool for client computers and devices

- Windows 10 Clients: Automatically assigned from this range
- Laptops/Mobile Devices: Any device joining the network
- Temporary Devices: Guest devices, testing equipment



Benefits of This Approach:

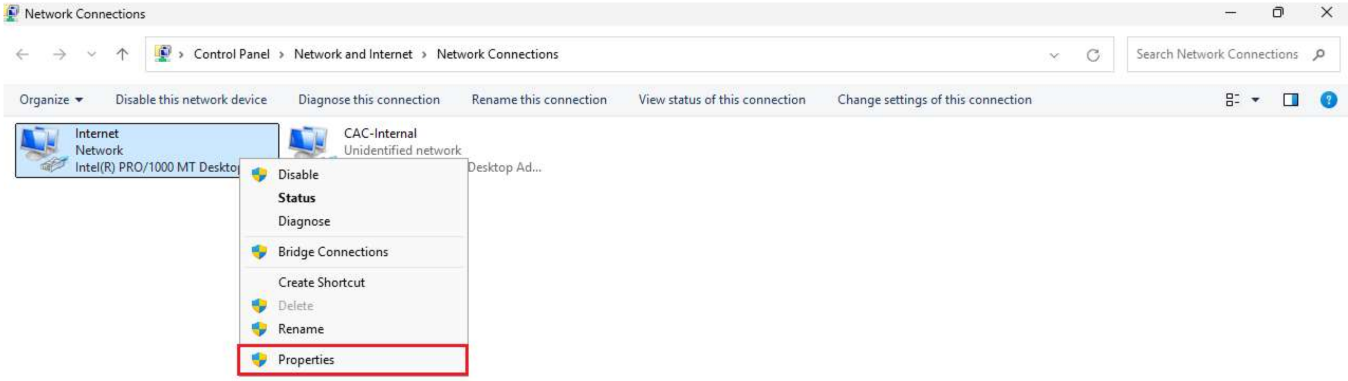
- Predictable Server Locations: Always know where to find servers
- Simplified Troubleshooting: Static devices have consistent addresses
- Network Documentation: Easy to document and maintain
- Scalability: Clear separation allows for network growth

DHCP Exclusions:

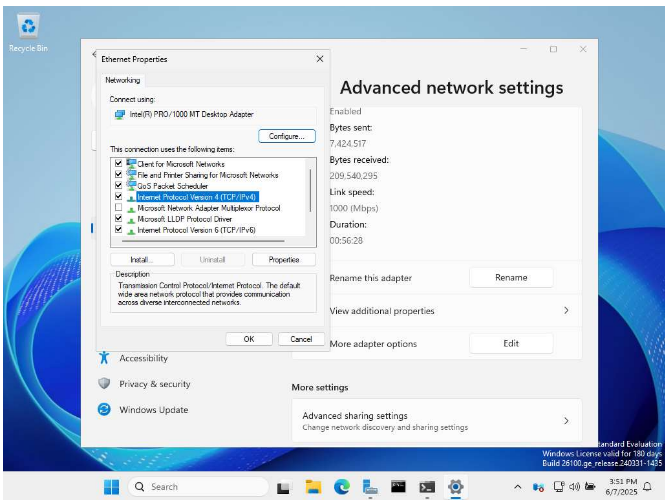
Configure DHCP to exclude the static range (192.168.2.1-192.168.2.100) to prevent conflicts.

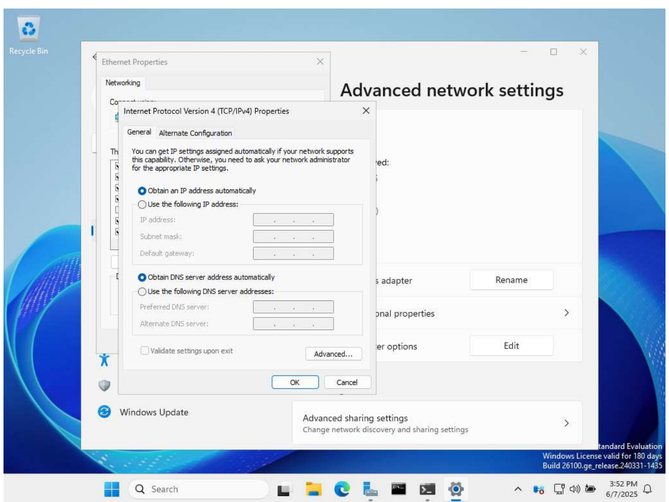
11. Setting Static IP Addresses

- Next, set the IP Settings on the Network. Right Click on Internet connection and click Properties.



- Double click Internet Protocol Version 4 to open settings.
- THIS WINDOW WILL OPEN.

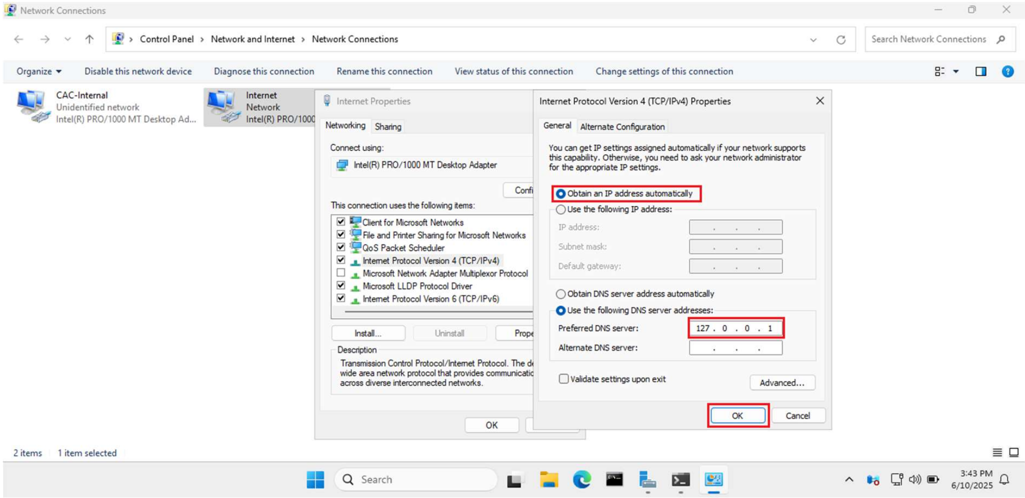


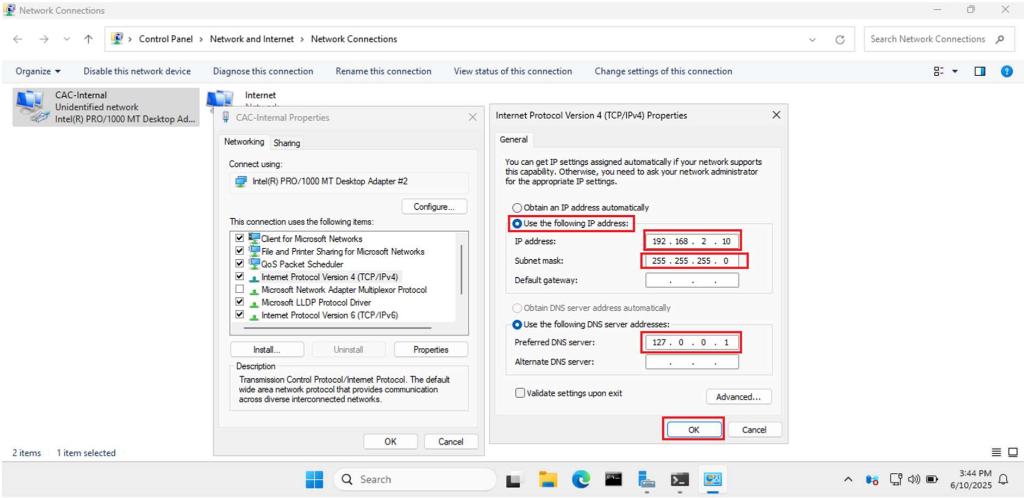


Select Obtain an IP address automatically and change the DNS server

address to 127.0.0.1 (loopback address).

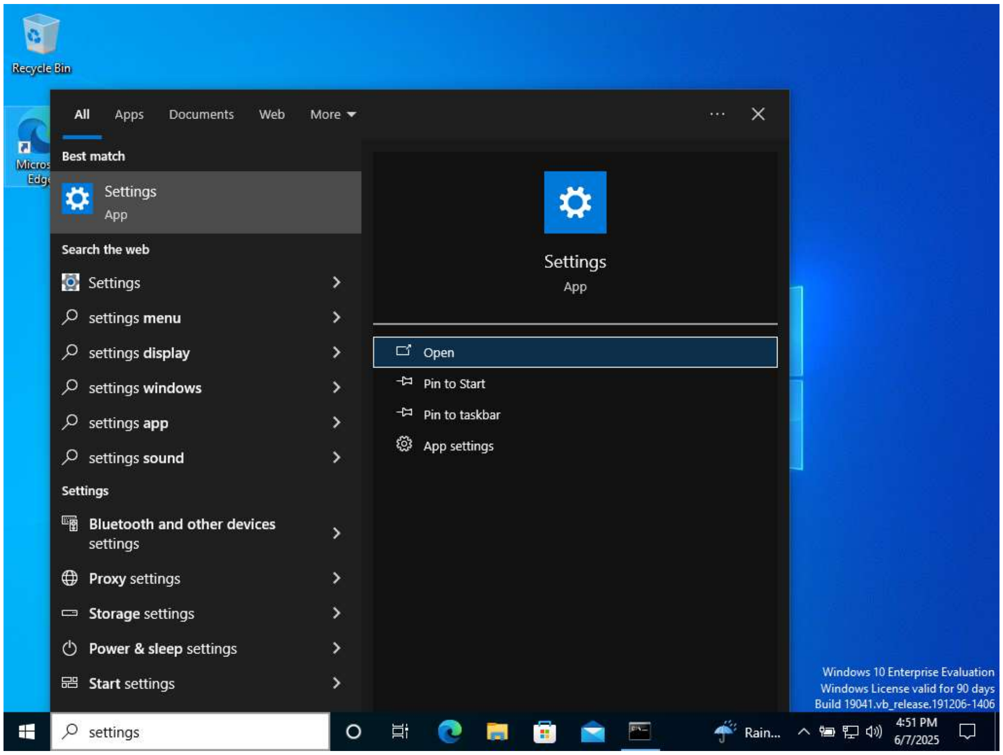
- Right Click on CAC-Internal connection and click Properties and set the following. For this lab I set a static IPV4 address of 192.168.2.10, Subnet mask: 255.255.255.0, DNS: 127.0.0.1.





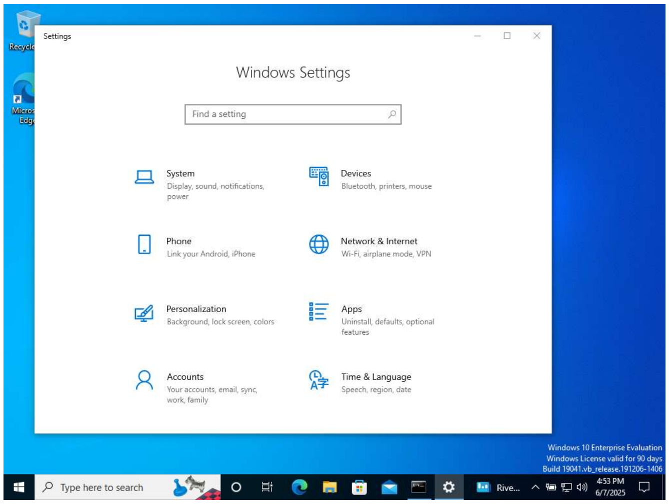
Windows 10

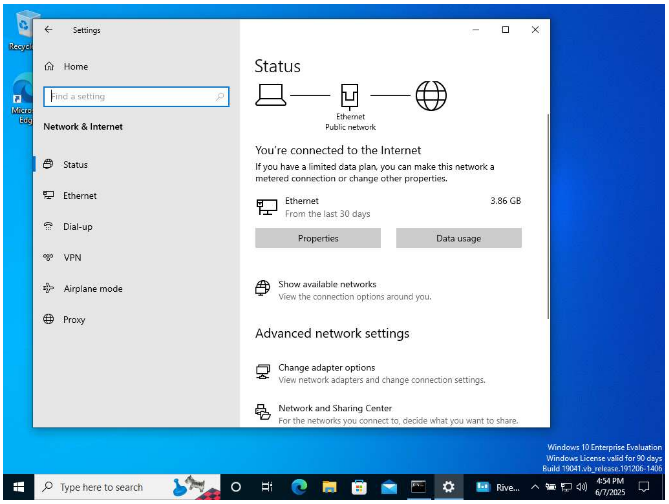
- Open Settings
- Select Network & Internet



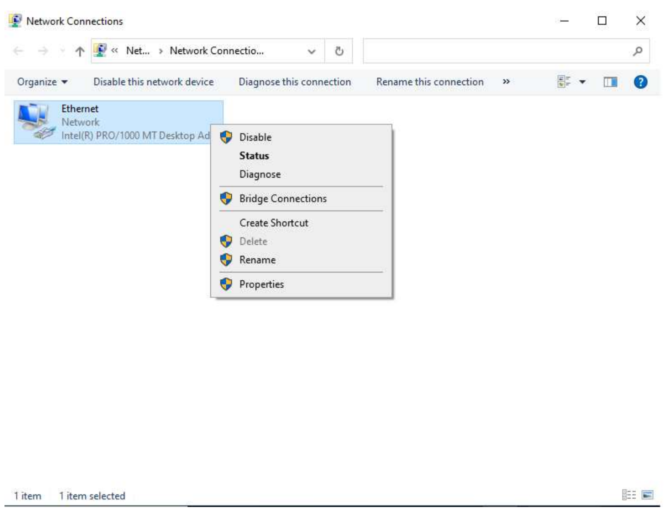
Change Adapter Settings

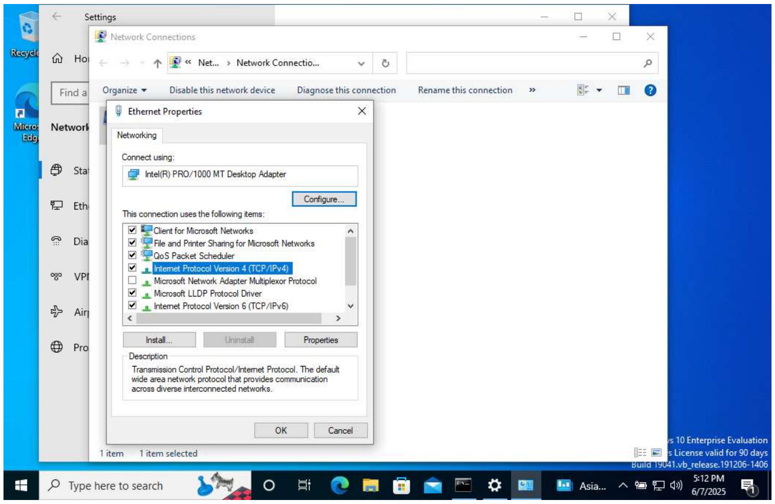
- Right click on Ethernet connection and click Properties.





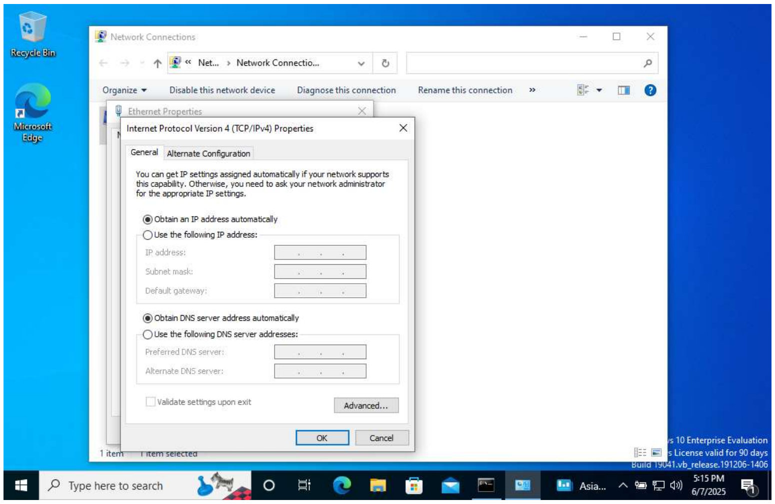
- Double click Internet Protocol Version 4 to set the Server's IP address.
- Enable, Use the following IP address.

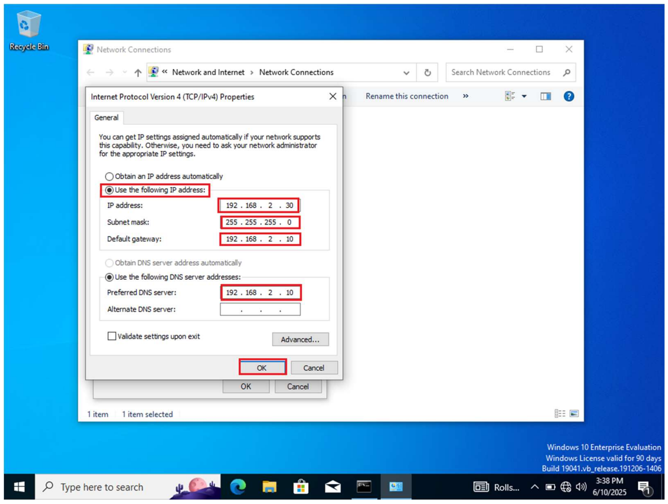




- For this lab I assigned my server a static IP address with the following settings.

12. What is Active Directory? (For Beginners)





Understanding Active Directory

Active Directory (AD) is Microsoft's directory service that acts as a centralized database for managing users, computers, and resources in a Windows network environment. Think of it as a digital phone book for your entire organization.

Key Concepts Explained Simply:

Domain: A logical grouping of computers, users, and resources that share the same security policies and user database. In our lab, we created "cac.local" as our domain name.

Domain Controller (DC): A server that holds a copy of the Active Directory database and handles authentication requests. When you log into a domain computer, you're actually logging into the Domain Controller.

Authentication: The process of verifying who you are. Instead of having separate user accounts on each computer, AD allows you to log in once and access all authorized resources.

Centralized Management: Instead of managing users and permissions on each individual computer, AD lets administrators control everything from one central location.

Why Use Active Directory?

- Single Sign-On: Users log in once and access all their authorized resources
- Centralized User Management: Create, modify, or delete users from one location
- Security Policies: Apply consistent security settings across all computers
- Resource Sharing: Easily share files, printers, and applications
- Scalability: Can grow from small offices to enterprise-level organizations

Real-World Example:

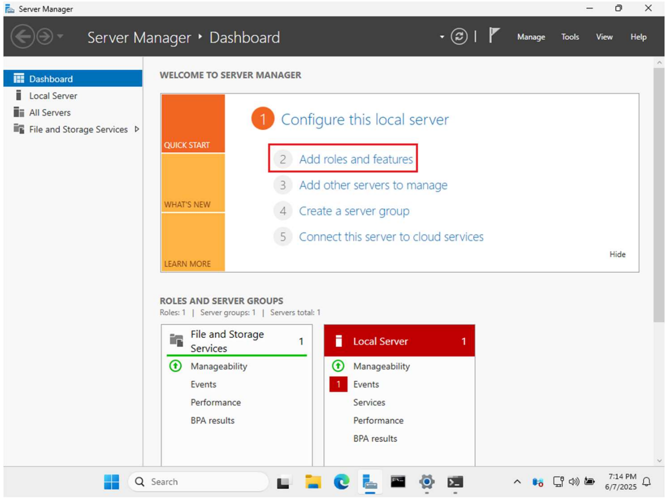
Imagine a company with 100 employees. Without AD, you'd need to create 100 user accounts on each computer they need to access. With AD, you create each user once, and they can access any authorized computer in the network.

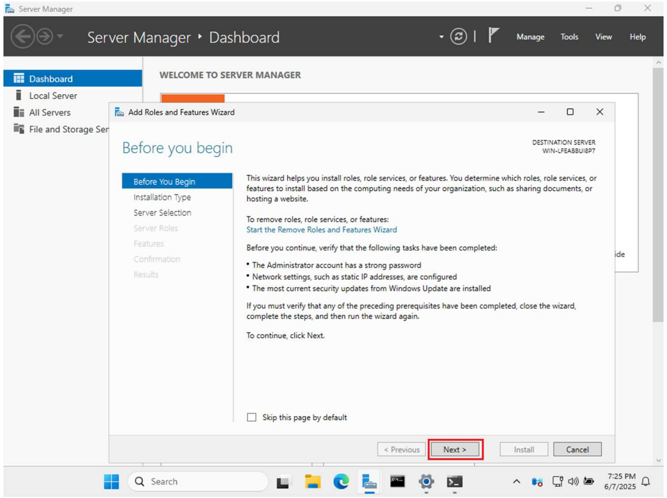
13. Deploying Active Directory

You should have noticed an application called Server Manager loaded

when we first logged into Windows Server. Click Add roles and features.

- Click Next.
- Click Role based or feature based installation and click Next.

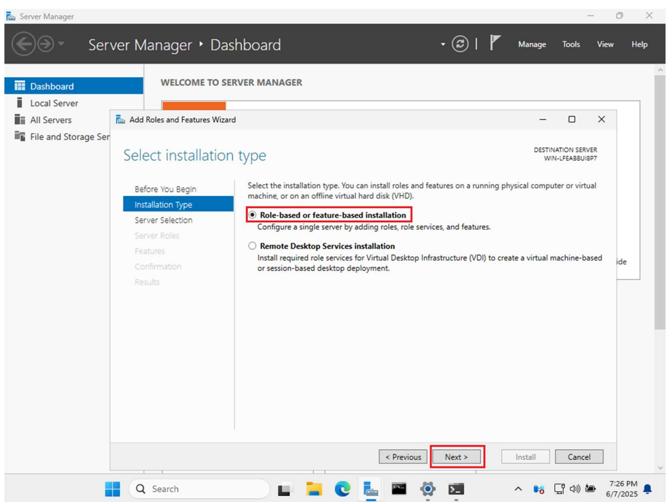


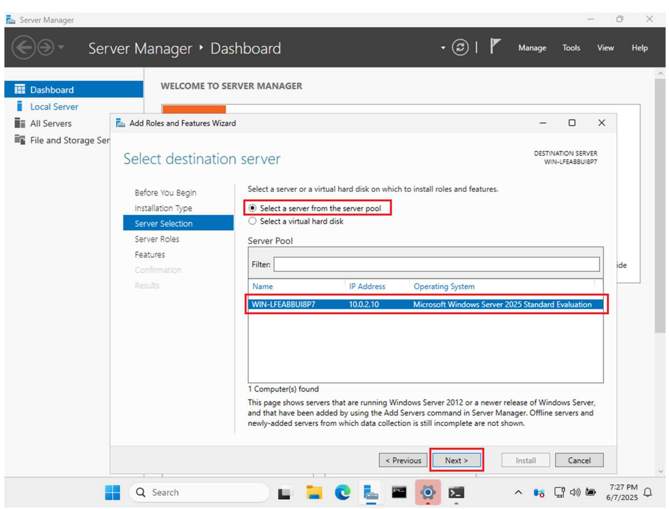


- Click Select a sever from the server pool, keep the defaults and click

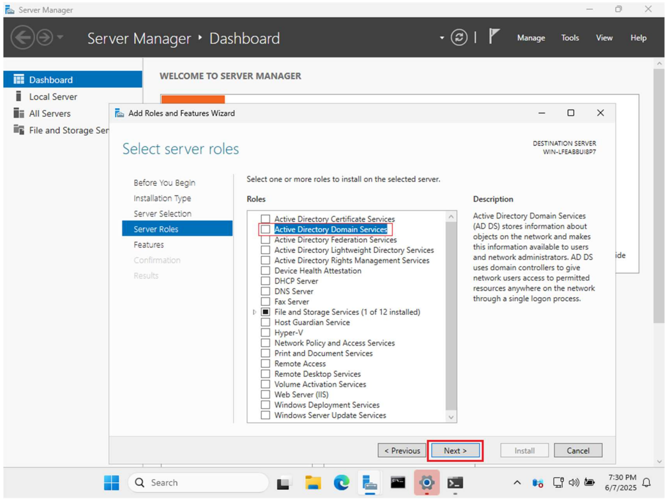
Next.

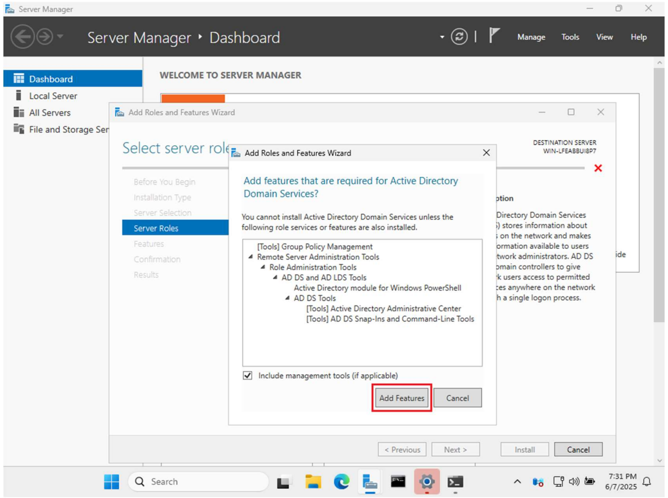
- Select Active Directory Domain Services and click Next.





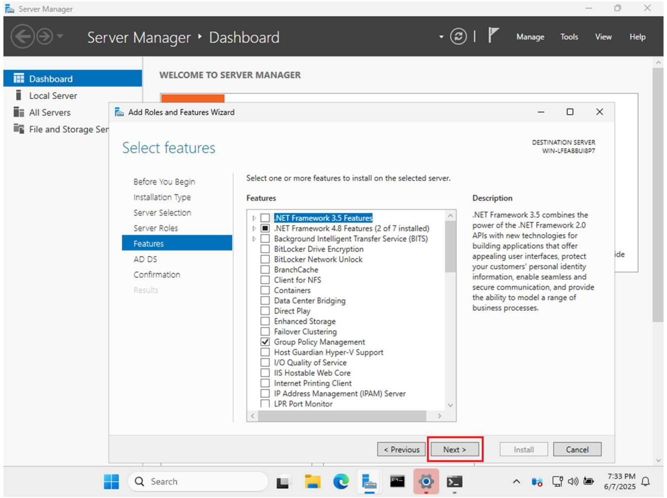
- Click Add Features and click Next.
- Click Next to continue with the defaults.

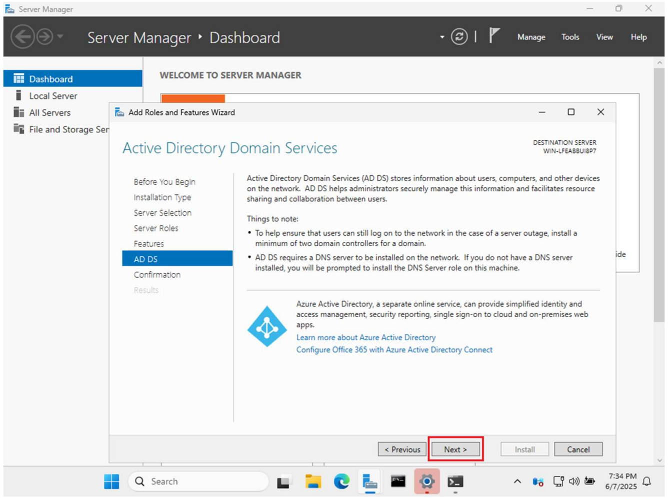




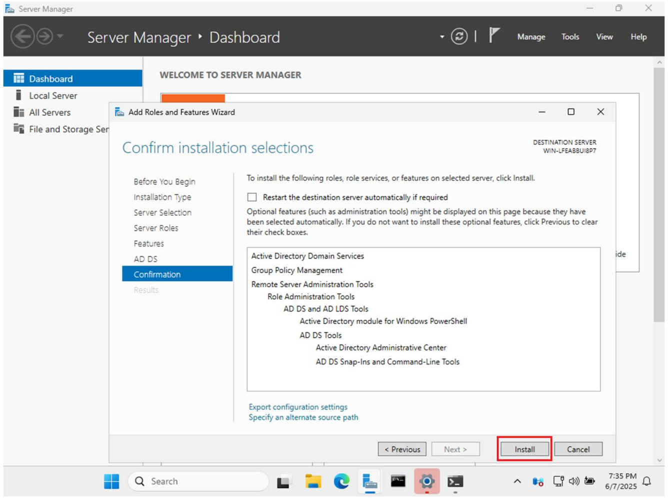
Click Next at the Azure Information Page

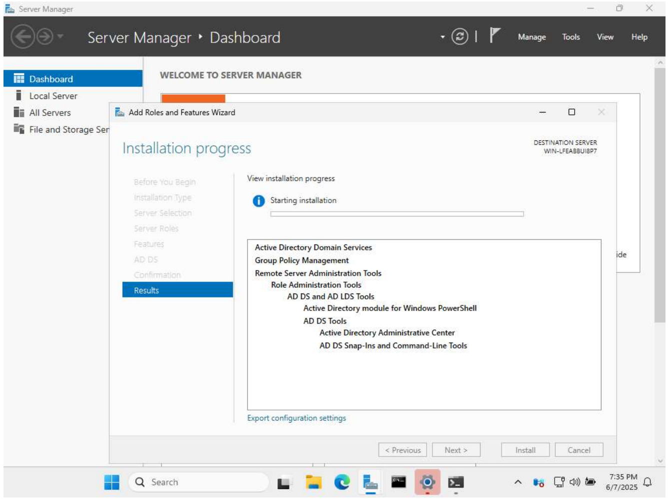
- Click Install.



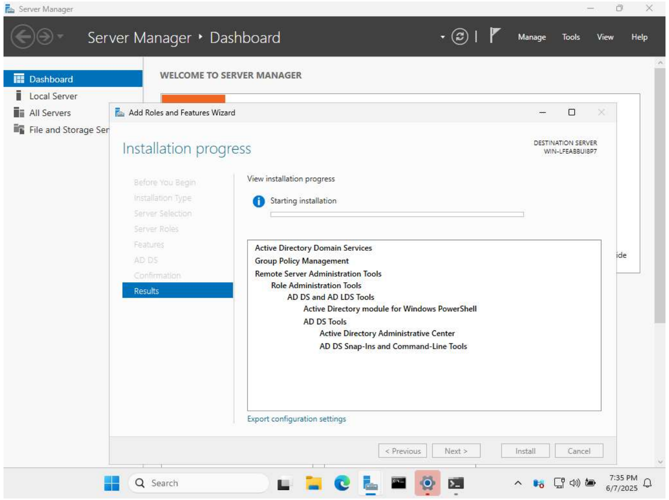


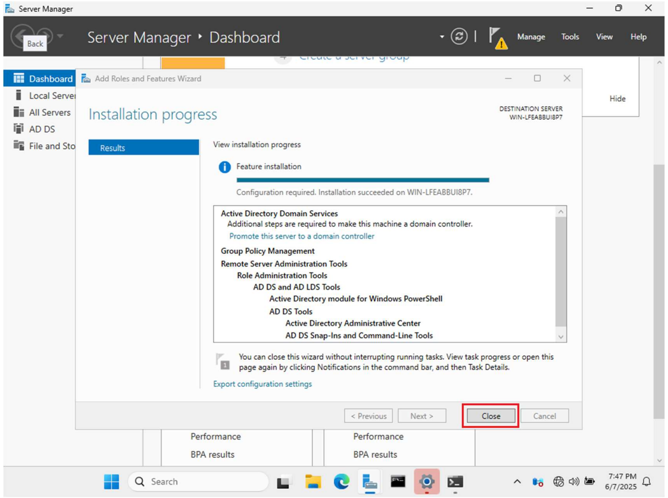
- Installation in progress.





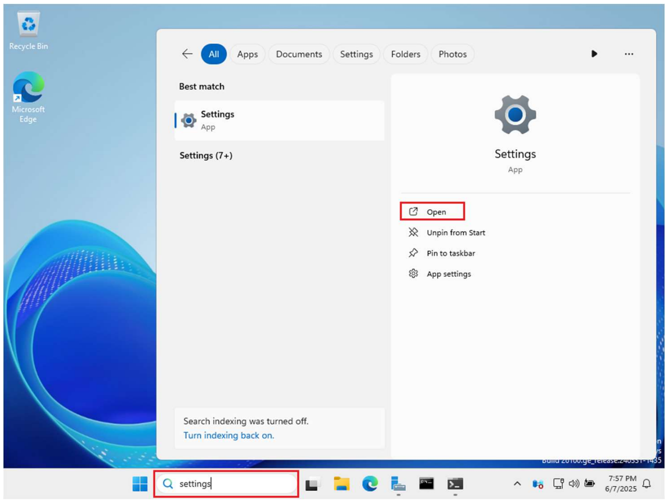
- Installation in progress.
- Installation completed, click Close.

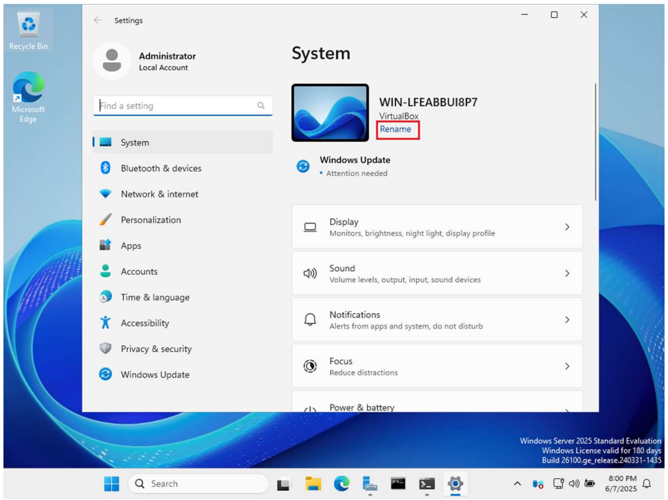




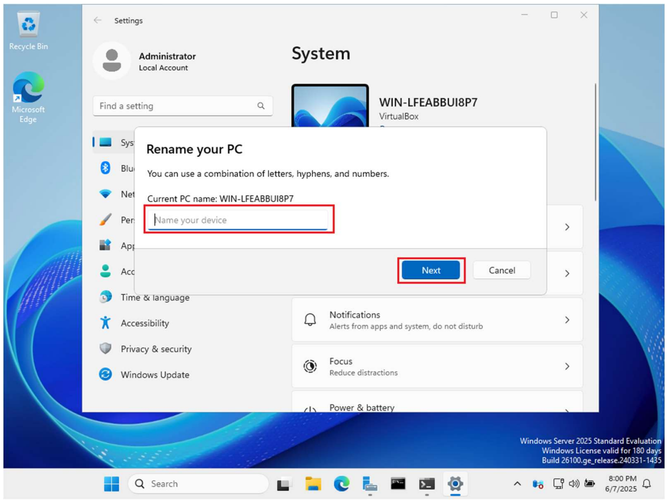
14. Promoting Server to Domain Controller

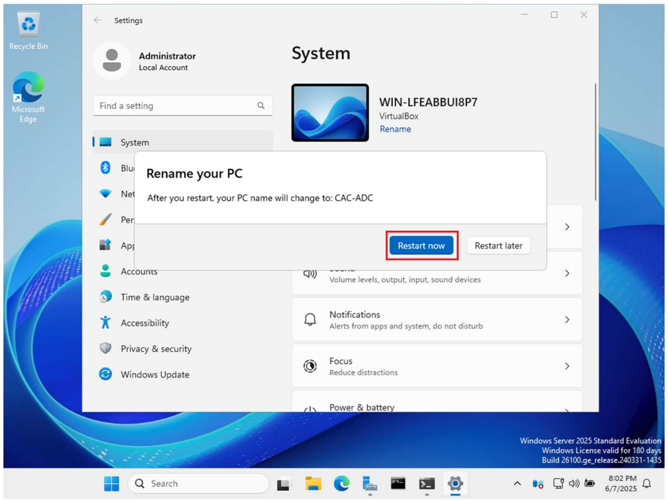
- Before promoting your server to a domain controller, it is important to give it a meaningful name.
- Click Rename.



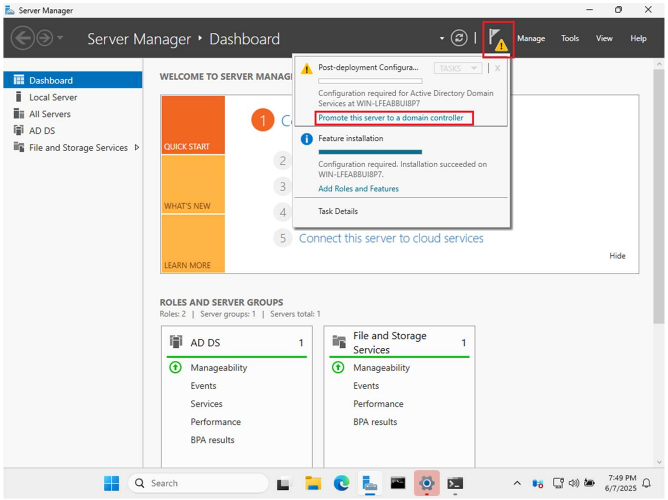


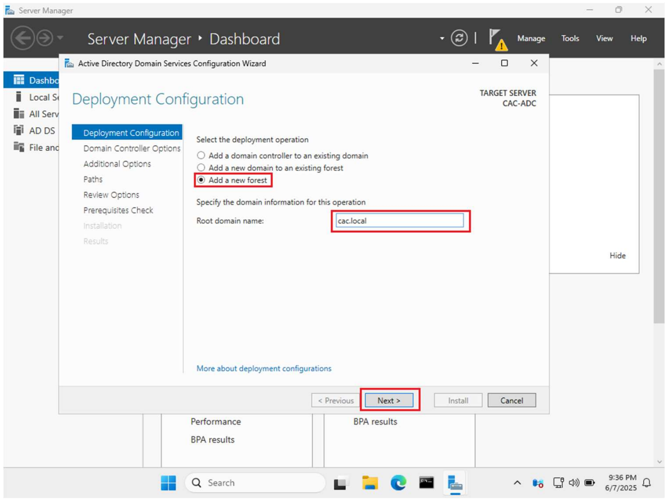
- Name your PC.
- Restart PC.



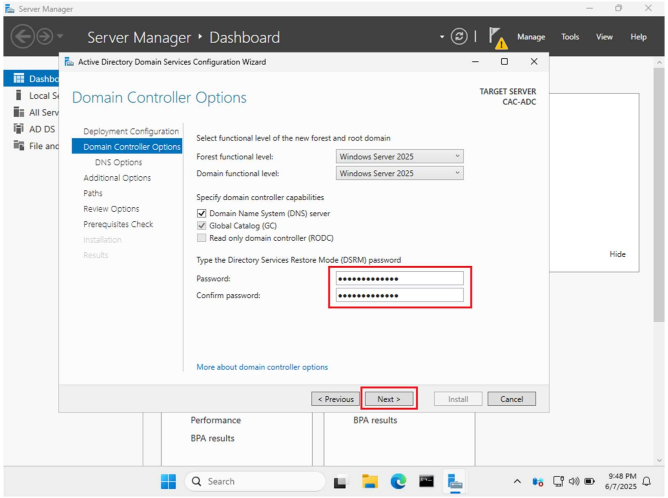


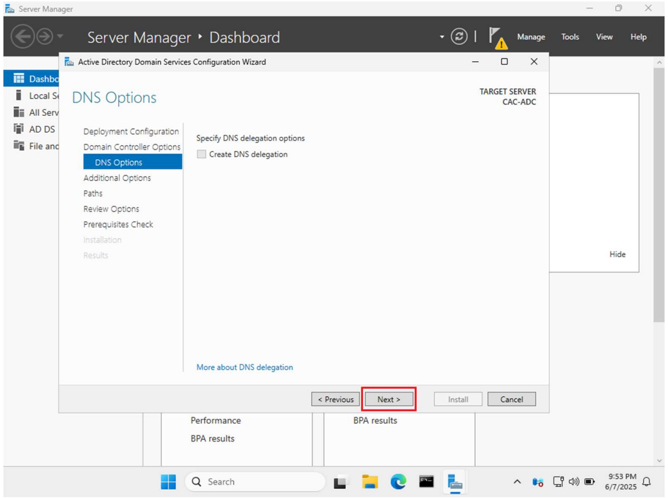
- Click the flag with yellow triangle in the right corner and click Promote this server to a domain controller.
- A window will appear, Add a new forest, assign a domain name (name.local) and click Next. I named mine cac.local



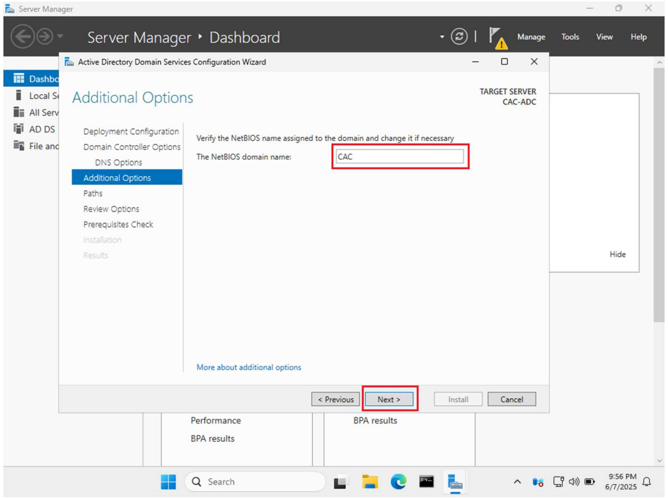


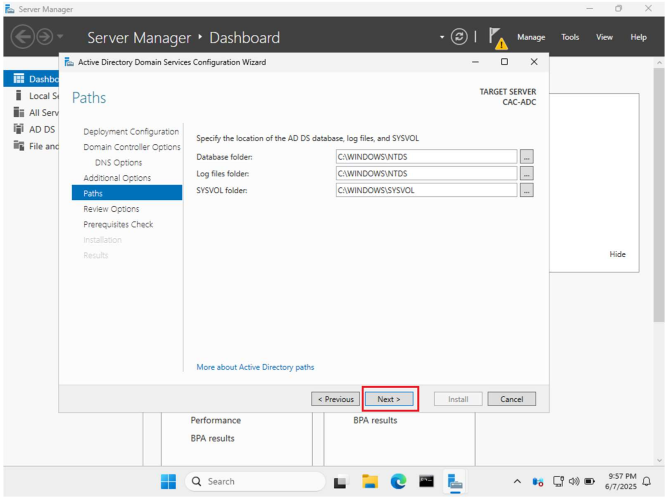
- Keep the defaults, set a DSRM password and click Next.
- Click Next.





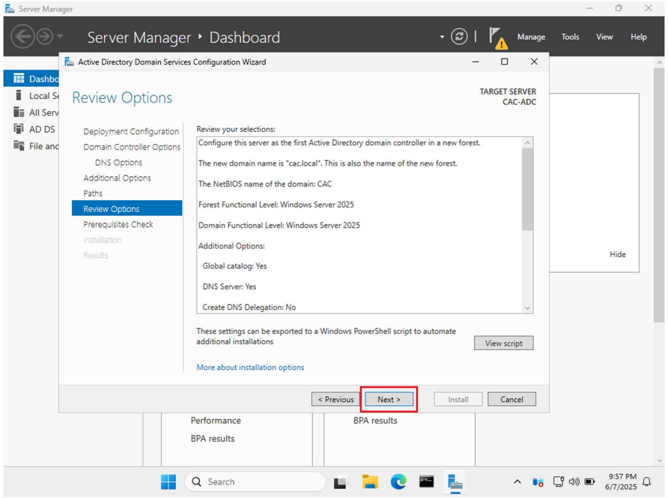
- Click Next.
- Click Next.

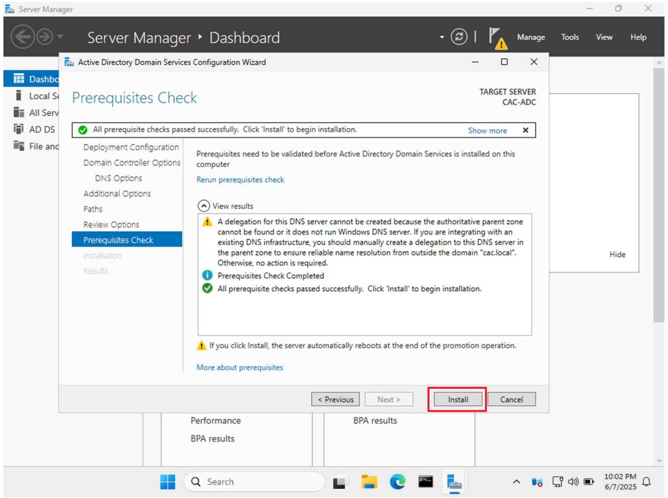




- Click Next.
- Click Next to start the installation. Once finished the VM will

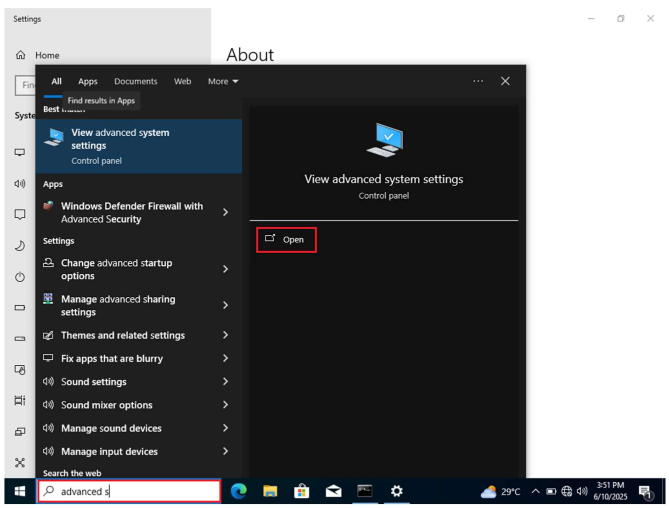
automatically restart.

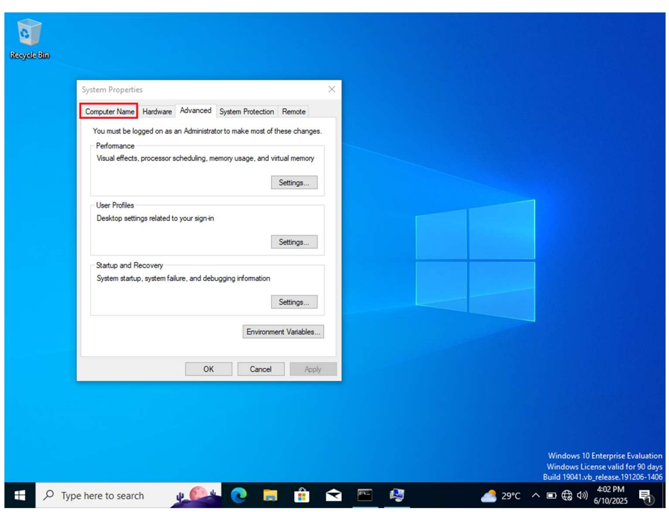




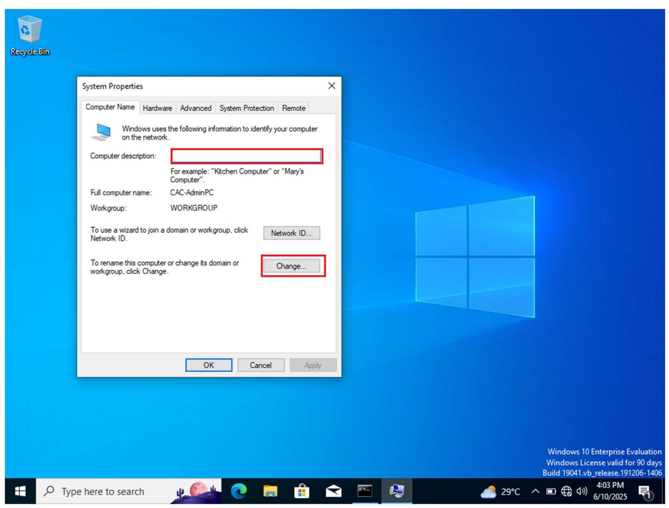
15. Adding Windows 10 Client to the Domain

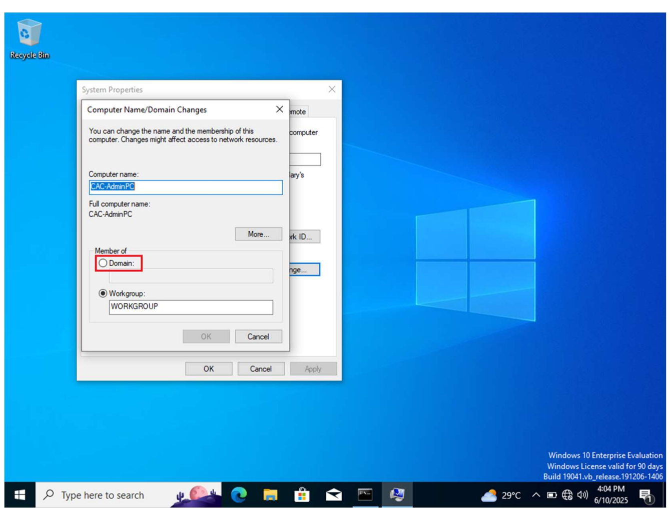
- Open View Advanced System Settings on Window 10 VM.
- Select Computer Name.



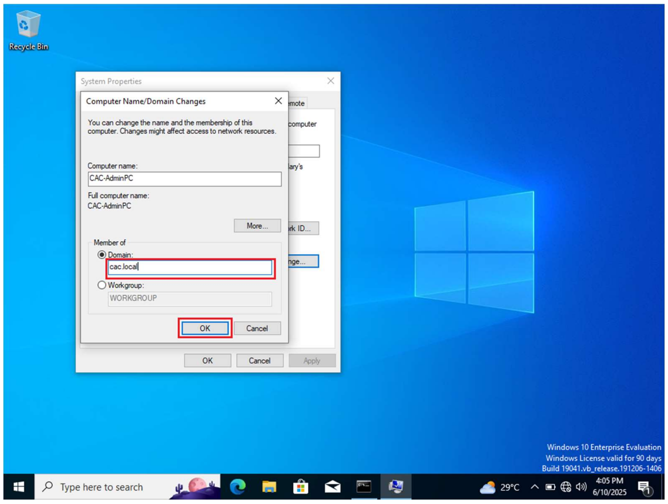


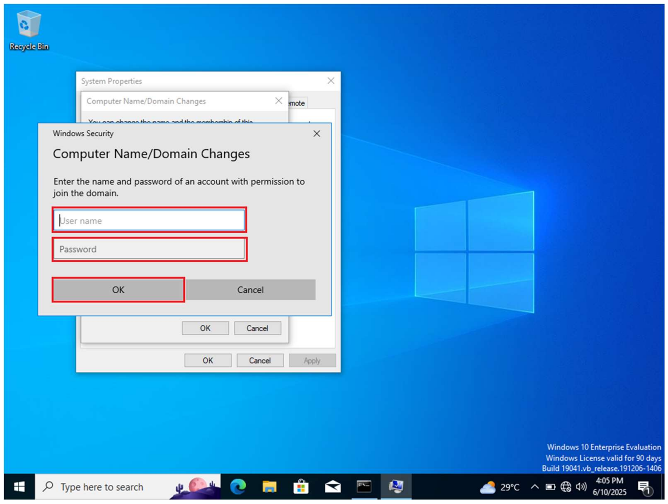
- Click Change.
- Click Domain.



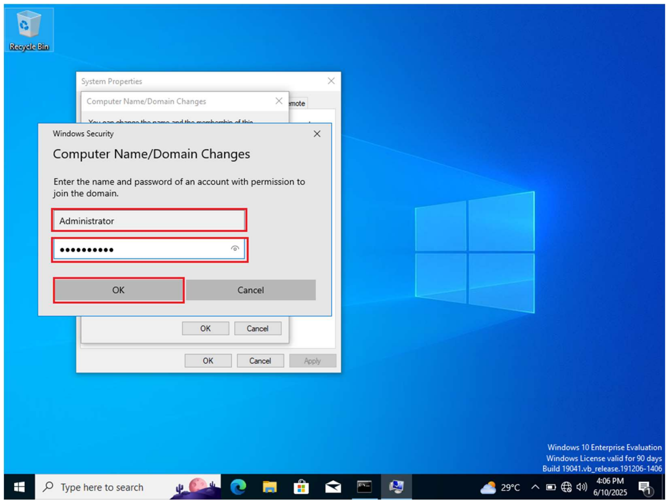


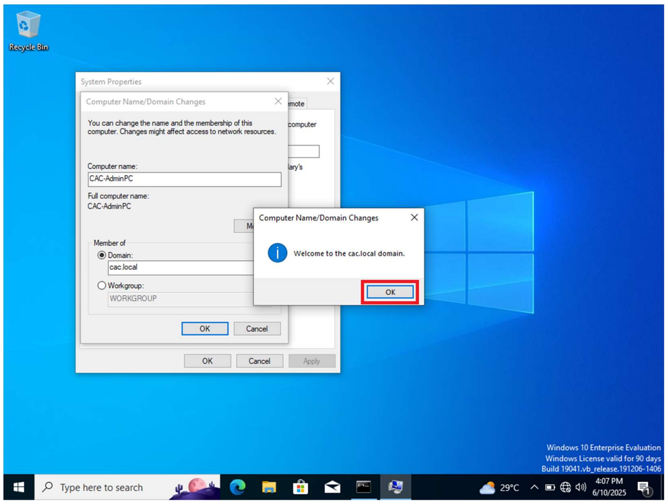
- Enter the Domain name and click OK. (thenameyouchoose.local) I chose CAC.
- Type the Administrator account and password you created for the Domain Controller (username is Administrator) and click OK.





Successfully joined domain



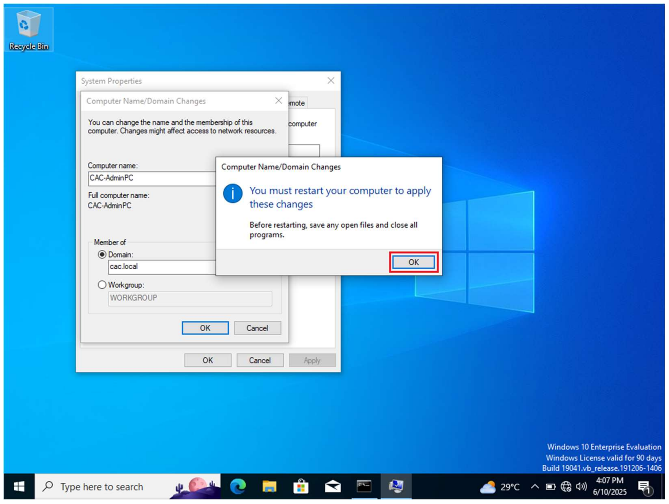


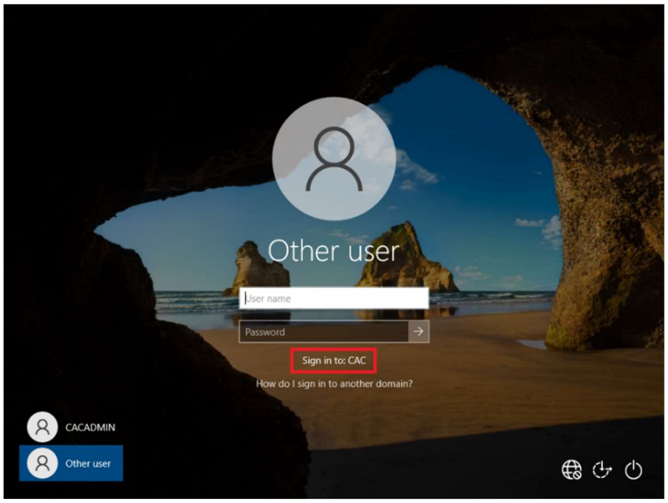
- Click OK to restart.
- After restart and you click other user, you will see Sign in to CAC which is

the name of the domain we created.

You sign in by typing: CAC\Administrator or administrator@cac.local or by

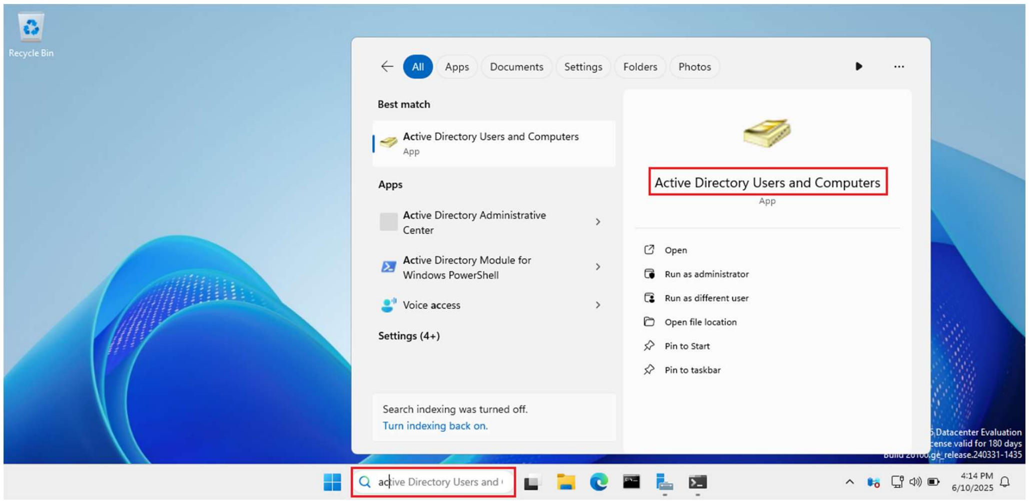
replacing Administrator with any other user on the domain

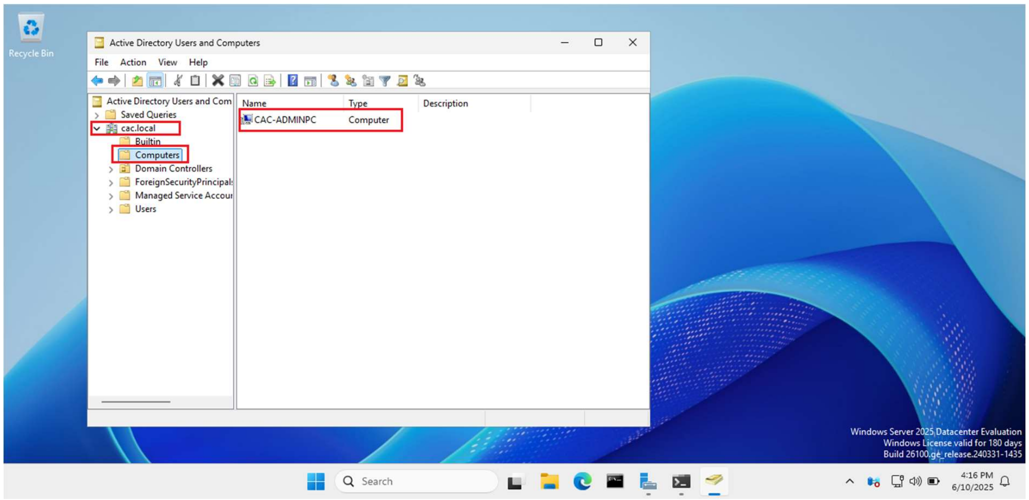




Another way to confirm that you successfully joined the domain is to

Check the computers OU on your Active Directory Domain Controller.





16. Remote Access and NAT Setup for Internet

Connectivity

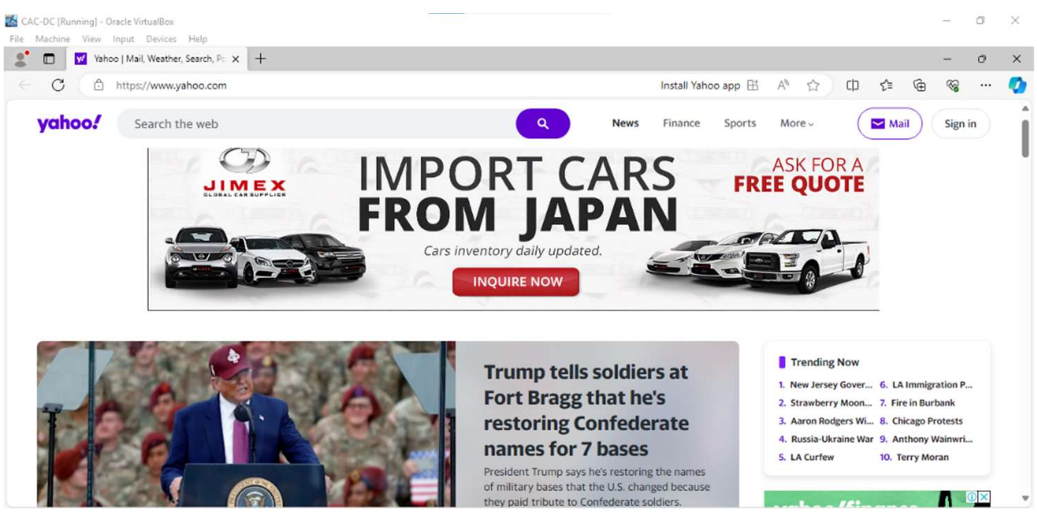
Remote Access, Routing for NAT and Internet Connectivity

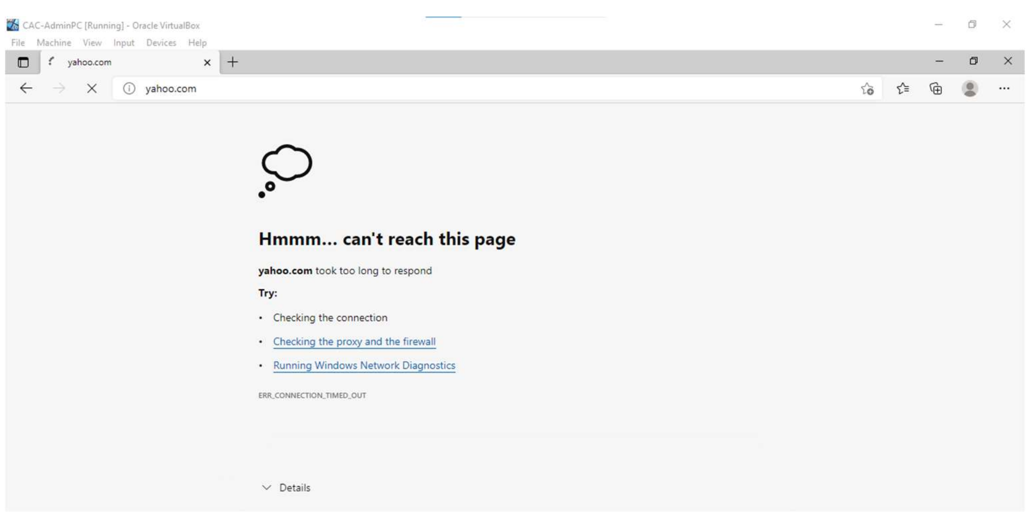
- If you tried to access a website on the domain controller it is possible

but we can not access the internet from the Windows 10 VM.

- This is because we need to setup up routing so that the Windows 10 VM

can access the internet through the DC.





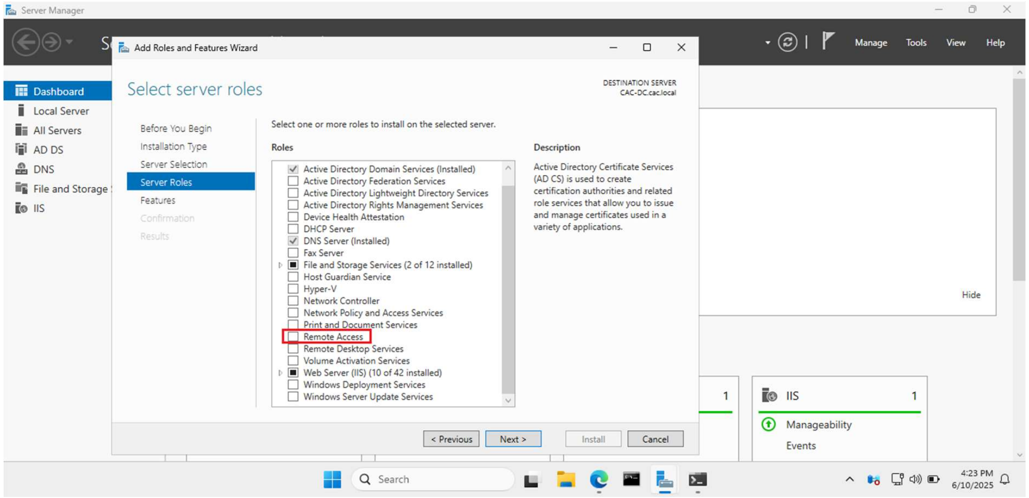
Install Remote Access and Routing

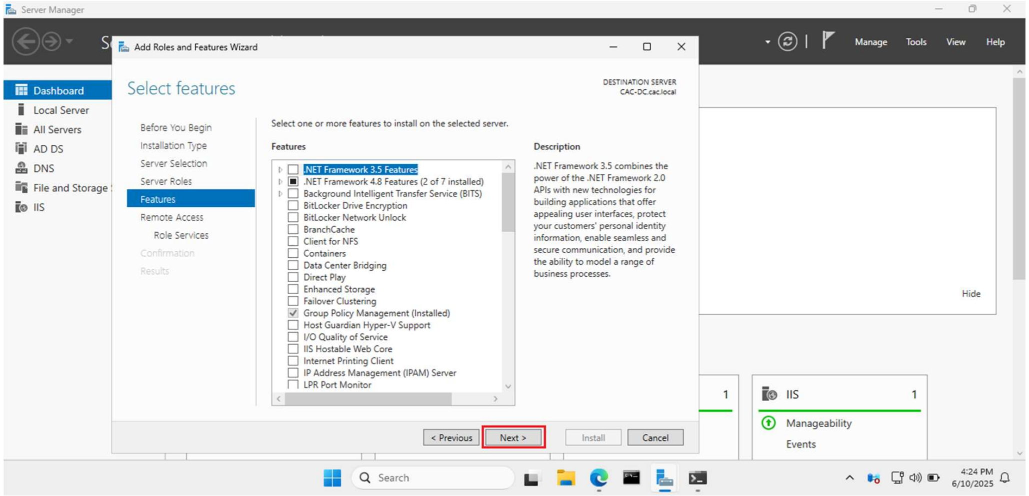
- Open Sever Manager, navigate to Manage and click Add Roles and

Features. Continue to click Next until you reach the Features page.

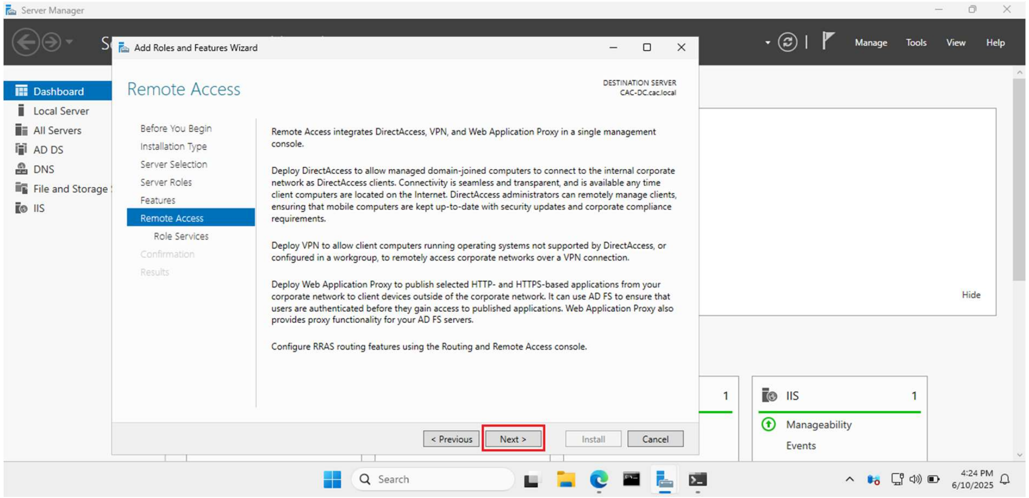
Select Remote Access.

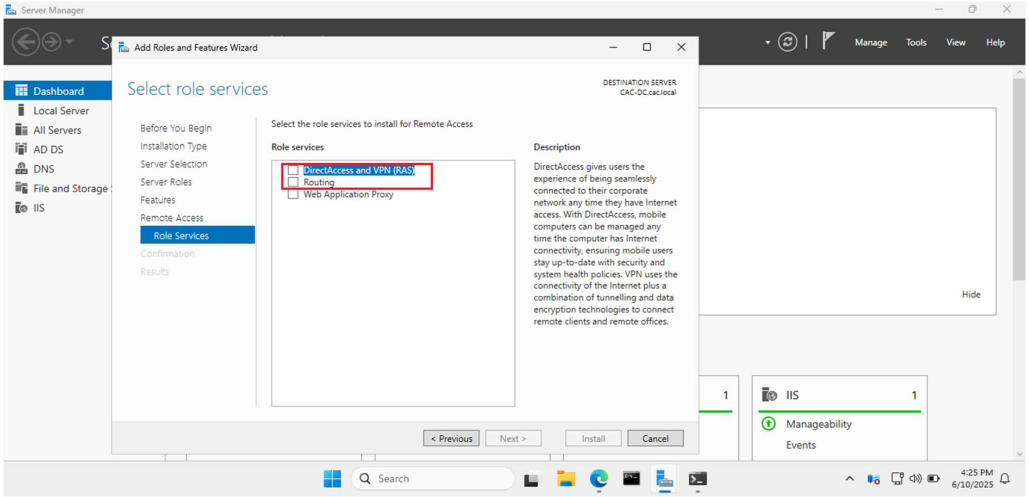
- Click Next.





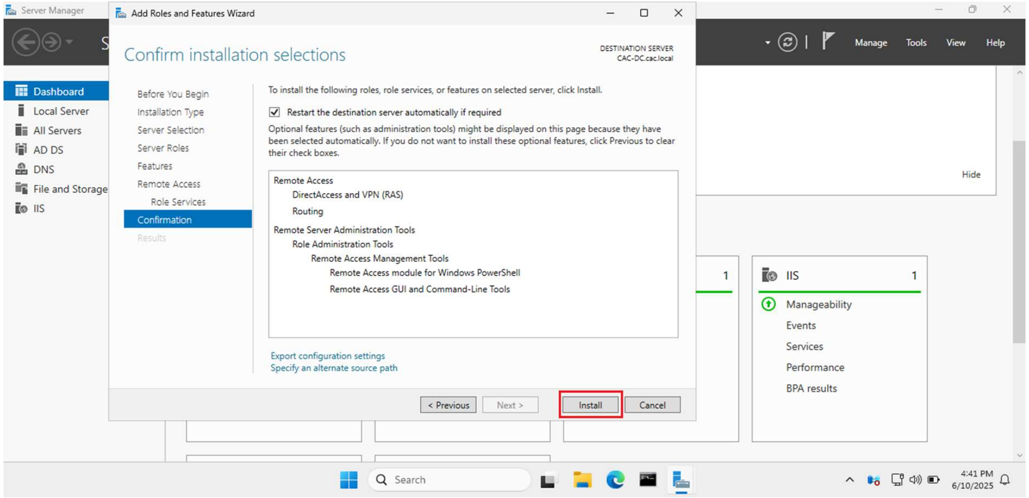
- Click Next.
- Click Select DirectAccess and VPN(RAS) and Routing.

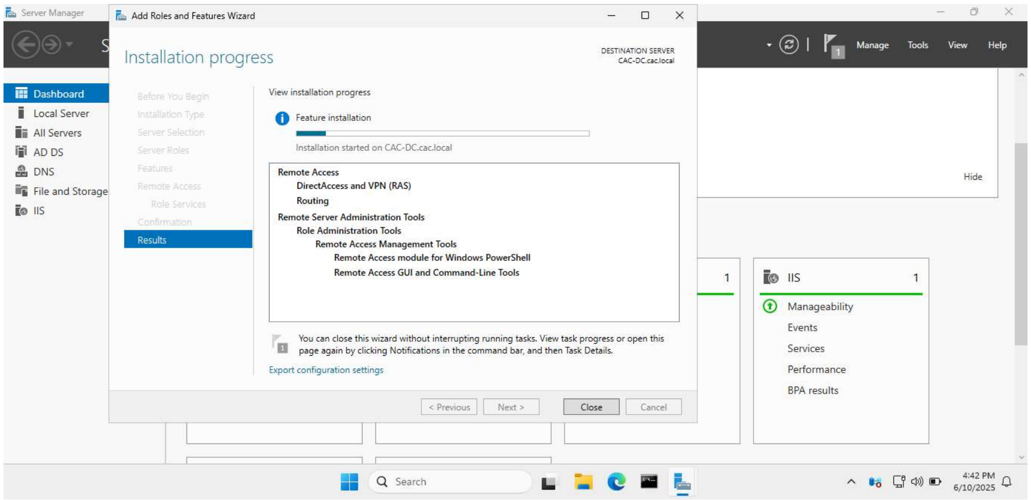




Click Install

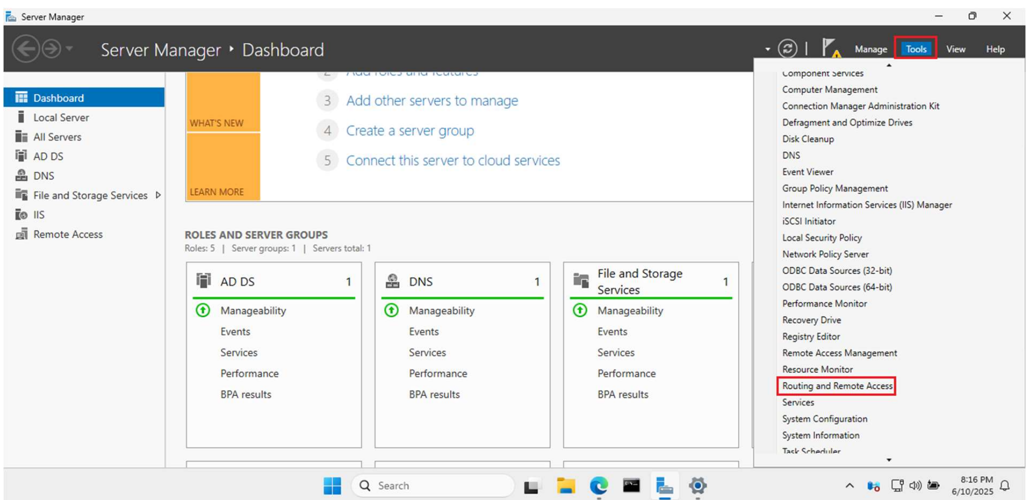
- Installing. After the install is complete Restart your server.

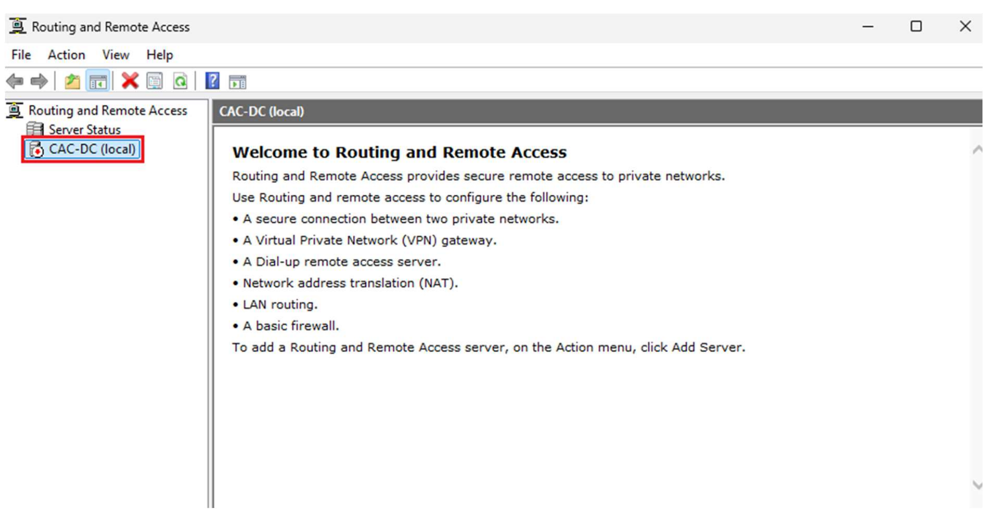




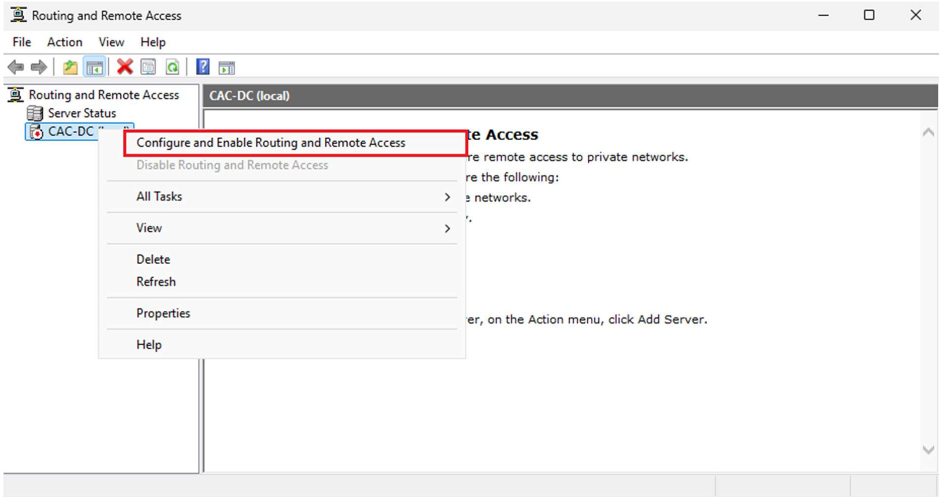
Configure Remote Access and Routing

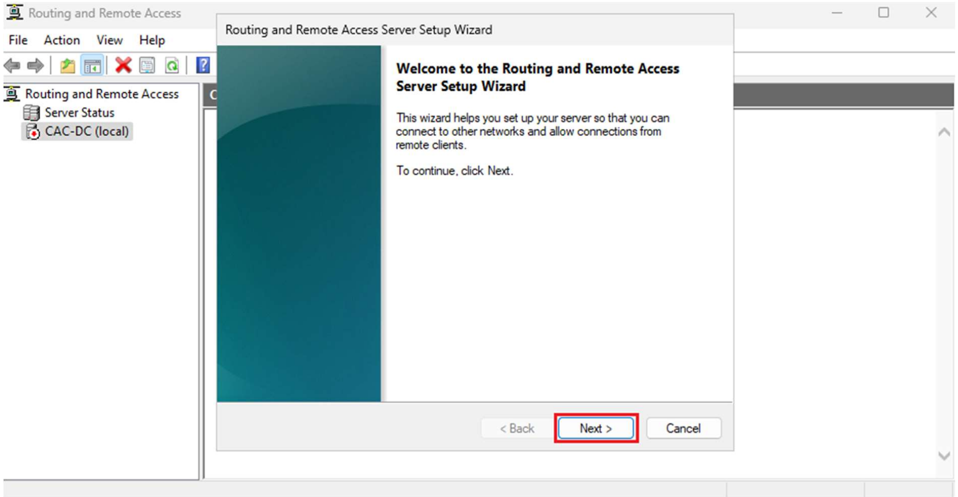
- Open Sever Manager, navigate to Tools and Select Remote and Remote Access.
- Right Click on your Domain Controller.





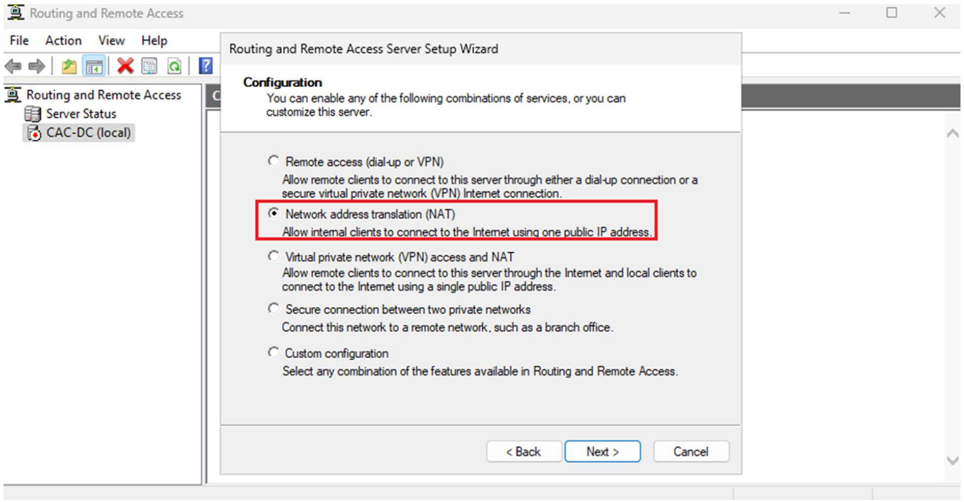
- Select Configure and Enable Routing and Remote Access.
- Click Next.

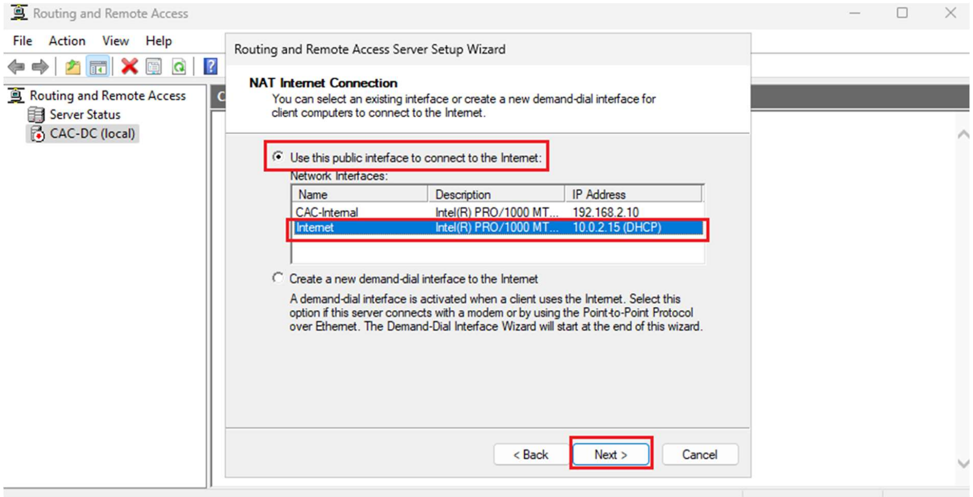




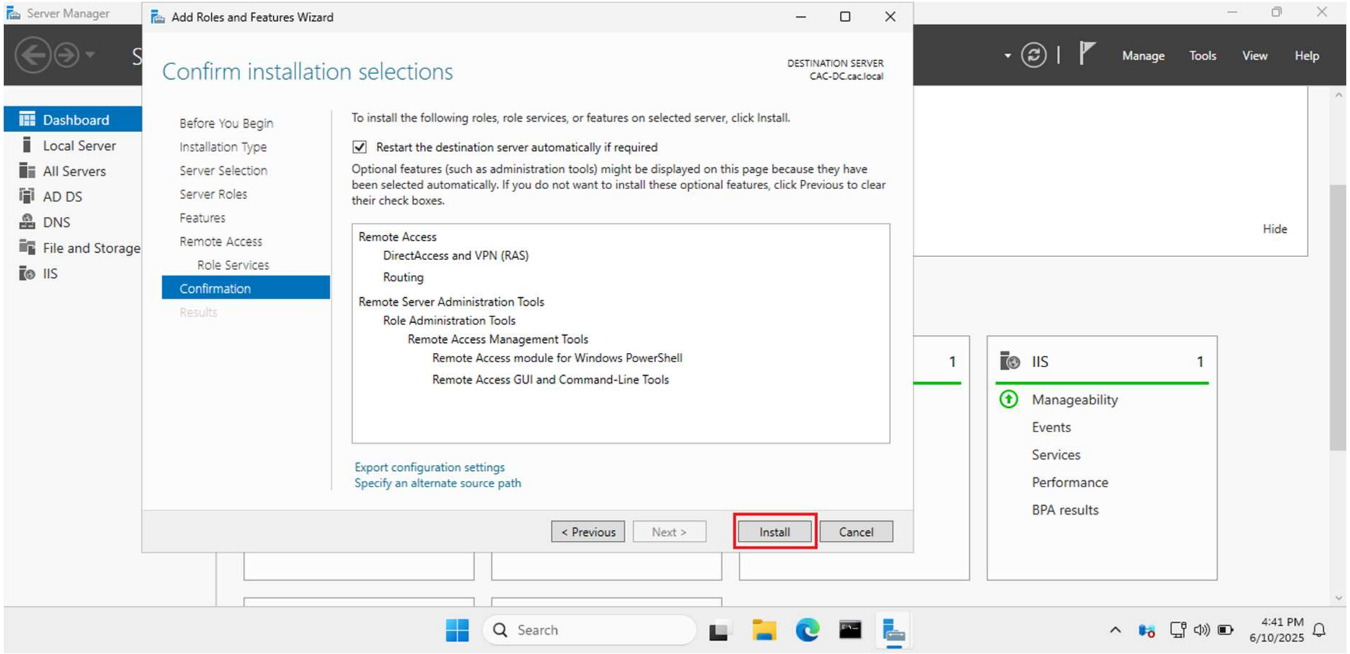
- Select Network address translation (NAT) and click Next.
- Select Use this public interface to the Internet, select your Internet

connection (Ethernet1) and click Next.





- Click Install.



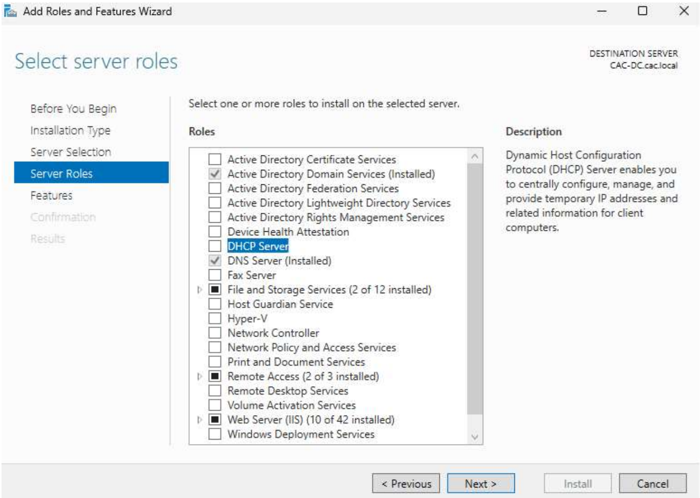
17. Installing and Configuring DHCP Server

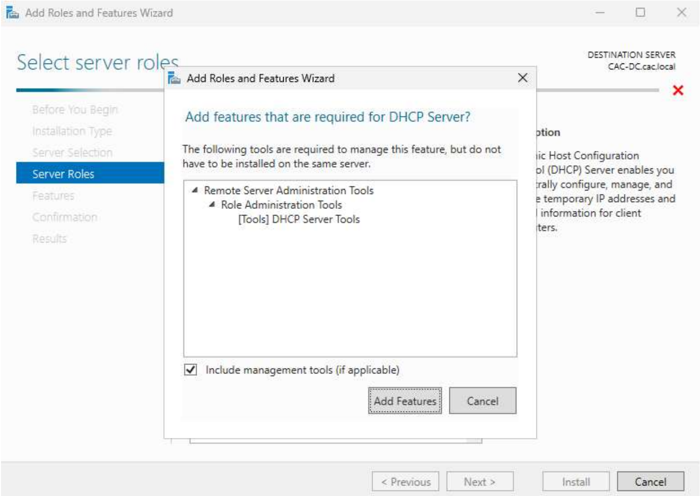
- Open Sever Manager, navigate to Manage and click Add Roles and

Features. Continue to click Next until you reach the Features page.

Select DHCP Server.

- Click Add Features.

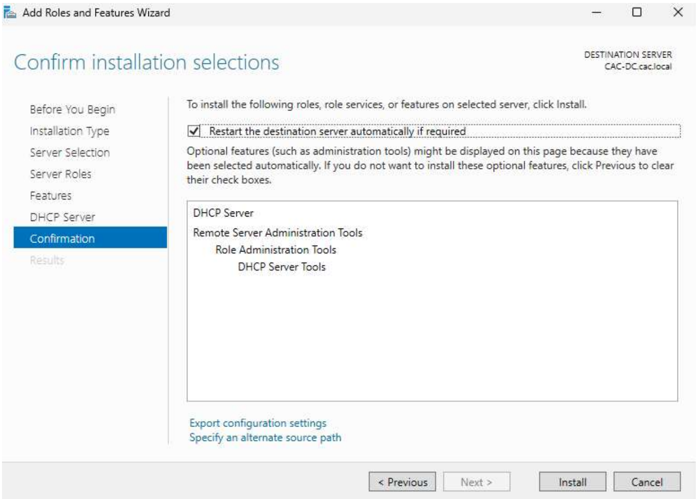


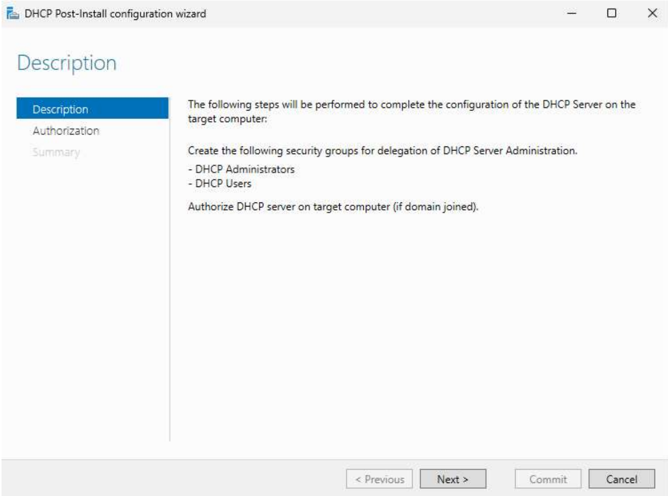


- Click Install.

DHCP Post Installation

- Click Add Features.

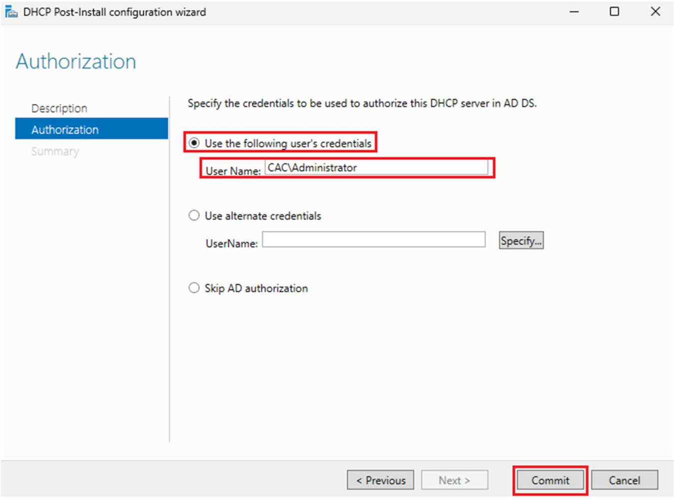


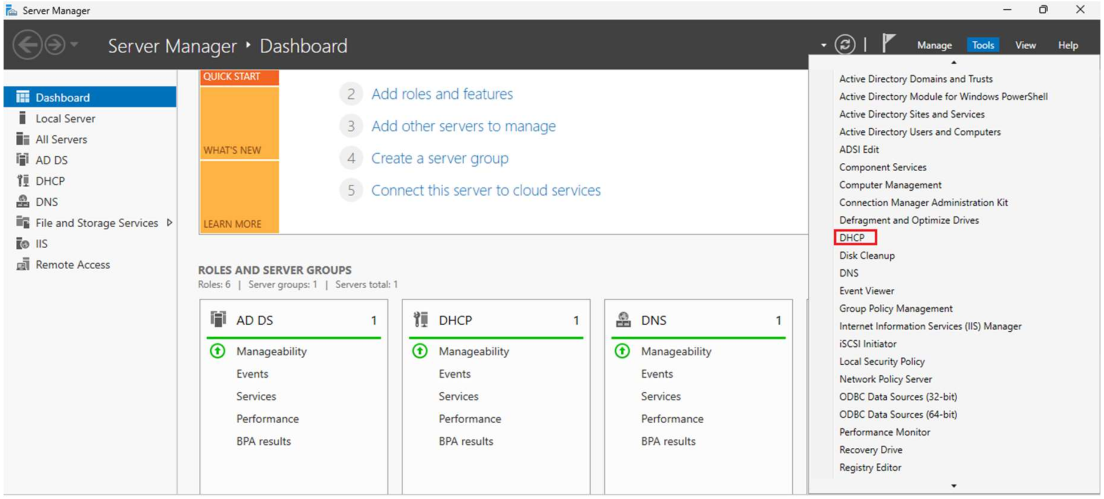


Select Use the following credentials. Username, Select Administrator account and click Commit.

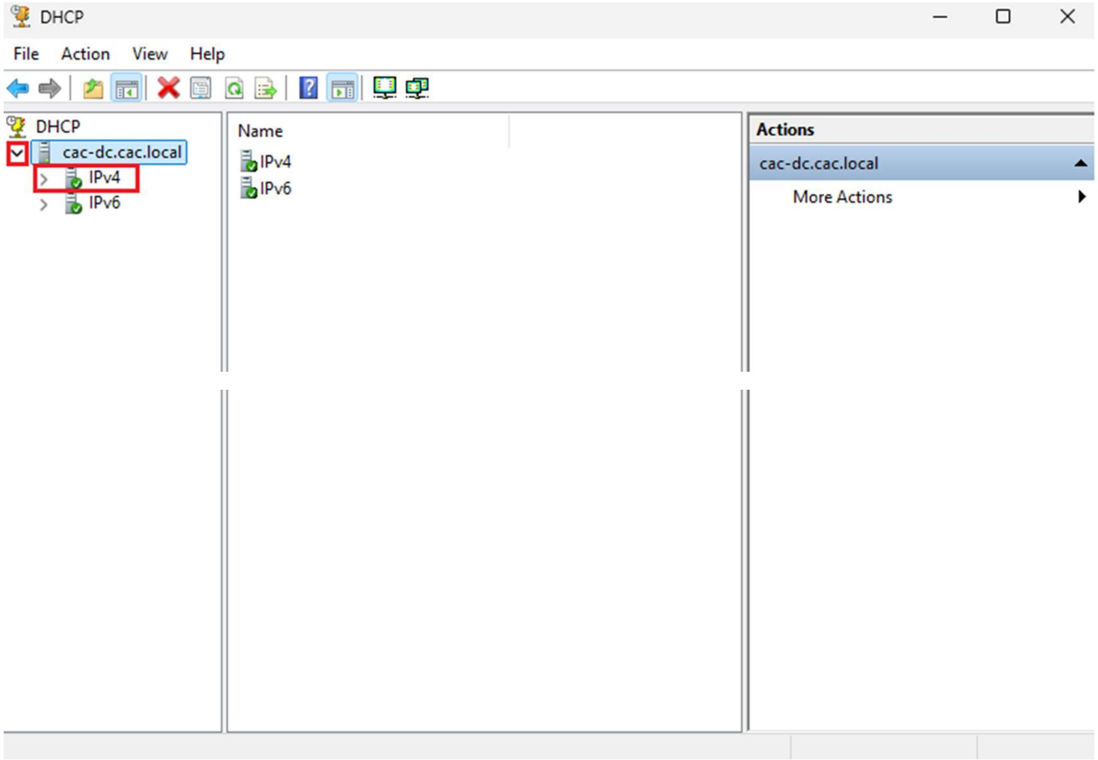
DHCP Configuration

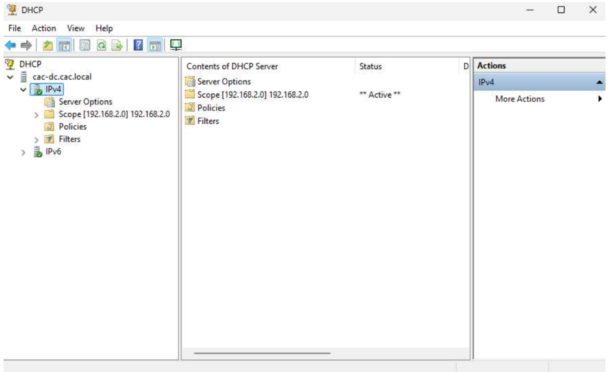
- Open Sever Manager, navigate to Tools and Select DHCP.



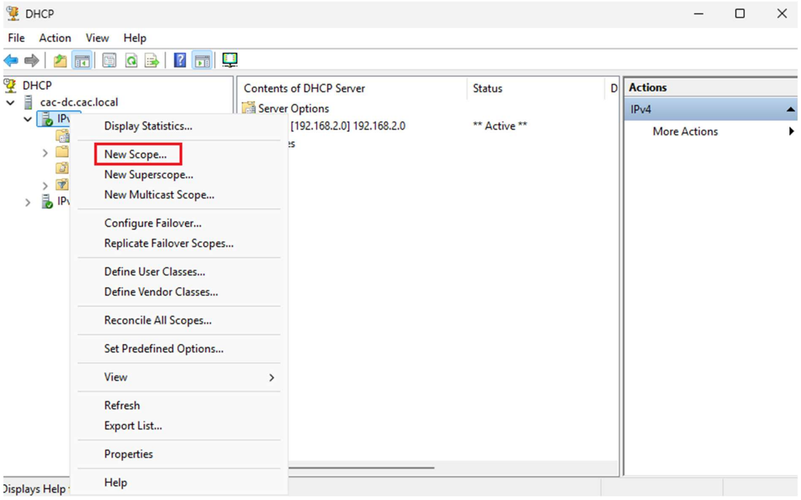


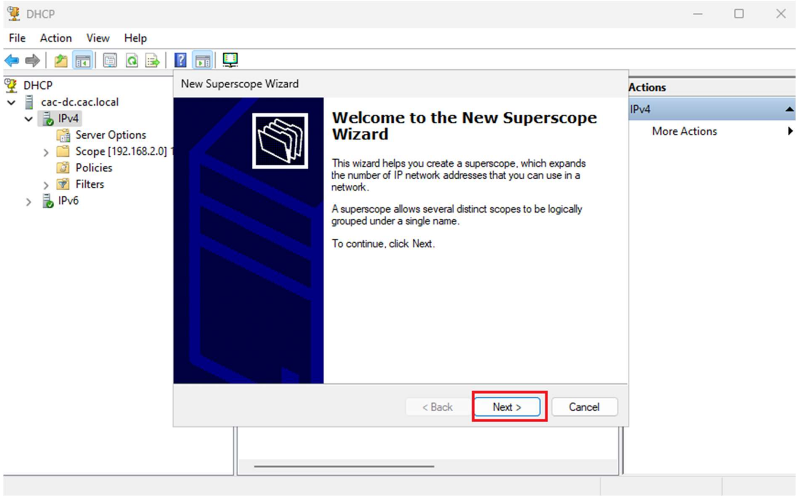
- Click the cross arrow beside the machine name to open up the menu.
- Click on IPV4.





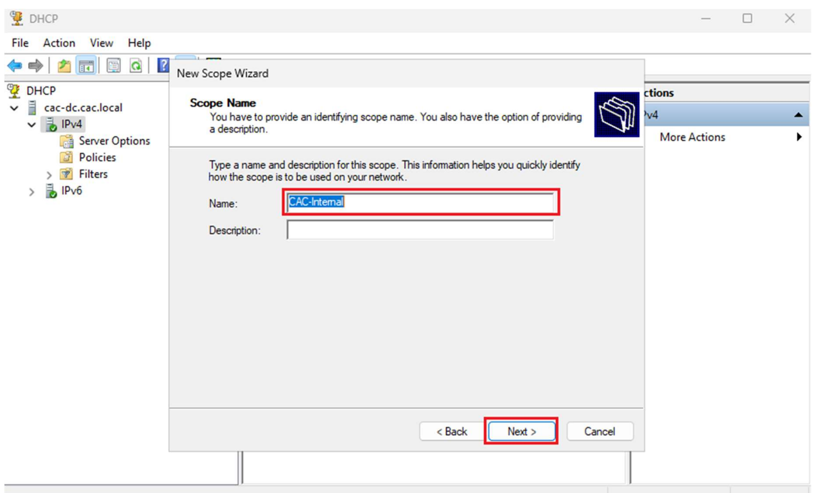
- Right click on IPV4 and click New Scope.
- The New Superscope Wizard will open click Next.

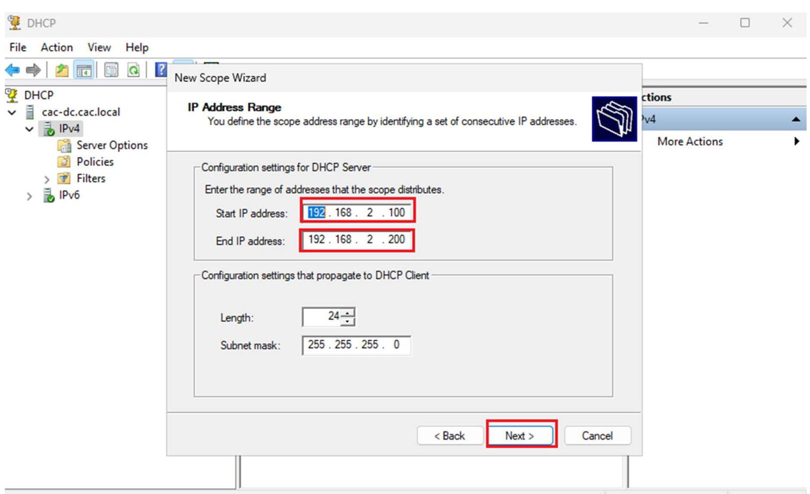




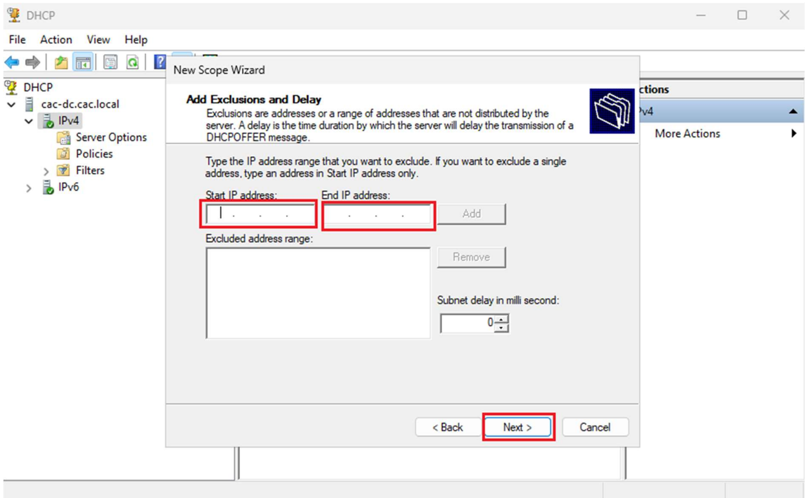
- Name your scope and click Next.
- Set the range of the IP addresses that you want the server to distribute

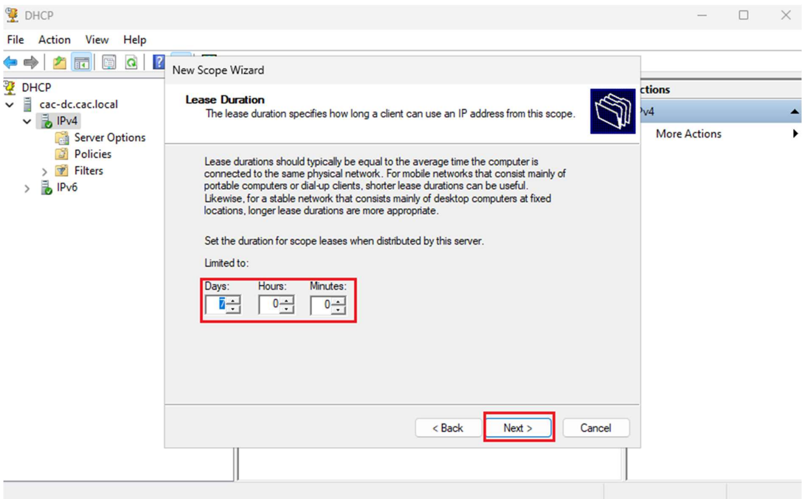
and click Next.





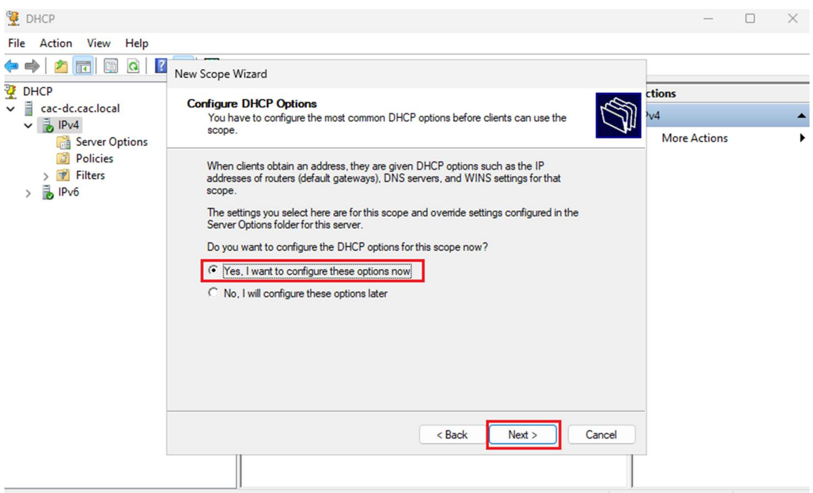
- Add any IP addresses you want the server to exclude and click Next.
- Set the Lease Duration and click Next.

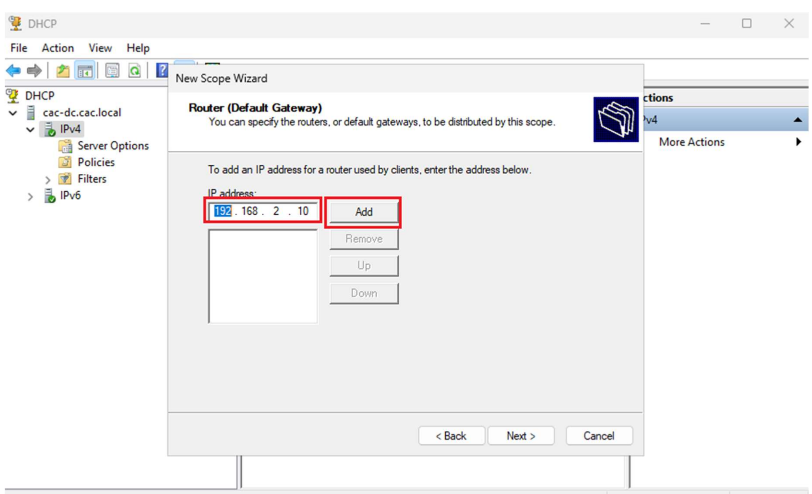




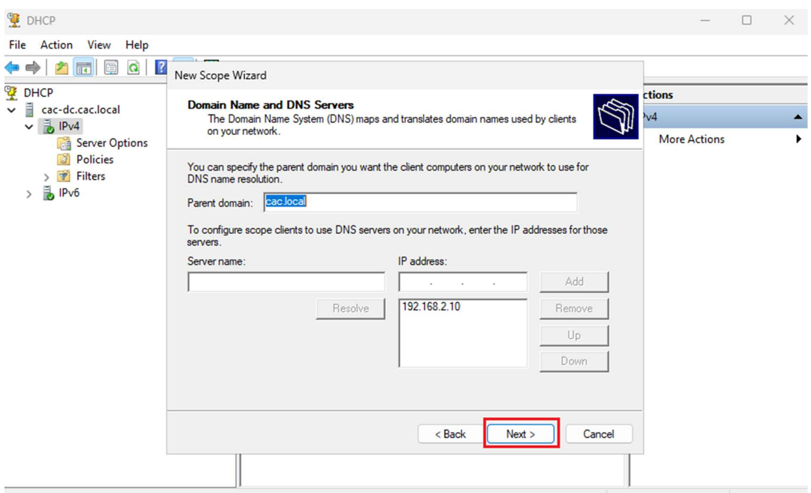
- Select Yes, I want to configure these options now and click Next.
- Add the IP address of the router to be used by clients and click Add and

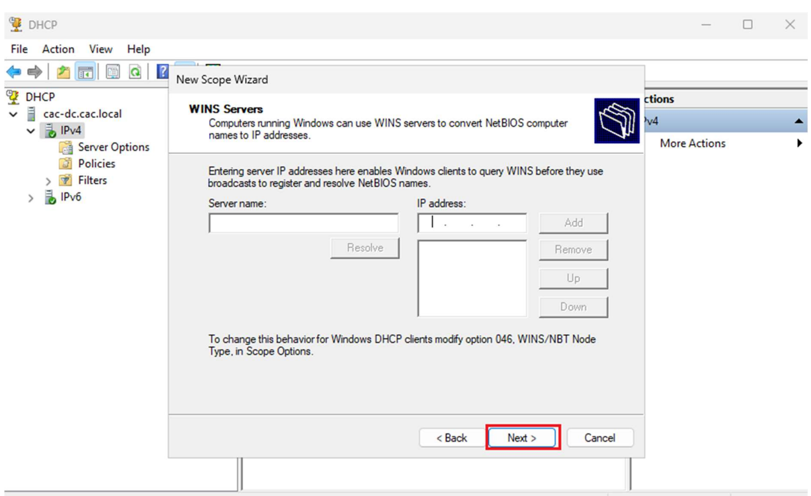
Next.



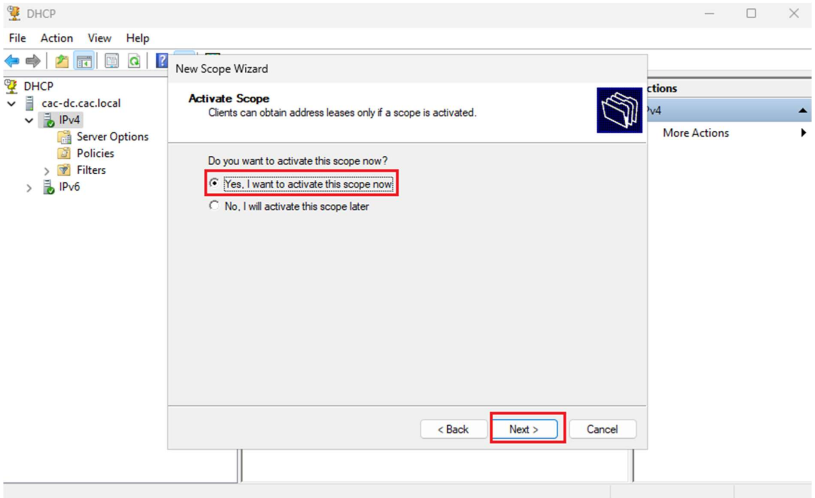


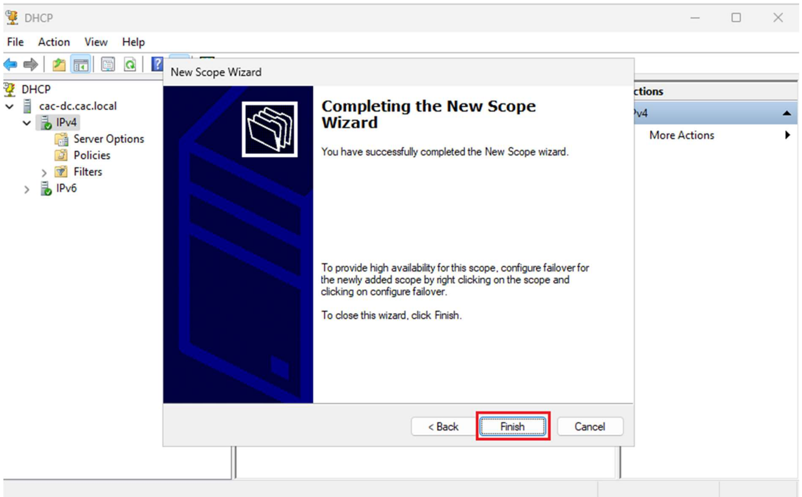
- Click Next.
- Click Next.





- Select Yes, I want to configure these options now and click Next.
- Click Finish.





18. Testing and Results

1. Domain Controller Verification

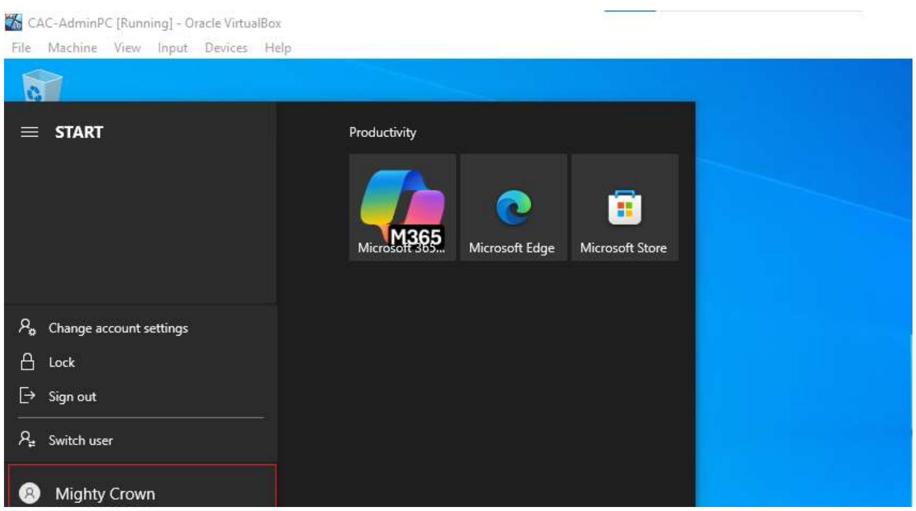
a) Log into Windows Server 2025

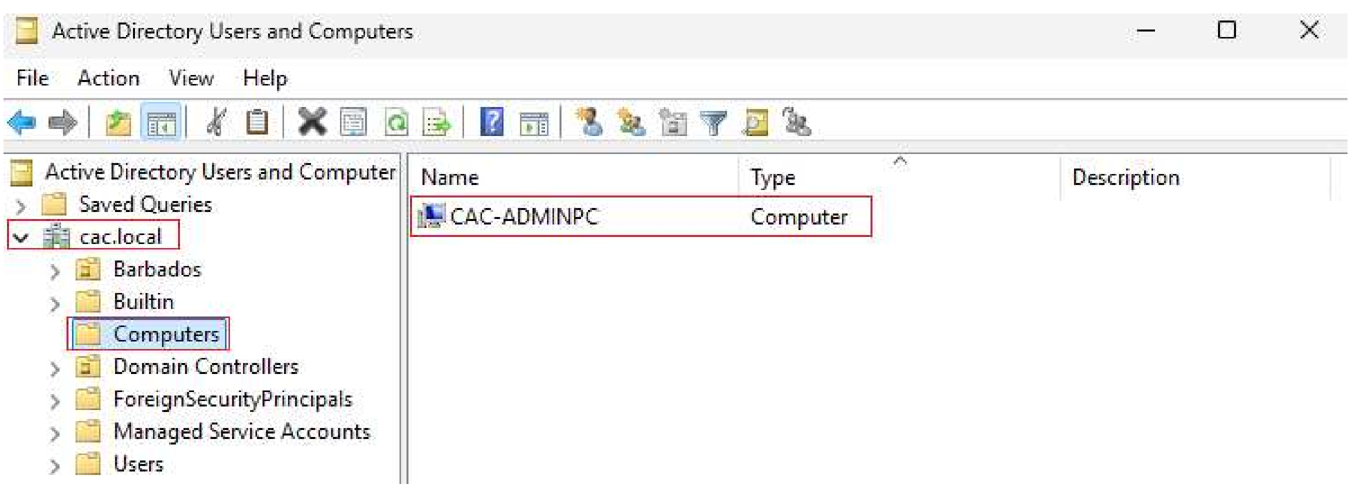
b) Check Active Directory

Open Server Manager >> Tools >> Active Directory Users and Computers.

Confirm the domain is listed.

Confirm that your Windows 10 client computer appear under the Computers (OU).



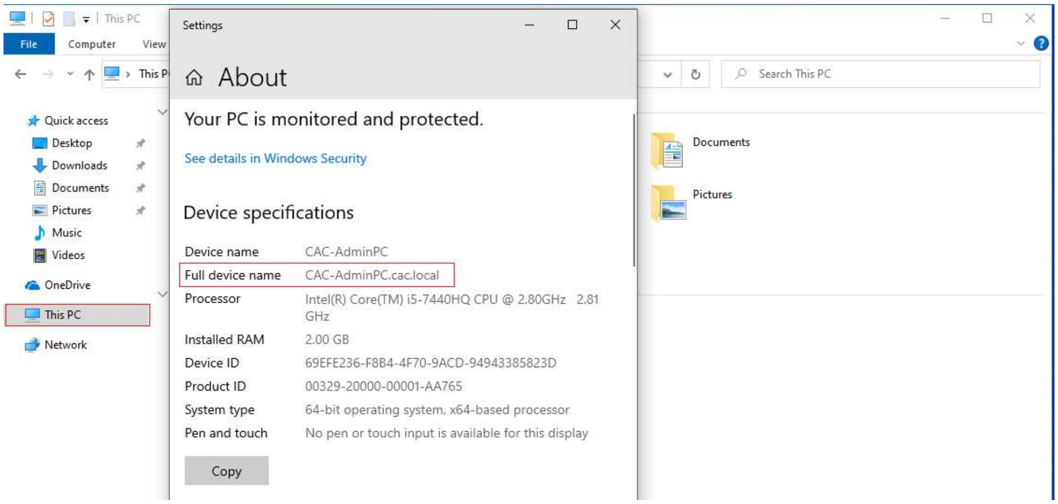


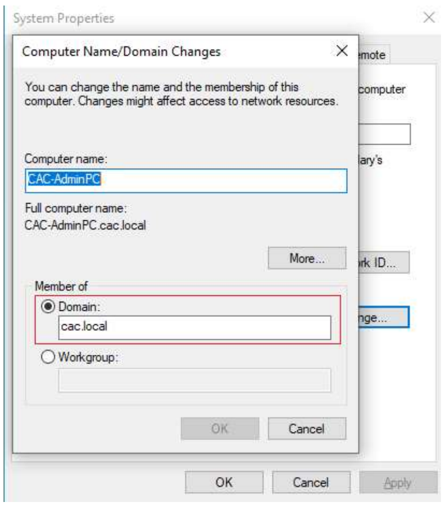
2. Client Join Verification

Log in to Windows 10 VM and check domain membership

Right click This PC >> Properties >> confirm the domain name

Go to Settings >> Advanced >> Computer Name >> Change, to verify.





3. Network Communication

Open Command Prompt on Windows 10 and run

ping cac.local

ping 192.168.2.10 (Servers IP address)

Results

Open Command Prompt on the Server and run

ping cac.local

ping 192.168.2.30 (Client's IP address)

Results

```
C:\Users\crownm>ping cac.local

Pinging cac.local [192.168.2.10] with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time<1ms TTL=128
Reply from 192.168.2.10: bytes=32 time=1ms TTL=128
Reply from 192.168.2.10: bytes=32 time=1ms TTL=128
Reply from 192.168.2.10: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\crownm>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time<1ms TTL=128
Reply from 192.168.2.10: bytes=32 time<1ms TTL=128
Reply from 192.168.2.10: bytes=32 time<1ms TTL=128
Reply from 192.168.2.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\Users\Administrator>ping cac.local

Pinging cac.local [10.0.2.15] with 32 bytes of data:
Reply from 10.0.2.15: bytes=32 time<1ms TTL=128
Reply from 10.0.2.15: bytes=32 time<1ms TTL=128
Reply from 10.0.2.15: bytes=32 time<1ms TTL=128
Reply from 10.0.2.15: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.2.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>ping 192.168.2.30

Pinging 192.168.2.30 with 32 bytes of data:
Reply from 192.168.2.30: bytes=32 time<1ms TTL=128
Reply from 192.168.2.30: bytes=32 time=1ms TTL=128
Reply from 192.168.2.30: bytes=32 time=1ms TTL=128
Reply from 192.168.2.30: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

4. DHCP Functionality

On Windows 10, open cmd and run: Ipconfig/all

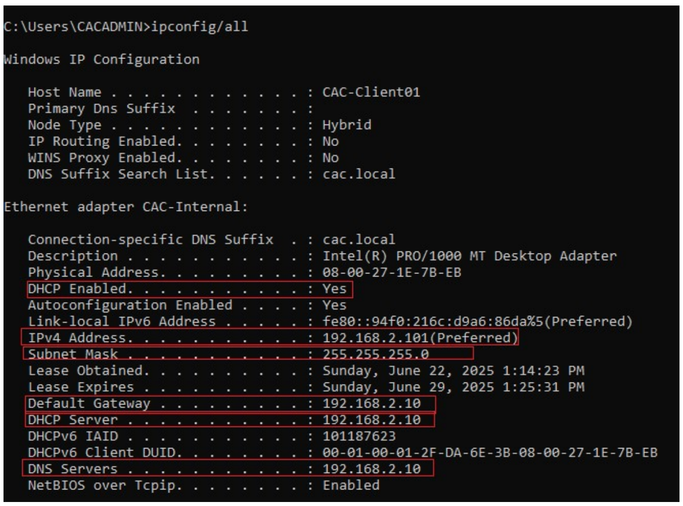
This verify that the IP, Subnet Mask, Gateway, and DNS were assigned by the

DHCP Server.

Results

In Server Manager >> DHCP >> IPV4 >> Scope >> Address Leases

Results



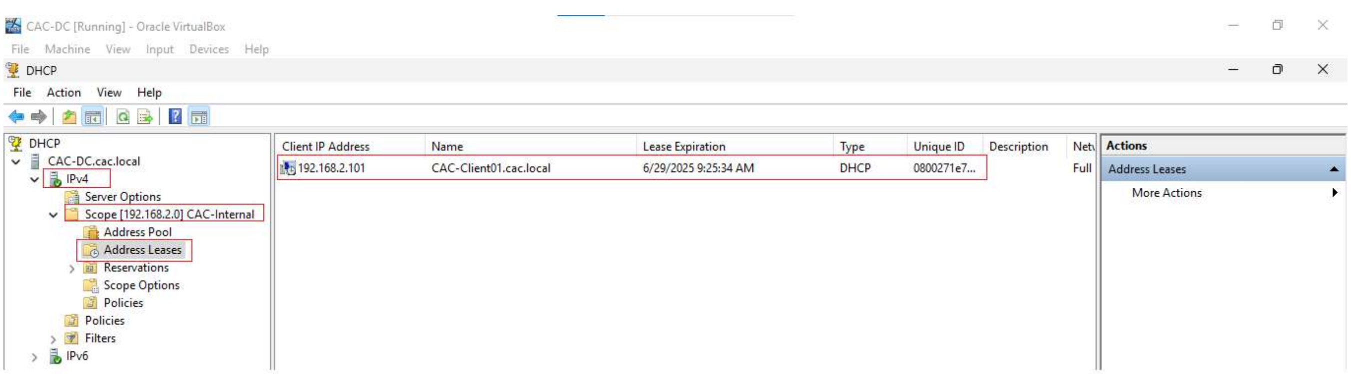
```
C:\Users\CACADMIN>ipconfig/all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : CAC-Client01
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : cac.local

Ethernet adapter CAC-Internal:

   Connection-specific DNS Suffix  . : cac.local
   Description . . . . . . . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
   Physical Address. . . . . . . . . : 08-00-27-1E-7B-EB
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::94f0:216c:d9a6:86da%5(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.2.101(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Sunday, June 22, 2025 1:14:23 PM
   Lease Expires . . . . . . . . . . : Sunday, June 29, 2025 1:25:31 PM
   Default Gateway . . . . . . . . . : 192.168.2.10
   DHCP Server . . . . . . . . . . . : 192.168.2.10
   DHCPv6 IAID . . . . . . . . . . . : 101187623
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2F-DA-6E-3B-08-00-27-1E-7B-EB
   DNS Servers . . . . . . . . . . . : 192.168.2.10
   NetBIOS over Tcpip. . . . . . . . : Enabled
```



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5. DNS Resolution

From Window 10 VM run: nslookup cac.local

Results

This verifies that DNS is resolving.

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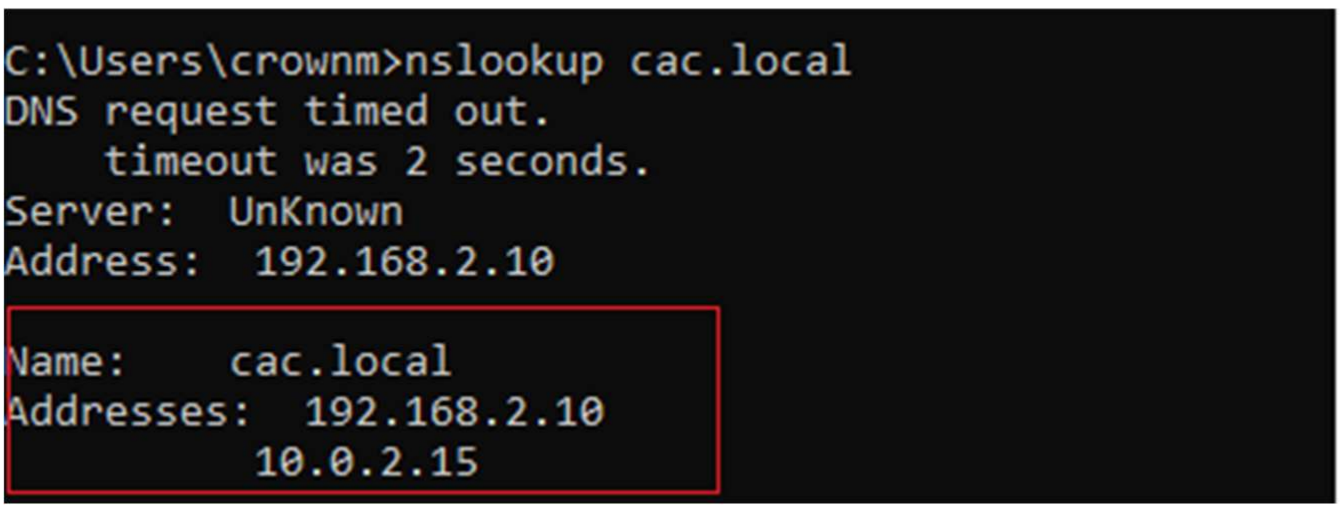
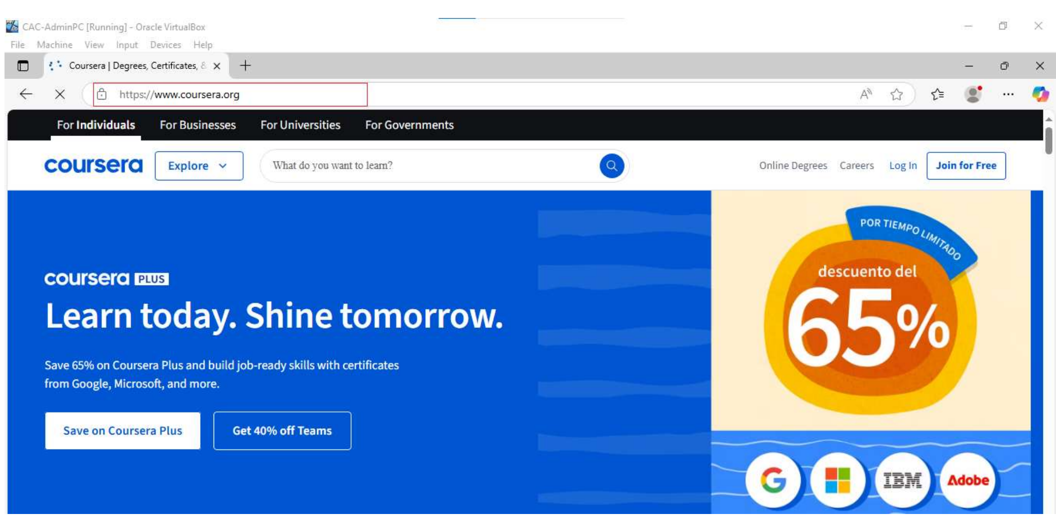
6. Internet Access via NAT (Windows 10)

Open browser on Windows 10 machines: Navigate to a website

Results





7. Group Policy Test

To test group policy, I am going to set a desktop background.

Create a Shared Folder on the Windows Server and place the image in the folder. Right click on folder Properties >> Sharing >> Advanced Sharing >> Permissions. Select Everyone >> Read >> OK.

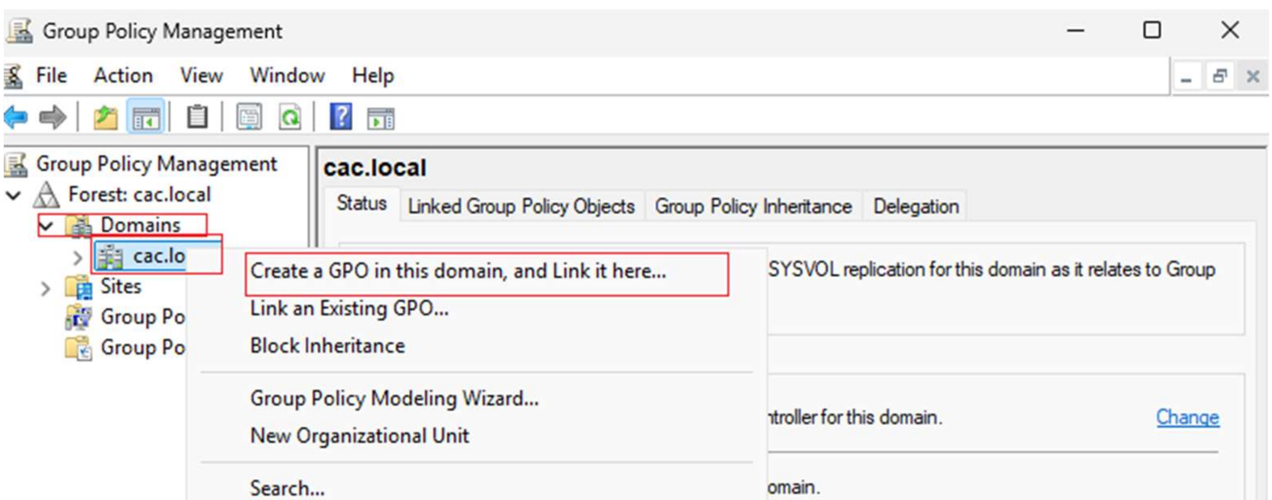
Next, go to Start >> Server Manager >> Tools >> Group Policy Management.

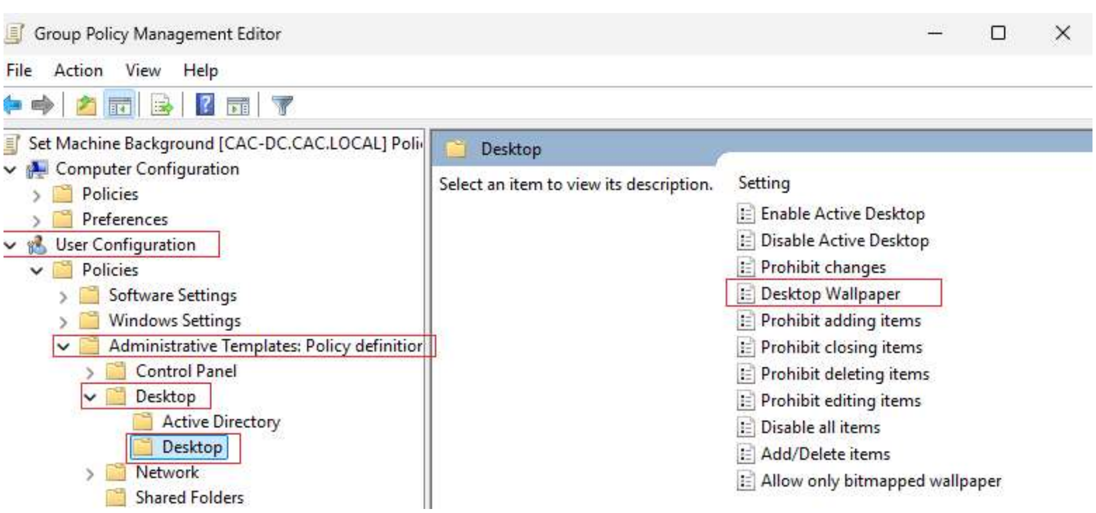
Right click on the domain and select Create GPO in this domain and link it here.

Name the Policy and click OK.

Right Click on the policy and click Edit. User Configuration >> Administrative Template >> Desktop >> Desktop, then double click on Desktop Wallpaper

Enable and enter the path of the wallpaper image. i.e (C:\Images\photo.jpg)





On windows 10 machine, open Command Prompt and run: gpupdate /force

Log out of the machine and log back in.

19. Troubleshooting Section

Common Issues and Solutions

Issue 1: Cannot Join Windows 10 to Domain

Symptoms: Error message when attempting to join domain, "The specified

domain either does not exist or could not be contacted"

Solutions:

1. Check DNS Settings:

o

Verify Windows 10 DNS points to Domain Controller IP

(192.168.2.10)

o

Run nslookup cac.local from Windows 10 - should resolve to DC IP

2. Verify Network Connectivity:

o

Ping Domain Controller: ping 192.168.2.10

o

Ping domain name: ping cac.local

3. Check Domain Controller Services:

o

Open Services.msc on DC

o

Verify "Active Directory Domain Services" is running

o

Verify "DNS Server" is running

Issue 2: No Internet Access on Windows 10

Symptoms: Can ping Domain Controller but cannot access websites

Solutions:

1. Check NAT Configuration:

- Verify Remote Access is installed and configured
- Check that NAT is using the correct external interface

(Internet/NAT adapter)

2. Verify DHCP Settings:

- Check that DHCP scope includes correct gateway (192.168.2.10)

3. DNS Resolution:

Verify DNS forwarders are configured on DC

- Test with nslookup google.com

Issue 3: DHCP Not Assigning IP Addresses

Symptoms: Windows 10 gets 169.x.x.x address instead of 192.168.2.x

Solutions:

1. Activate DHCP Scope:

- Open DHCP console
- Right-click scope → Ac(cid:415)vate

2. Check DHCP Authorization:

- Right-click DHCP server → Authorize
- Verify server shows green arrow (authorized)

3. Verify Scope Configuration:

- Ensure scope range doesn't conflict with static IPs
- Check exclusions are properly configured

Issue 4: Virtual Machine Won't Start

Symptoms: VM fails to boot or shows errors

Solutions:

1. Check Virtualization:

- Verify VT-x/AMD-V is enabled in BIOS
- Restart VirtualBox if needed

2. Memory Issues:

- Verify sufficient RAM is available on host
- Reduce VM memory allocation if necessary

3. Storage Issues:

Check available disk space

- Verify VM disk files aren't corrupted

Issue 5: Guest Additions Installation Fails

Symptoms: Cannot install Guest Additions, features don't work

Solutions:

1. Prerequisites:

- Ensure VM is fully updated
- Install as Administrator

2. Clean Installation:

- Uninstall old Guest Additions first
- Restart VM before installing new version

3. Manual Installation:

- Download Guest Additions ISO separately
- Mount manually if auto-mount fails

General Troubleshooting Steps:

1. Check Event Logs: Windows Event Viewer often contains detailed error information

2. Verify Network Configuration: Use ipconfig /all to check network settings

3. Test Connectivity: Use ping, telnet, and nslookup for network diagnostics

4. Restart Services: Many issues resolve with service restarts

Check Firewall: Windows Firewall can block domain communications